18.4 Error Codes Table

Diagnosis display	Abnormality/Protection control	Abnormality judgement	Primary location to verify
H00	No abnormality detected	—	—
H12	Indoor/Outdoor capacity unmatched	90s after power supply	• Indoor/outdoor connection wire • Indoor/outdoor PCB • Specification and combination table in catalogue
H15	Outdoor compressor temperature sensor abnormality	Continue for 5 sec.	• Compressor temperature sensor (defective or disconnected)
H20	Water pump abnormality	Continue for 10 sec.	• Indoor PCB • Water pump (malfunction)
H23	Indoor refrigerant liquid temperature sensor abnormality	Continue for 5 sec.	• Refrigerant liquid temperature sensor (defective or disconnected)
H27	Service valve error	Continue for 5 minutes	• High pressure sensor (defective or disconnected)
H28	Abnormal solar sensor	Continue for 5 sec.	• Solar temperature sensor (defective or disconnected)
H31	Abnormal swimming pool sensor	Continue for 5 sec.	• Pool temperature sensor (defective or disconnected)
H36	Abnormal buffer tank sensor	Continue for 5 sec.	• Buffer tank sensor (defective or disconnected)
H38	Brand code not match	When indoor and outdoor brand code not same	—
H42	Compressor low pressure abnormality	—	• Outdoor pipe temperature sensor • Clogged expansion valve or strainer • Insufficient refrigerant • Outdoor PCB • Compressor
H43	Abnormal Zone 1 sensor	Continue for 5 sec.	• Water temperature Zone 1 sensor
H44	Abnormal Zone 2 sensor	Continue for 5 sec.	• Water temperature Zone 2 sensor
H62	Water flow switch abnormality	Continue for 1 min.	• Water flow switch
H63	Abnormal low pressure sensor	4 times in 20 minutes	• Low pressure sensor (defective or disconnect)
H64	Refrigerant high pressure abnormality	Continue for 5 sec.	• Outdoor high pressure sensor (defective or disconnected)
H65	Abnormal deice water circulation	water flow > 7 l/min continuously for 20 seconds during anti freeze deice	• Water pump
H67	Abnormal External Thermistor 1	Continue for 5 sec.	• Room temperature Zone 1 sensor
H68	Abnormal External Thermistor 2	Continue for 5 sec.	• Room temperature Zone 2 sensor
H70	Back-up heater OLP abnormality	Continue for 60 sec.	• Back-up heater OLP (Disconnection or activated)
H72	Tank sensor abnormal	Continue for 5 sec.	• Tank sensor
H74	PCB communication error	Communication or transfer error	• Indoor main PCB and Sub PCB
H75	Low water temperature control	Room heater disable and deice request to operate under low water temperature	• Heater operation must enable to increase water temperature
H76	Indoor - control panel communication abnormality	—	• Indoor - control panel (defective or disconnected)
H90	Indoor/outdoor abnormal communication	> 1 min after starting operation	• Internal/external cable connections • Indoor/Outdoor PCB
H91	Tank heater OLP abnormality	Continue for 60 sec.	• Tank heater OLP (Disconnection or activated)
H95	Indoor/Outdoor wrong connection	—	• Indoor/Outdoor supply voltage
H98 / F95	Outdoor high pressure overload protection Cooling high pressure overload protection	—	• Outdoor high pressure sensor • Water pump or water leakage • Clogged expansion valve or strainer • Excess refrigerant • Outdoor PCB
H99	Indoor heat exchanger freeze prevention	—	• Indoor heat exchanger • Refrigerant shortage
F12	Pressure switch activate	4 times occurrence within 30 minutes	• Pressure switch
F14	Outdoor compressor abnormal revolution	4 times occurrence within 20 minutes	• Outdoor compressor
F15	Outdoor fan motor lock abnormality	2 times occurrence within 30 minutes	• Outdoor PCB • Outdoor fan motor
F16	Total running current protection	3 times occurrence within 20 minutes	• Excess refrigerant • Outdoor PCB

Diagnosis display	Abnormality/Protection control	Abnormality judgement	Primary location to verify
F20	Outdoor compressor overheating protection	4 times occurrence within 30 minutes	• Compressor tank temperature sensor • Clogged expansion valve or strainer • Insufficient refrigerant • Outdoor PCB • Compressor
F22	IPM (power transistor) overheating protection	3 times occurrence within 30 minutes	• Improper heat exchange • IPM (Power transistor)
F23	Outdoor Direct Current (DC) peak detection	7 times occurrence continuously	• Outdoor PCB • Compressor
F24	Refrigeration cycle abnormality	2 times occurrence within 30 minutes	• Insufficient refrigerant • Outdoor PCB • Compressor low compression
F25	Cooling/Heating cycle changeover abnormality	4 times occurrence within 30 minutes	• 4-way valve • V-coil
F27	Pressure switch abnormality	Continue for 1 min.	• Pressure switch
F30	Water outlet sensor 2 abnormality	Continue for 5 sec.	• Water outlet sensor 2 (defective or disconnected)
F32	Abnormal Internal Thermostat	Continue for 5 sec.	• Control panel PCB thermostat
F36	Outdoor air temperature sensor abnormality	Continue for 5 sec.	• Outdoor air temperature sensor (defective or disconnected)
F37	Indoor water inlet temperature sensor abnormality	Continue for 5 sec.	• Water inlet temperature sensor (defective or disconnected)
F40	Outdoor discharge pipe temperature sensor abnormality	Continue for 5 sec.	• Outdoor discharge pipe temperature sensor (defective or disconnected)
F41	PFC control	4 times occurrence within 10 minutes	• Voltage at PFC
F42	Outdoor heat exchanger temperature sensor abnormality	Continue for 5 sec.	• Outdoor heat exchanger temperature sensor (defective or disconnected)
F43	Outdoor defrost sensor abnormality	Continue for 5 sec.	• Outdoor defrost sensor (defective or disconnected)
F45	Indoor water outlet temperature sensor abnormality	Continue for 5 sec.	• Water outlet temperature sensor (defective or disconnected)
F46	Outdoor Current Transformer open circuit	—	• Insufficient refrigerant • Outdoor PCB • Compressor low
F48	Outdoor EVA outlet temperature sensor abnormality	Continue for 5 sec.	• Outdoor EVA outlet temperature sensor (defective or disconnected)
F49	Outdoor bypass outlet temperature sensor abnormality	Continue for 5 sec.	• Outdoor bypass outlet temperature sensor (defective or disconnected)

18.5 Self-Diagnosis Method

18.5.1 Connection Capability Rank Abnormality (H12)

Malfunction Decision Conditions:

During startup operation of cooling and heating, the capability rank of indoor checked by the outdoor is used to determine connection capability rank abnormality.

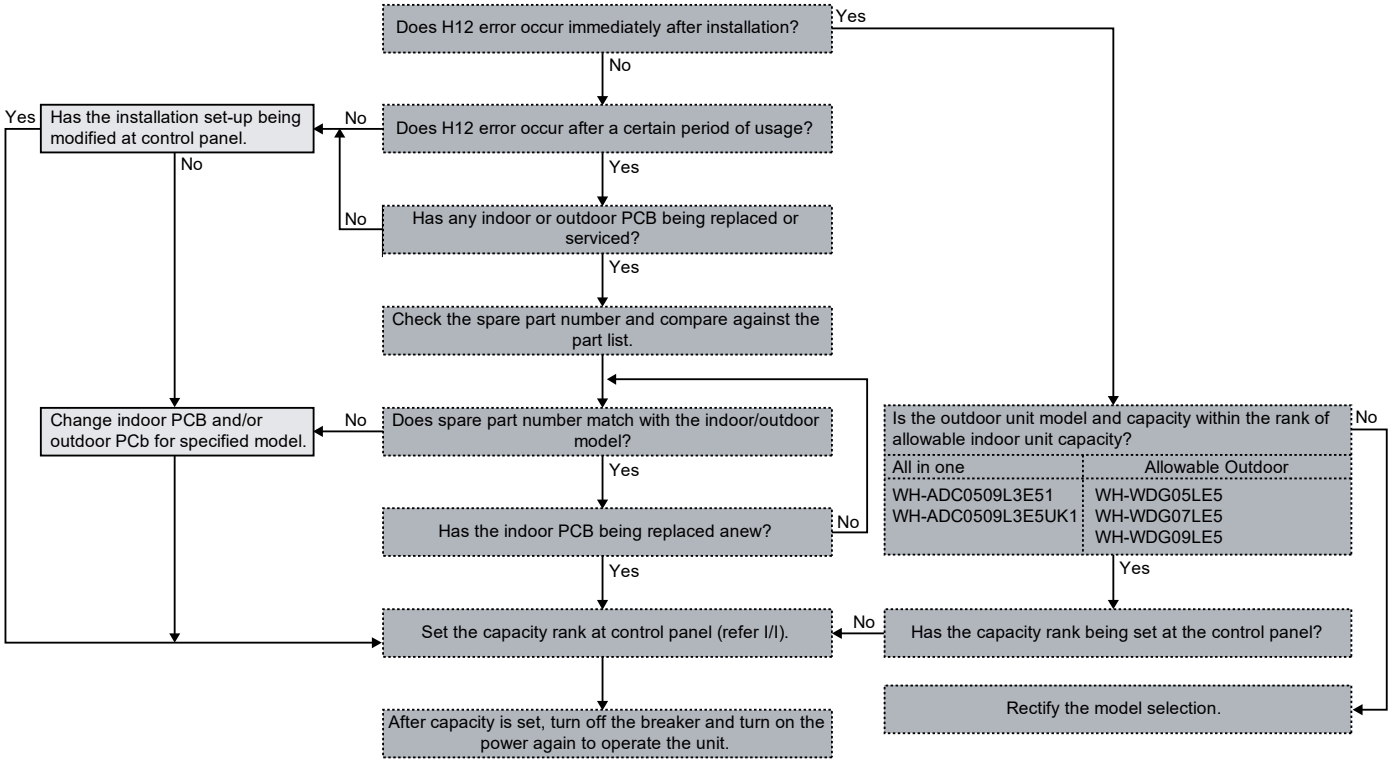
Malfunction Caused:

1 Wrong model interconnected.
2 Wrong indoor unit or outdoor unit PCB (main) used.
3 Faulty indoor unit or outdoor unit PCB (main).

Abnormality Judgment:

Continue for 90 seconds.

Troubleshooting:



18.5.2 Compressor Tank Temperature Sensor Abnormality (H15)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the compressor tank temperature sensor are used to determine sensor error.

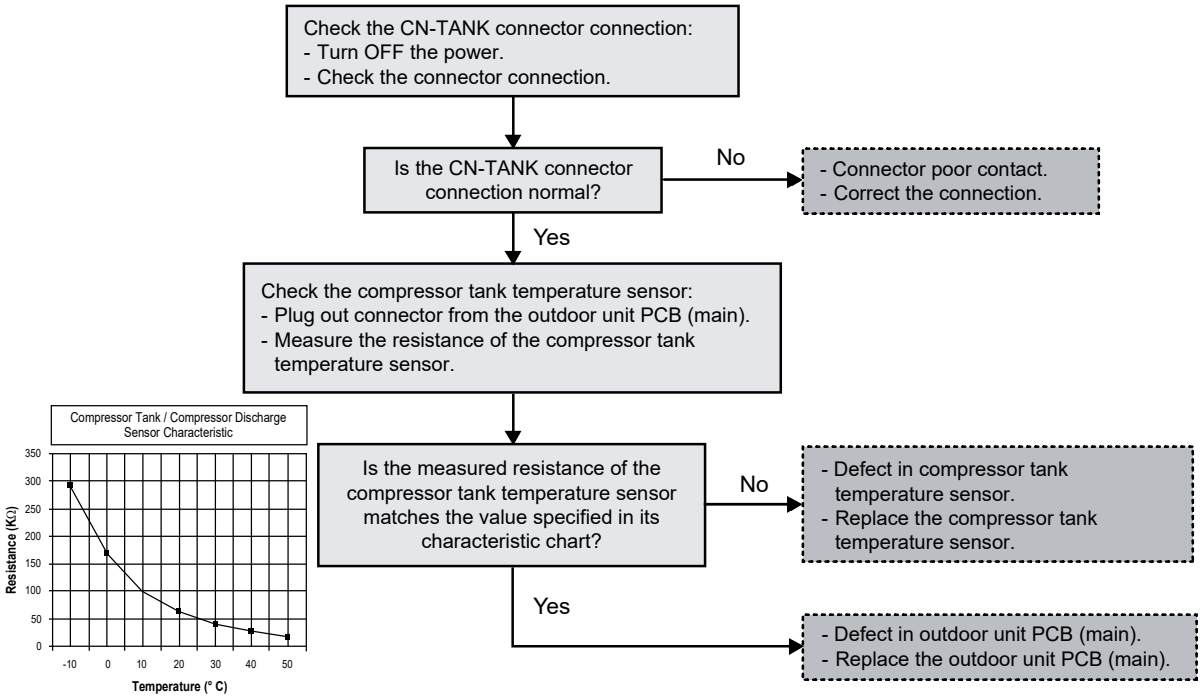
Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:



18.5.3 Water Pump Abnormality (H20)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the rotation speed detected by the IPM of water pump motor during water pump operation is used to determine abnormal water pump (feedback of rotation > 6,000rpm or < 1,000rpm).

Malfunction Caused:

1 Operation stop due to short circuit inside the water pump motor winding.
2 Operation stop due to breaking of wire inside the water pump motor.
3 Operation stop due to breaking of water pump lead wires.
4 Operation stop due to water pump motor IPM malfunction.
5 Operation error due to faulty indoor unit PCB.

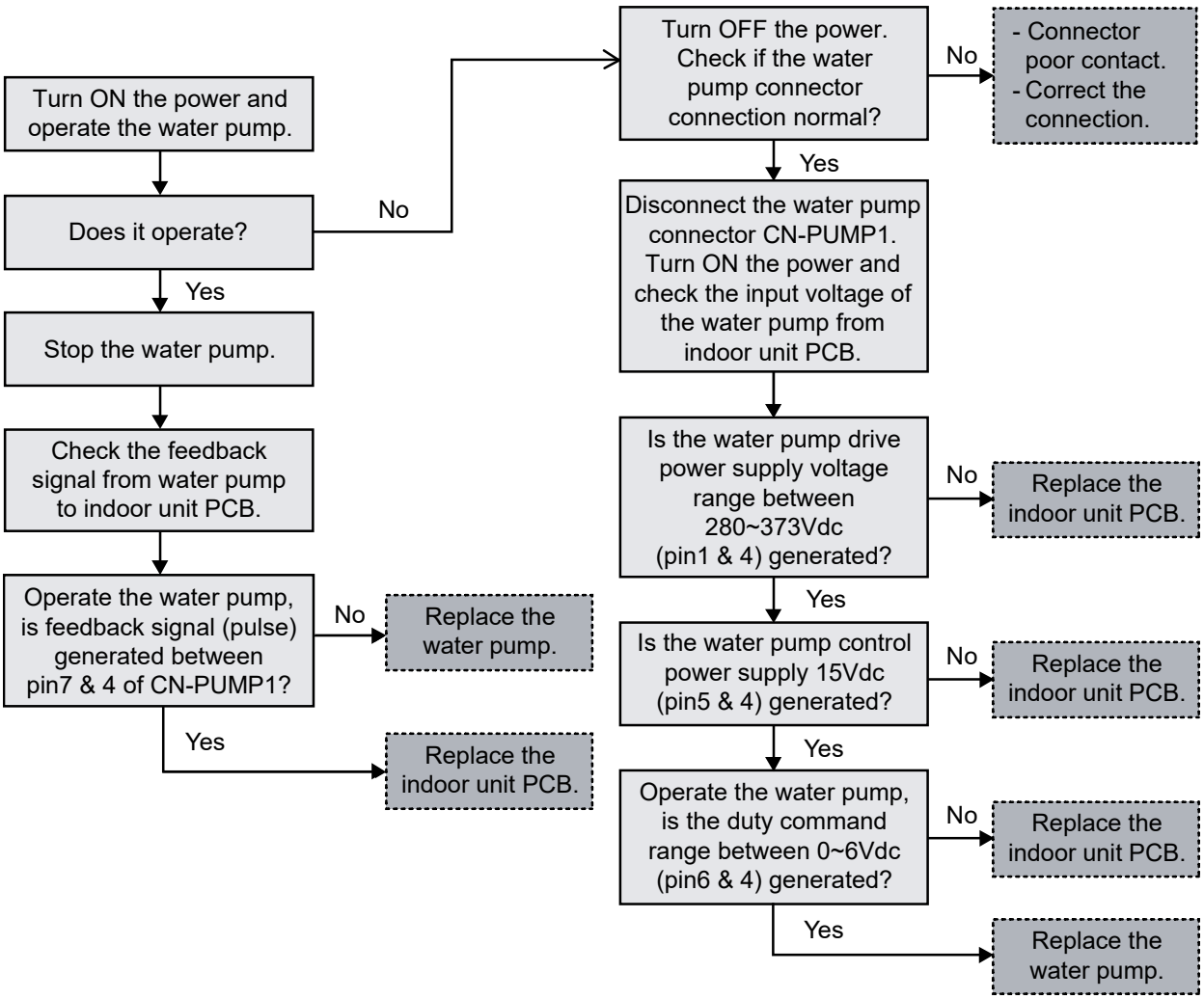
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.4 Indoor Refrigerant Liquid Temperature Sensor Abnormality (H23)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor refrigerant liquid temperature sensor are used to determine sensor error.

Malfunction Caused:

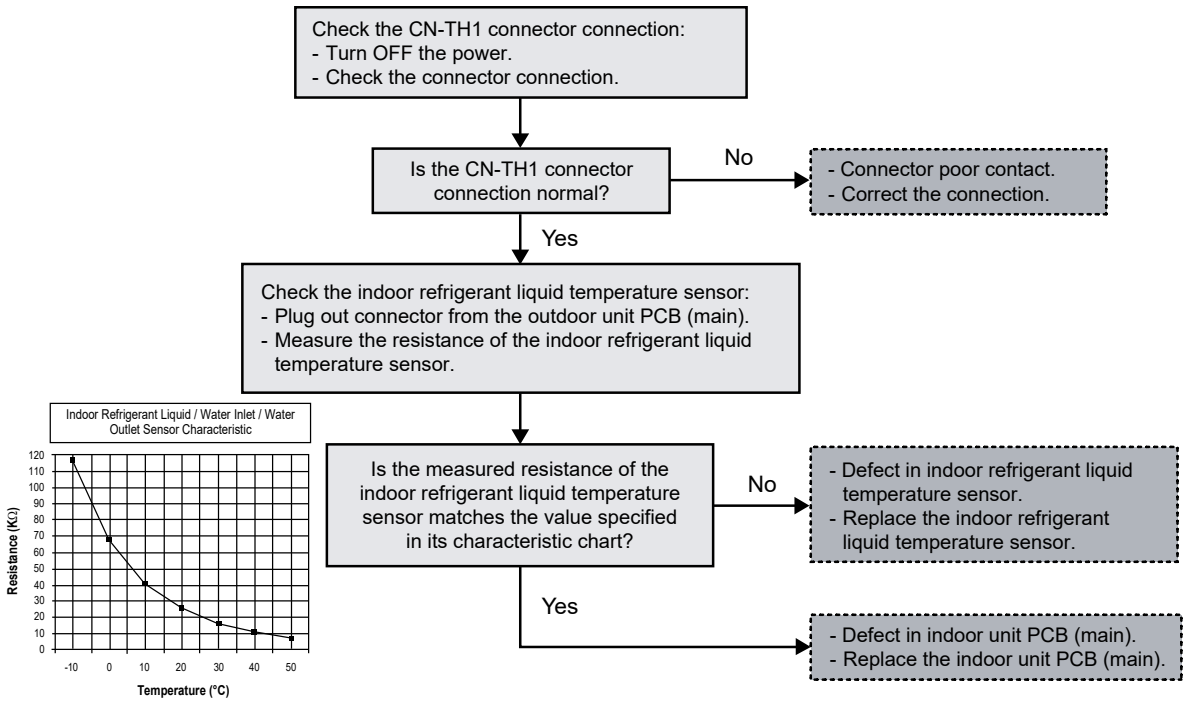
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.5 Service Valve Error (H27)

Malfunction Decision Conditions:

During cooling operation, when:-

[a] Indoor refrigerant pipe temperature at compressor startup - present indoor refrigerant pipe temperature < 2°C

[b] Present high pressure - high pressure at compressor startup < 5kg/cm^2

**Judgment only for first time cooling operation and not during pump down operation.

Malfunction Caused:

1 3 way valves closed.
2 Faulty high pressure sensor.
3 Faulty indoor refrigerant pipe temperature sensor
4 Faulty outdoor unit PCB (main).

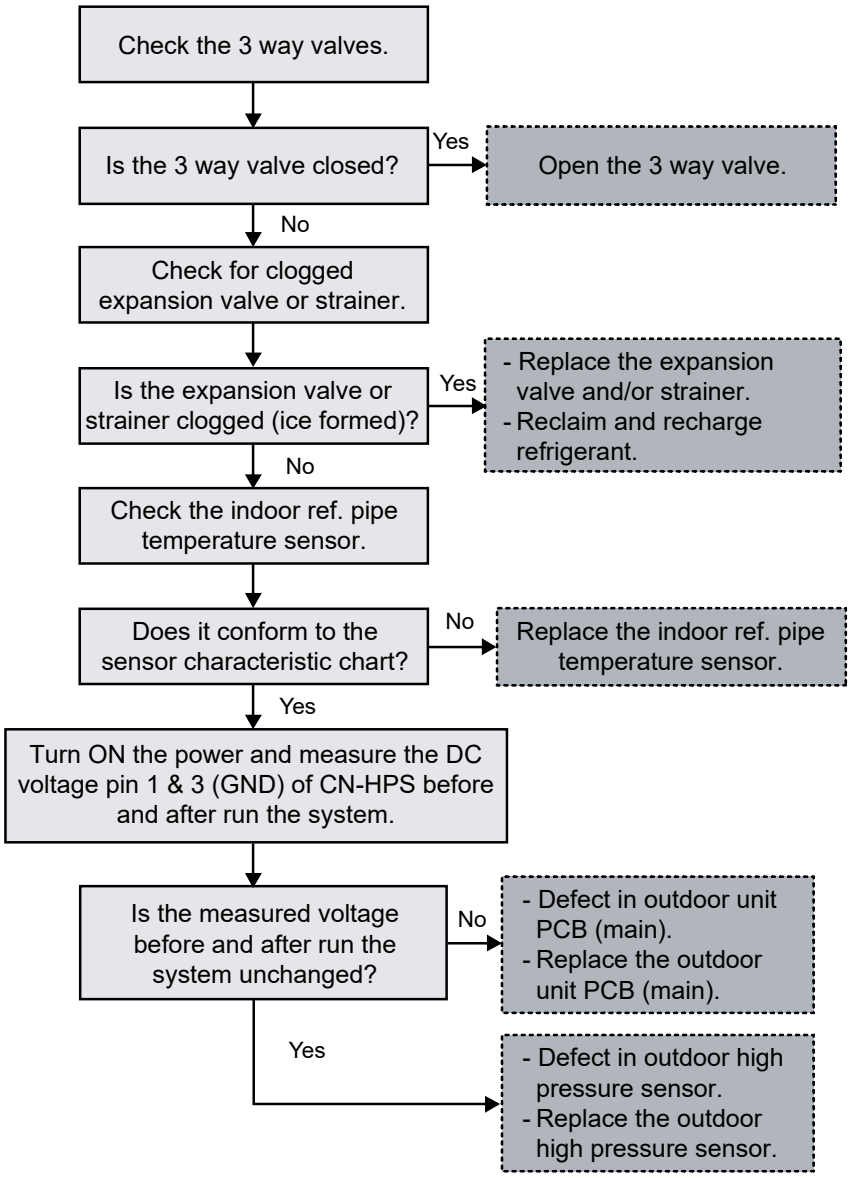
Abnormality Judgment:

Continue for 5 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.6 Abnormal Solar Sensor (H28)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty solar sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



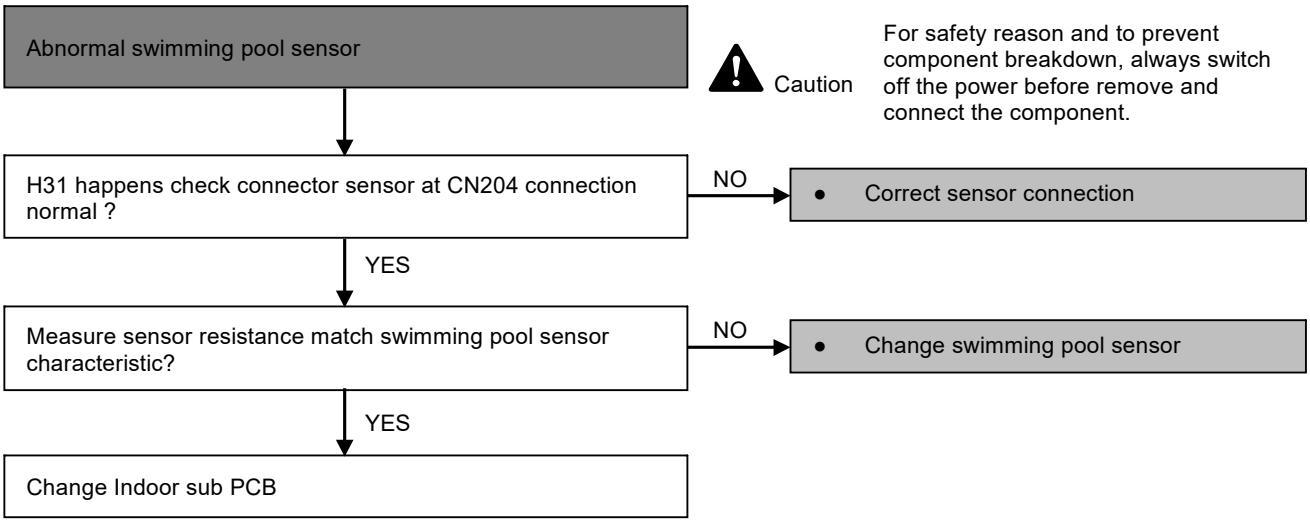
18.5.7 Abnormal Swimming Pool Sensor (H31)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty swimming pool sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



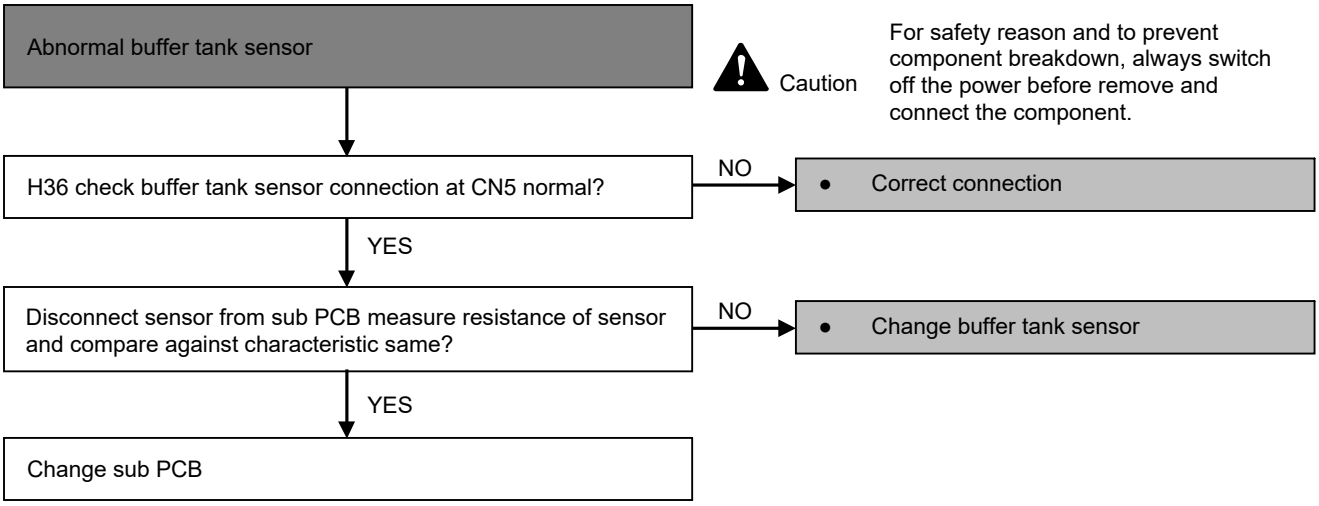
18.5.8 Abnormal Buffer Tank Sensor (H36)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

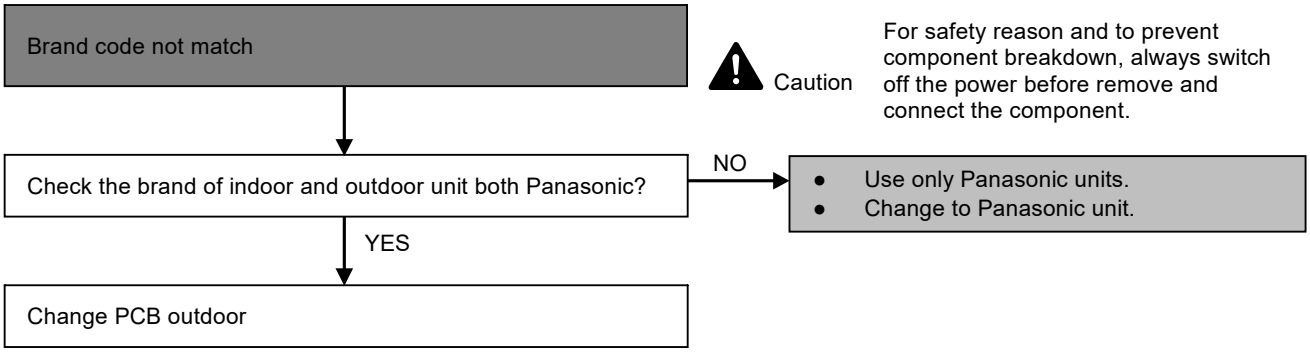
Continue for 5 seconds.



18.5.9 Brand Code Not Matching (H38)

Malfunction Caused:

1 Indoor and outdoor brand code not match.



18.5.10 Compressor Low Pressure Protection (H42)

Malfunction Decision Conditions:

During operation of heating and after 5 minutes compressor ON, when outdoor pipe temperature below -29°C or above 26°C is detected by the outdoor pipe temperature sensor.

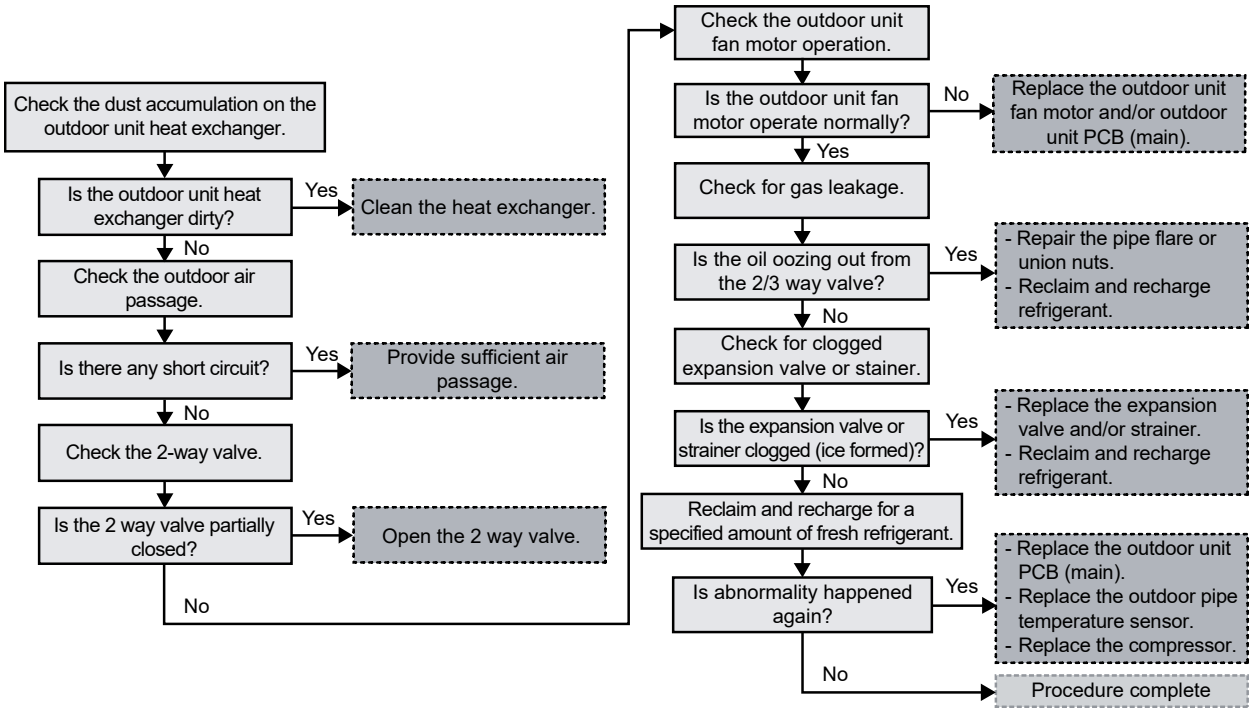
Malfunction Caused:

1 Dust accumulation on the outdoor unit heat exchanger.
2 Air short circuit at outdoor unit.
3 2 way valve partially closed.
4 Faulty outdoor unit fan motor.
5 Refrigerant shortage (refrigerant leakage).
6 Clogged expansion valve or strainer.
7 Faulty outdoor pipe temperature sensor.
8 Faulty outdoor unit main PCB (main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



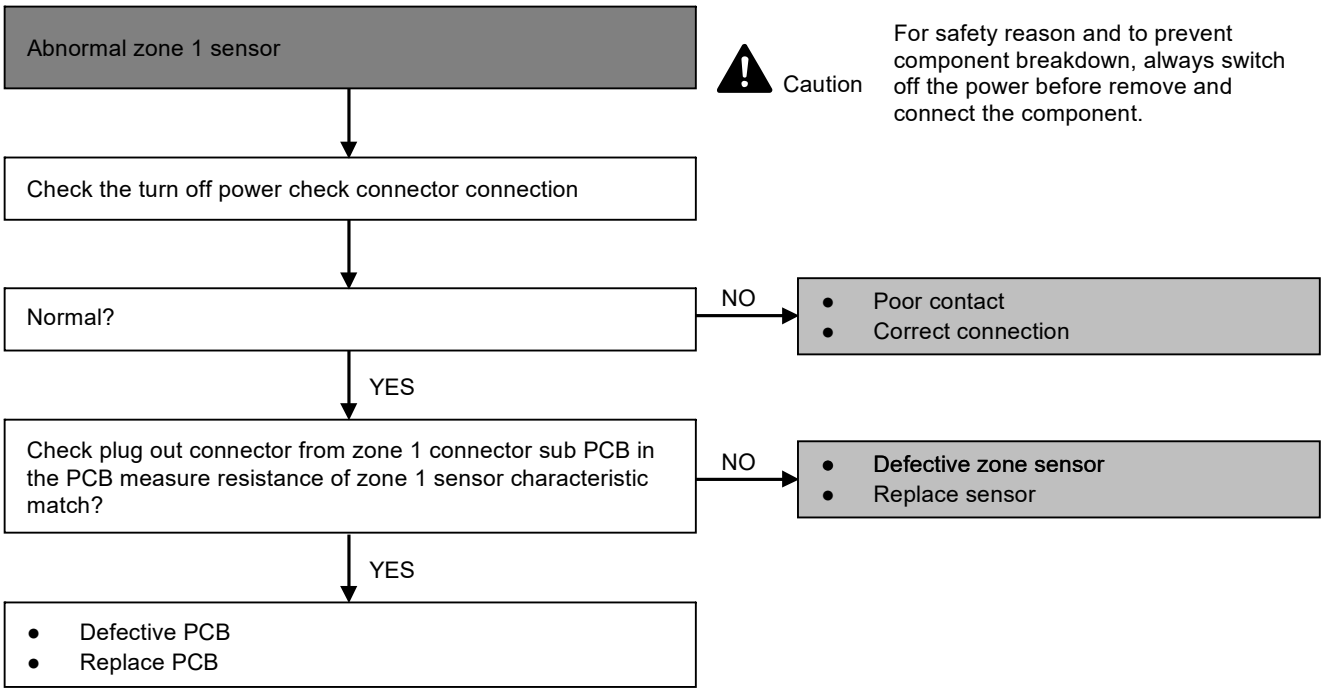
18.5.11 Abnormal Zone 1 Sensor (H43)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



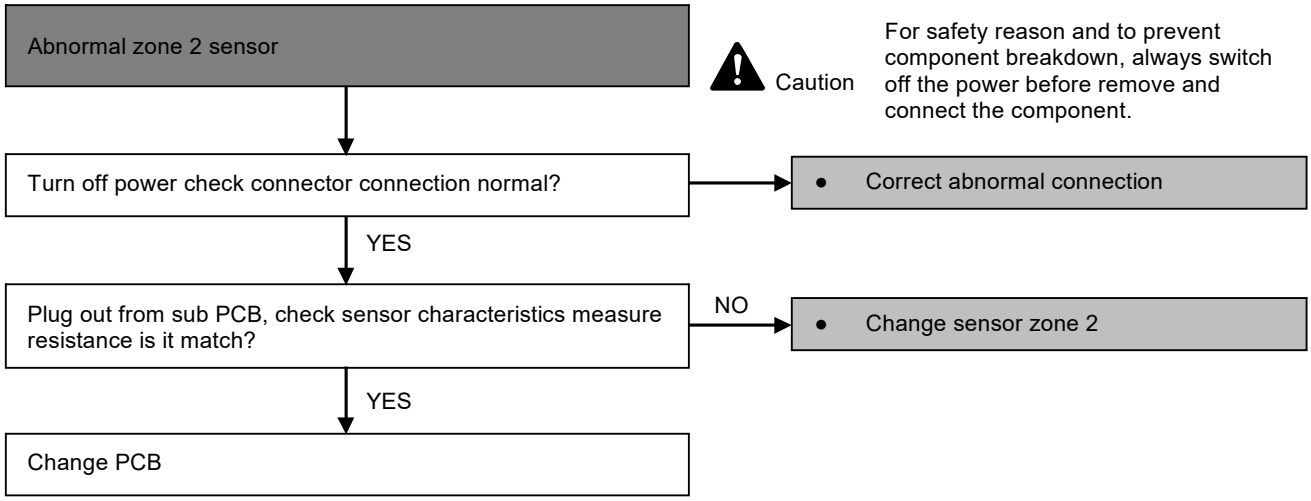
18.5.12 Abnormal Zone 2 Sensor (H44)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



18.5.13 Water Flow Switch Abnormality (H62)

Malfunction Decision Conditions:

During operation of cooling and heating, the water flow detected by the indoor water flow switch is used to determine water flow error.

Malfunction Caused:

1 Faulty water pump.
2 Water leak in system.
3 Faulty connector connection.
4 Faulty water flow switch.
5 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 10 seconds (but no judgment for 9 minutes after compressor startup/restart).

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.

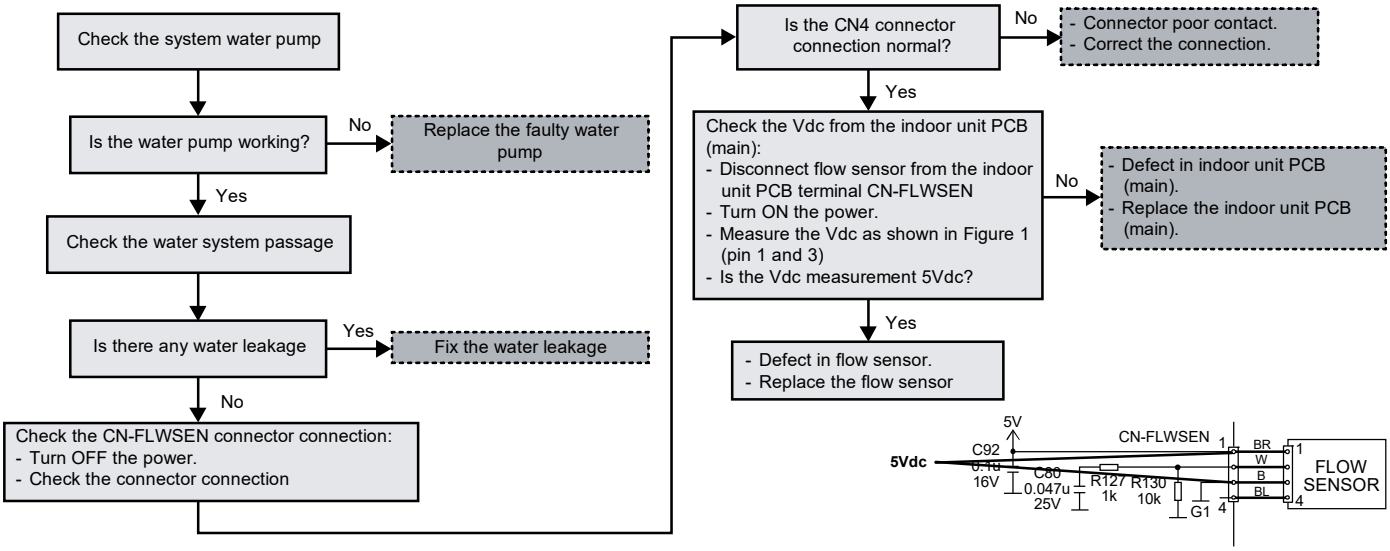


Figure 1

18.5.14 Outdoor High Pressure Abnormality (H64)

Malfunction Decision Conditions:

During operation of cooling and heating, when the outdoor high pressure sensor output signal is 0 Vdc or 5 Vdc.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

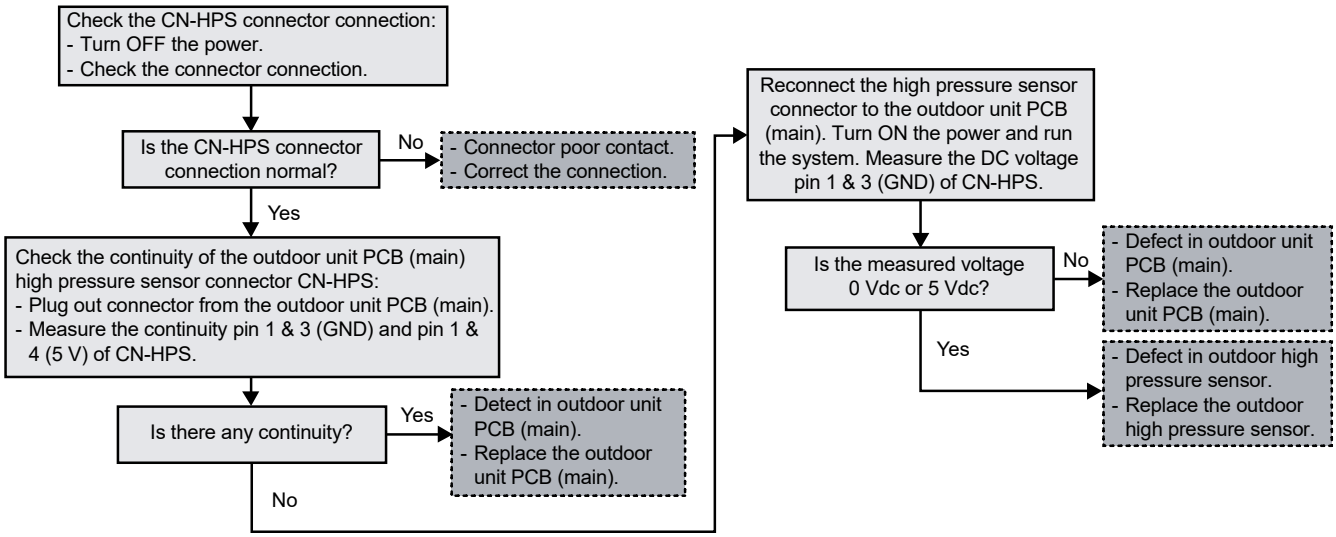
Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.15 Deice Circulation Error (H65)

Malfunction Decision Conditions:

During startup and operation of deice (mode 2), the water flow (> 10l/min) detected by the water flow switch is used to determine deice circulation error.

Malfunction Caused:

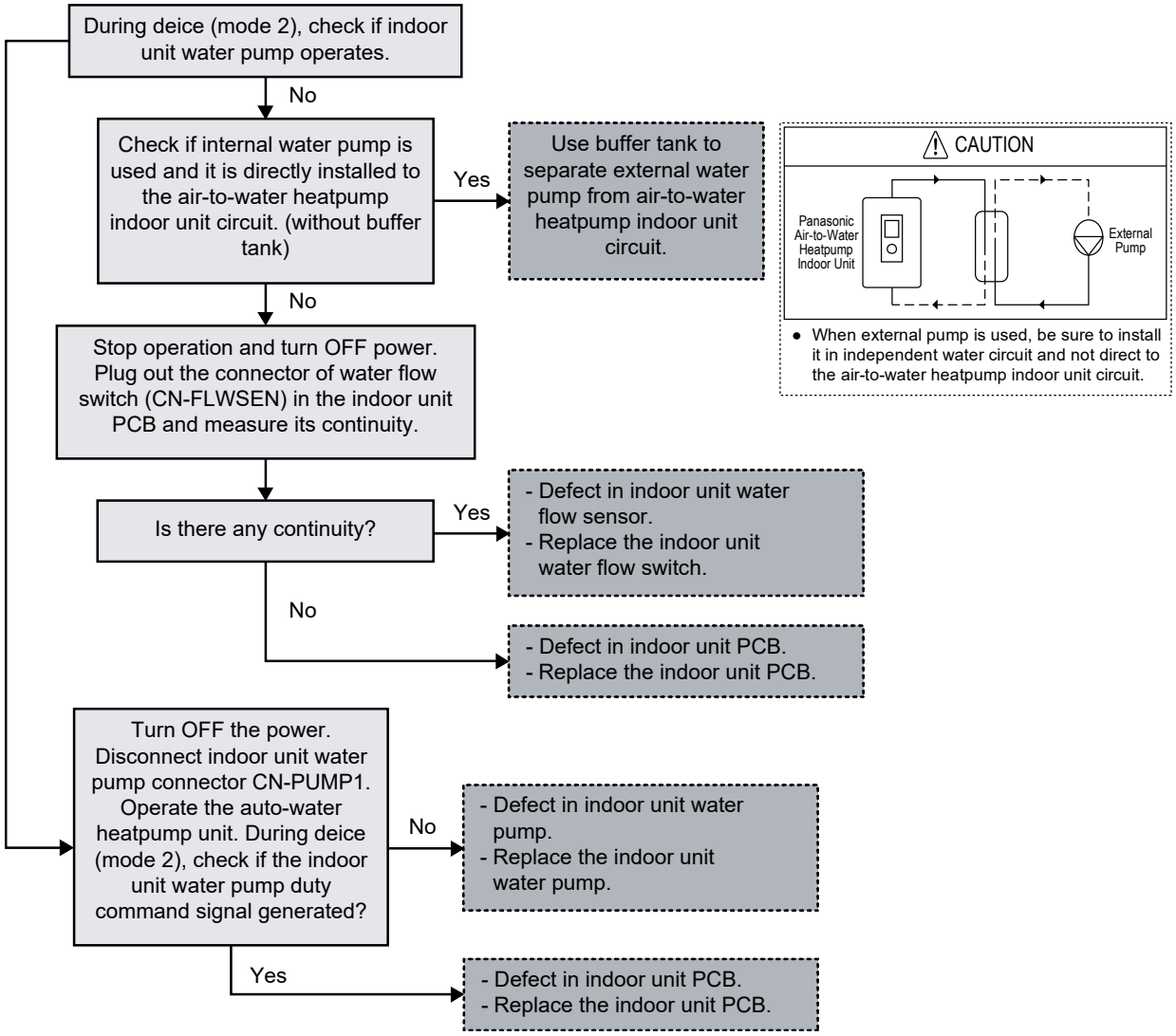
1 Water flow in air-to-water heatpump indoor unit circuitry.
2 Faulty indoor unit water flow switch.
3 Faulty indoor unit water pump.
4 Faulty indoor unit PCB.

Abnormality Judgment:

Continue for 10 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



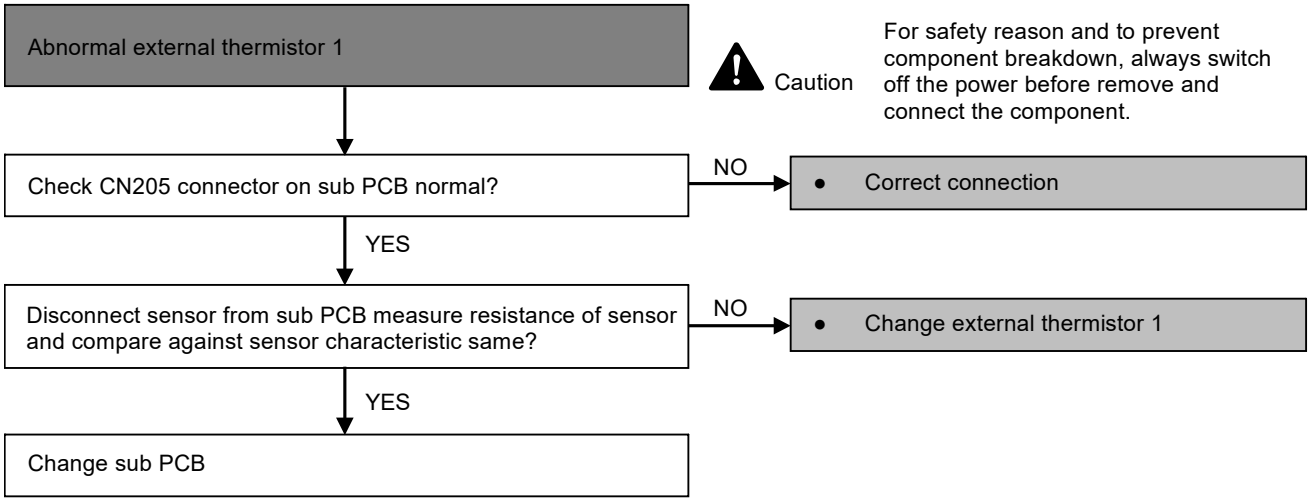
18.5.16 Abnormal External Thermistor 1 (H67)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty room temperature zone 1 sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



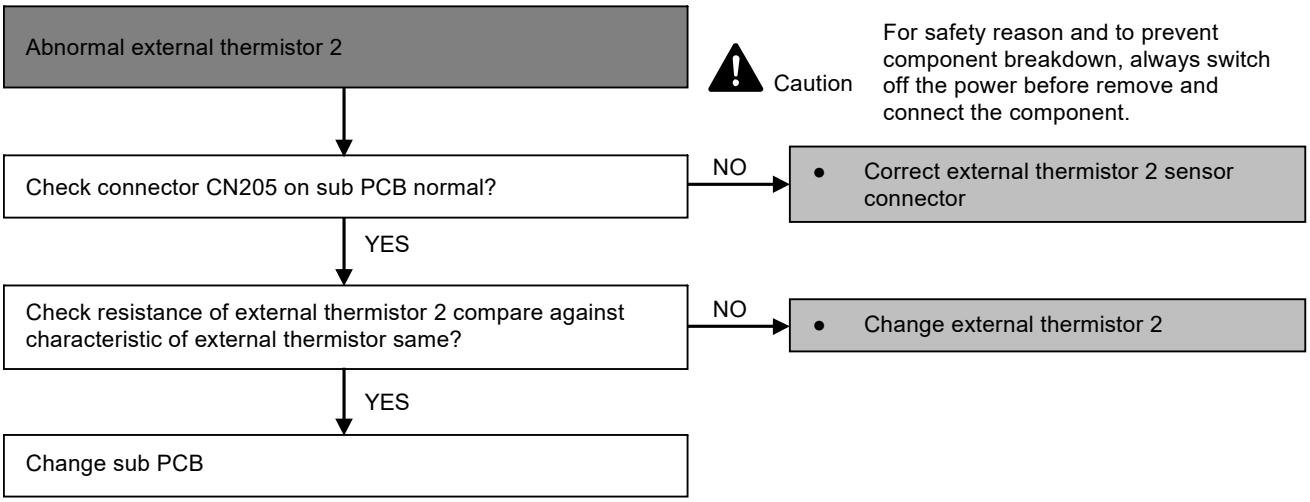
18.5.17 Abnormal External Thermistor 2 (H68)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty room temperature zone 2 sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



18.5.18 Indoor Backup Heater OLP Abnormality (H70)

Malfunction Decision Conditions:

During operation of indoor backup heater, when no power supplies to indoor backup heater or OLP open circuit.

Malfunction Caused:

1 Faulty power supply connector connection.
2 Faulty connector connection.
3 Faulty indoor backup heater overload protector (OLP).
4 Faulty indoor unit PCB (main).

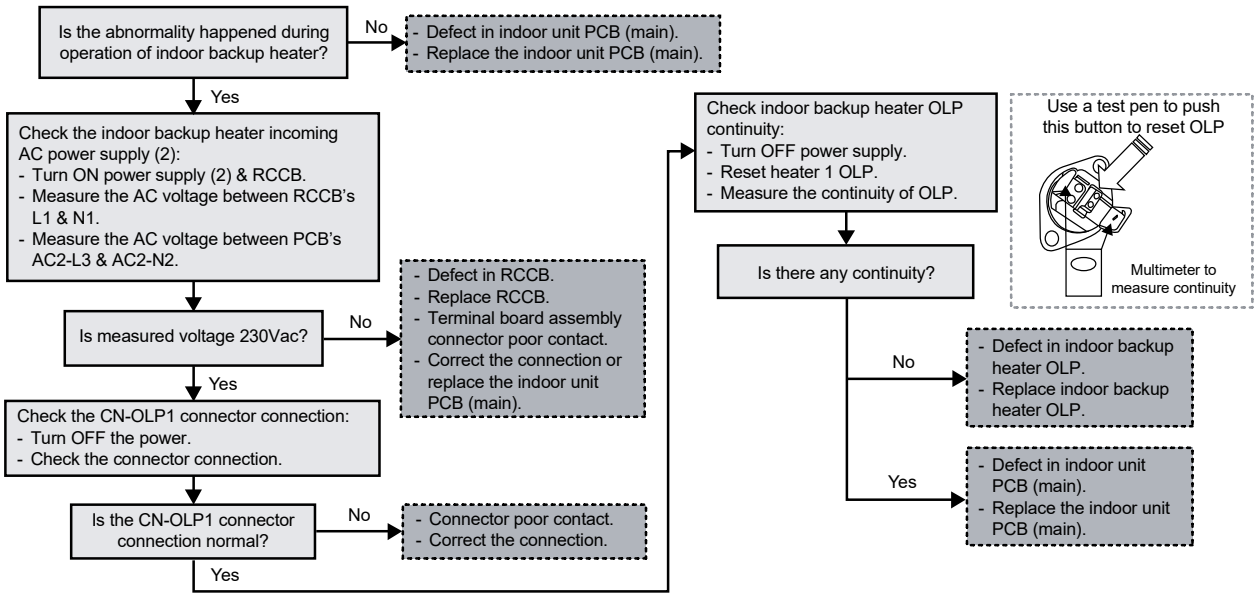
Abnormality Judgment:

Continue for 60 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.19 Tank Temperature Sensor Abnormality (H72)

Malfunction Decision Conditions:

When tank connection is set to ON, the temperatures detected by the tank temperature sensor are used to determine sensor error.

Malfunction Caused:

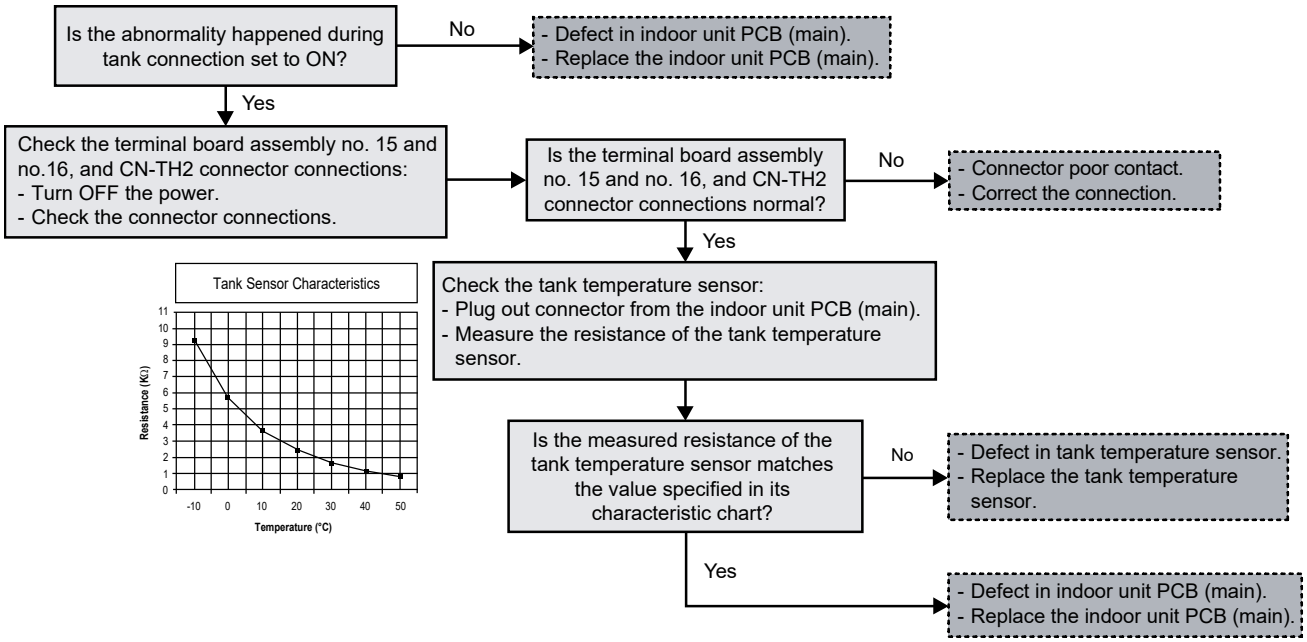
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.20 PCB Communication Error (H74)

Malfunction Decision Conditions:

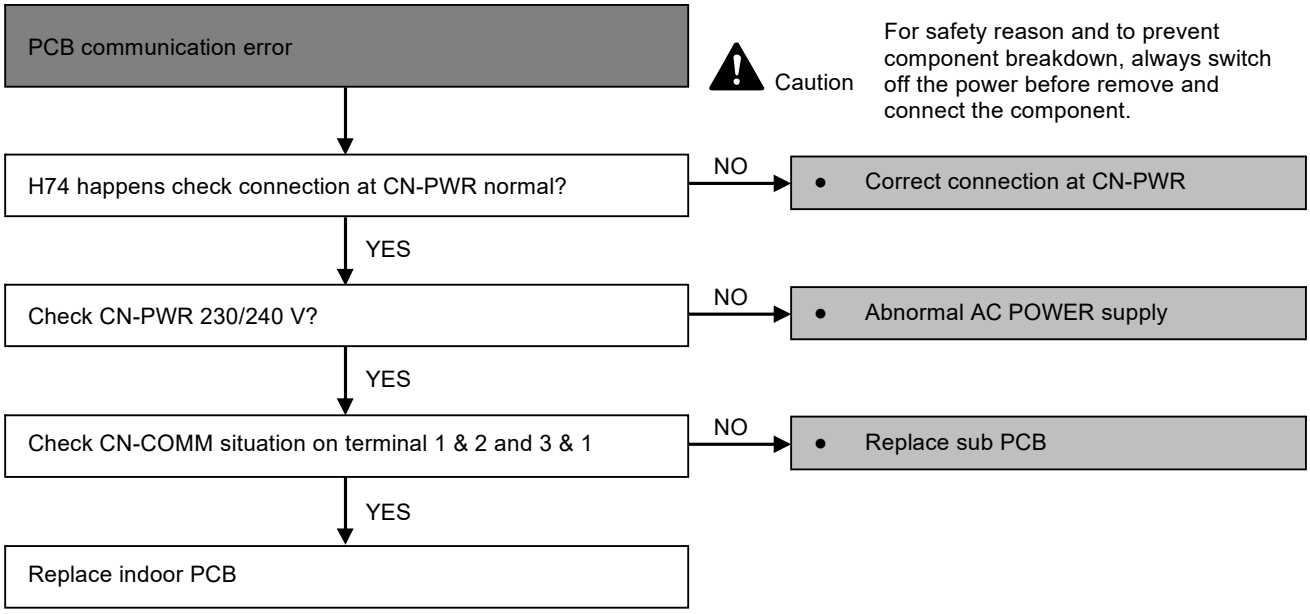
When External PCB connection is select “YES” and no communication with External PCB micon for 10 seconds and above.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty indoor PCB.
3 Faulty indoor sub PCB.

Abnormality Judgment:

After 1 minute operation started.



18.5.21 Indoor-Control Panel Communication Abnormality (H76)

Malfunction Decision Conditions:

During standby and operation of cooling and heating, indoor-control panel error occur.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty control panel.
3 Faulty indoor unit PCB (main).

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.

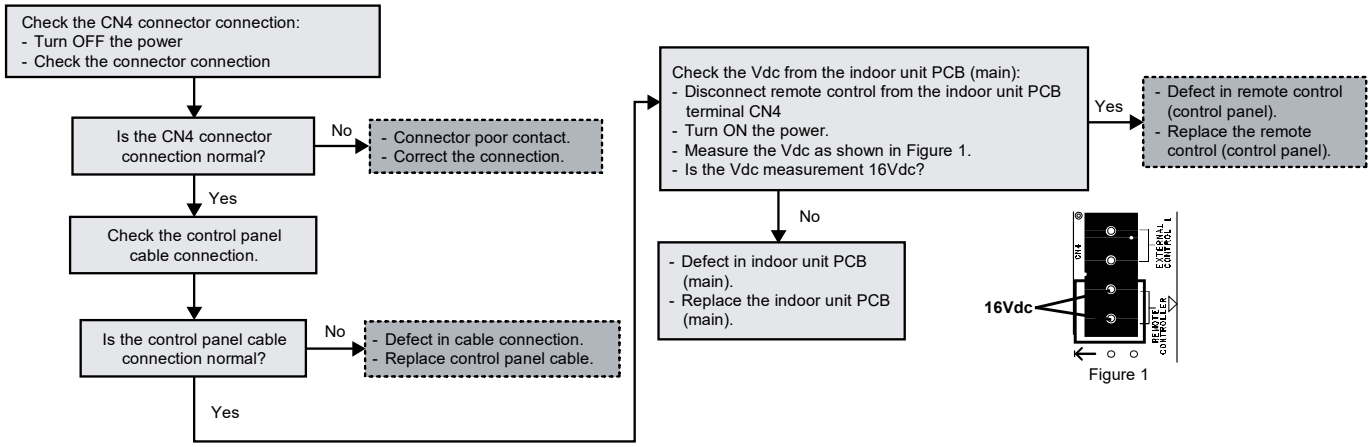


Figure 1

18.5.22 Indoor/Outdoor Abnormal Communication (H90)

Malfunction Decision Conditions:

During operation of cooling and heating, the data received from outdoor unit in indoor unit signal transmission is checked whether it is normal.

Malfunction Caused:

1 Faulty outdoor unit PCB (main).
2 Faulty indoor unit PCB (main).
3 Indoor-outdoor signal transmission error due to wrong wiring.
4 Indoor-outdoor signal transmission error due to breaking of wire in the connection wires between the indoor and outdoor units.
5 Indoor-outdoor signal transmission error due to disturbed power supply waveform.

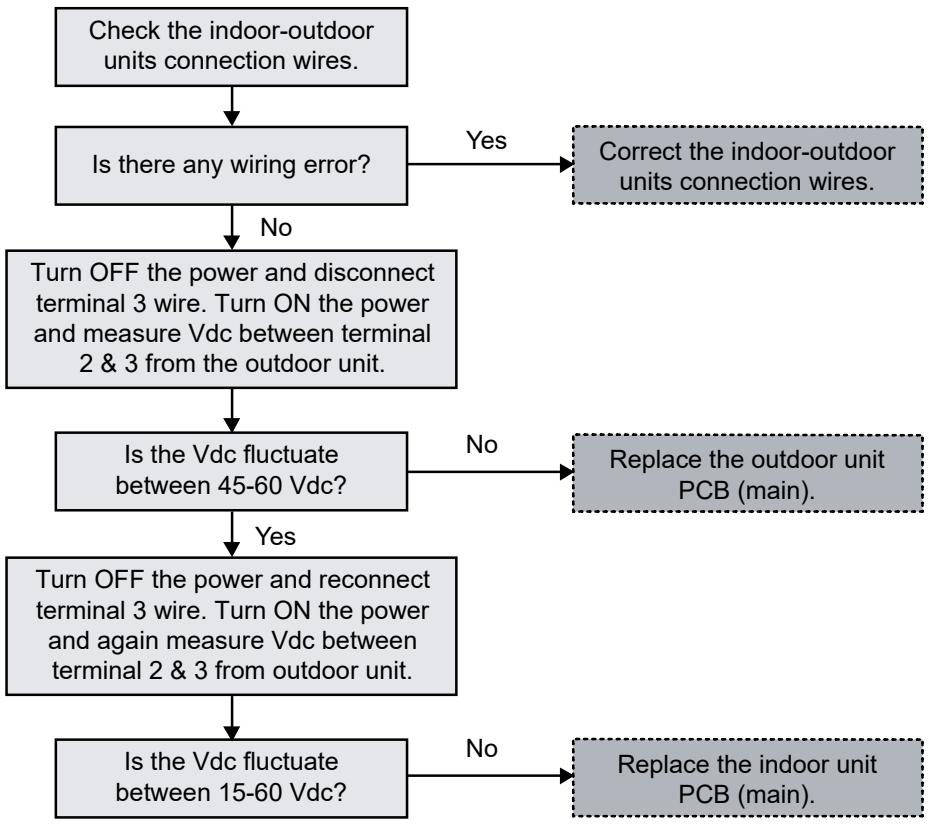
Abnormality Judgment:

Continue for 1 minute after operation.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.23 Tank Booster Heater OLP Abnormality (H91)

Malfunction Decision Conditions:

During operation of tank booster heater, and tank booster heater OLP open circuit.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty tank booster heater overload protector (OLP).
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 60 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.

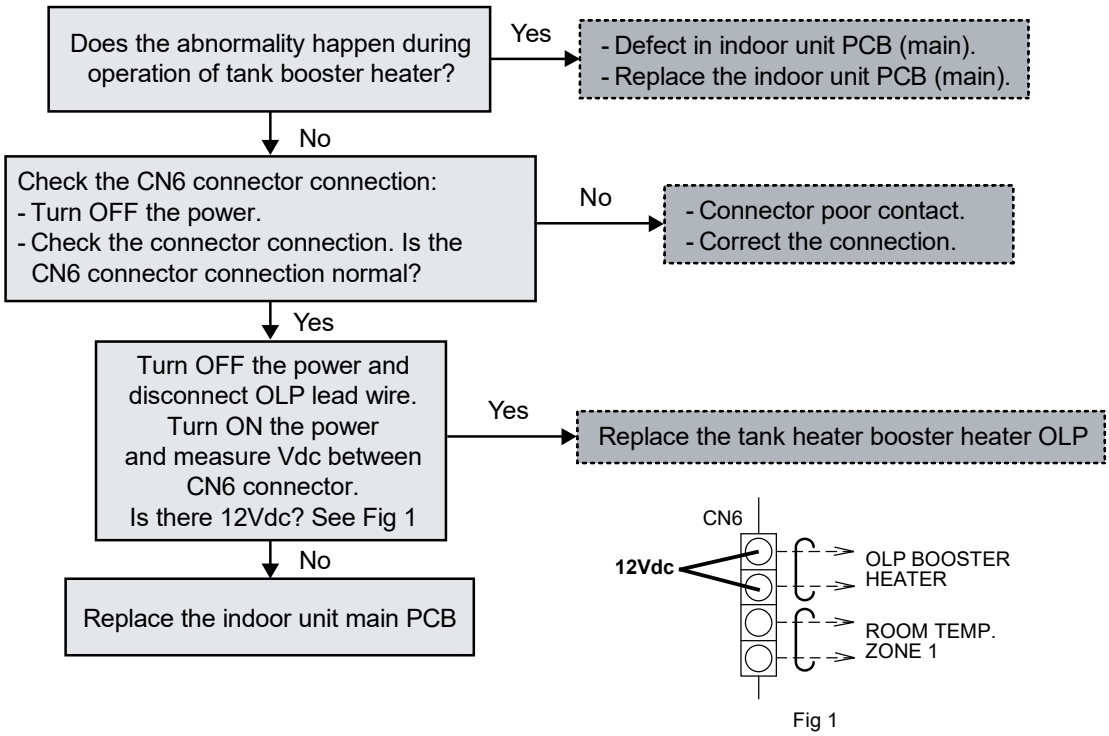


Fig 1

18.5.24 Unspecified Voltage between Indoor and Outdoor (H95)

Malfunction Decision Conditions:

The supply power is detected for its requirement by the indoor/outdoor transmission.

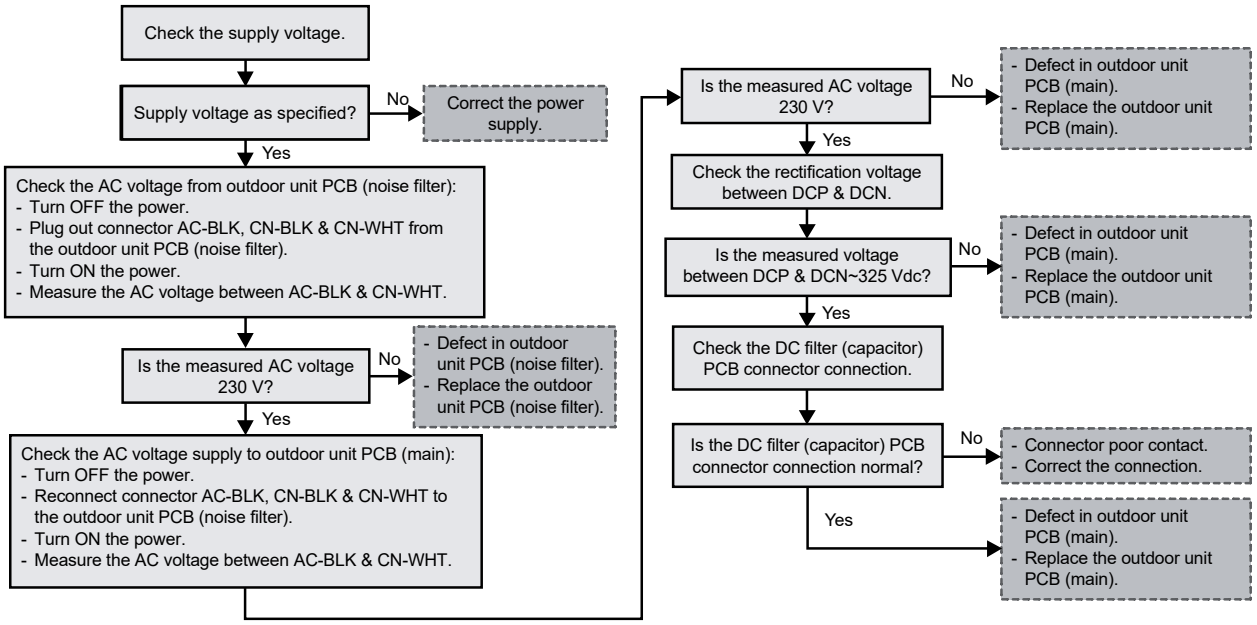
Malfunction Caused:

1 Insufficient power supply.
2 Faulty outdoor unit PCB (noise filter/main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.25 Outdoor High Pressure Protection (H98 / F95)

Malfunction Decision Conditions:

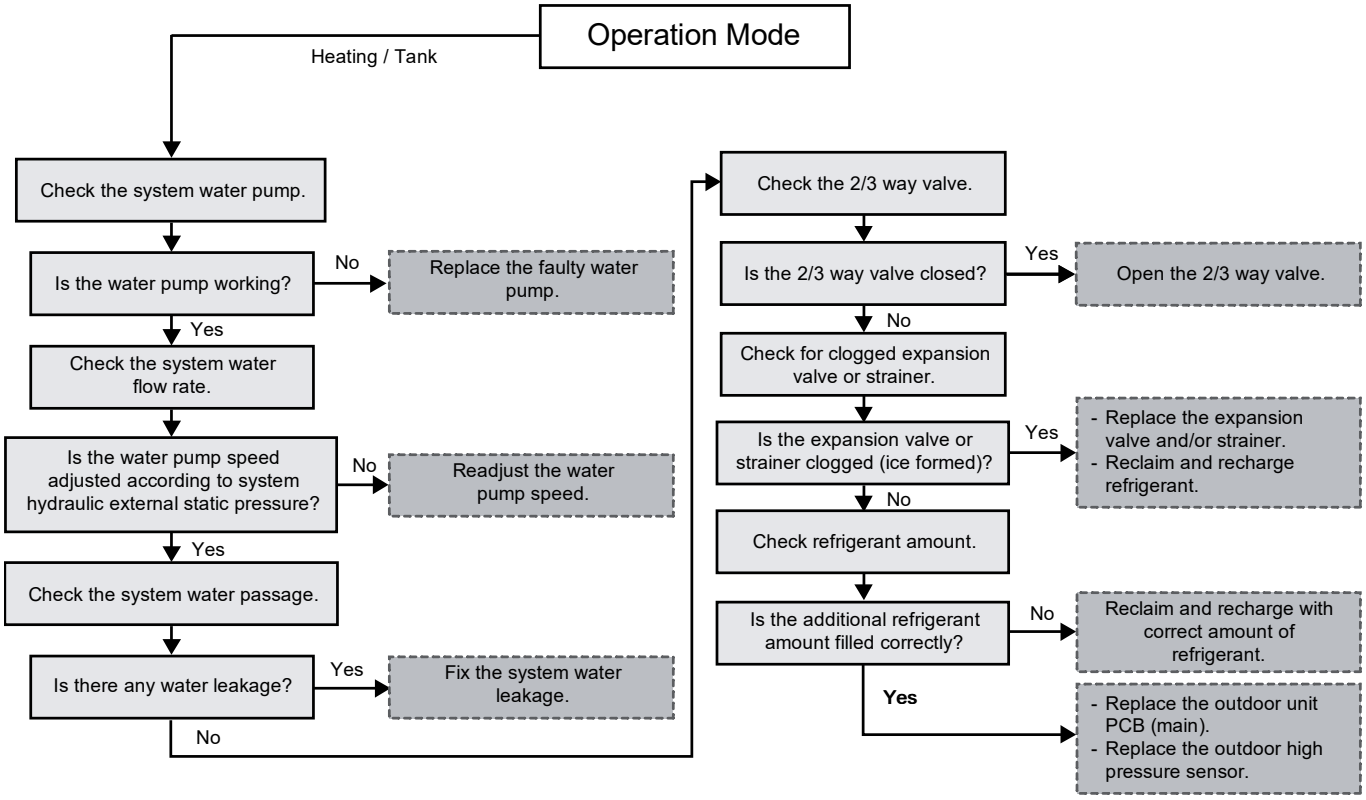
During operation of cooling / heating, when pressure 4.2 MPa and above is detected by outdoor high pressure sensor.

Malfunction Caused:

1 Faulty water pump.
2 Insufficient water flow rate in system.
3 Water leak in system.
4 Dust accumulation in the outdoor unit heat exchanger.
5 Air short circuit at outdoor.
6 Faulty outdoor unit fan motor.
7 2/3 way closed.
8 Clogged expansion valve or strainer.
9 Excessive refrigerant.
10 Faulty outdoor high pressure sensor.
11 Faulty outdoor unit PCB (main).

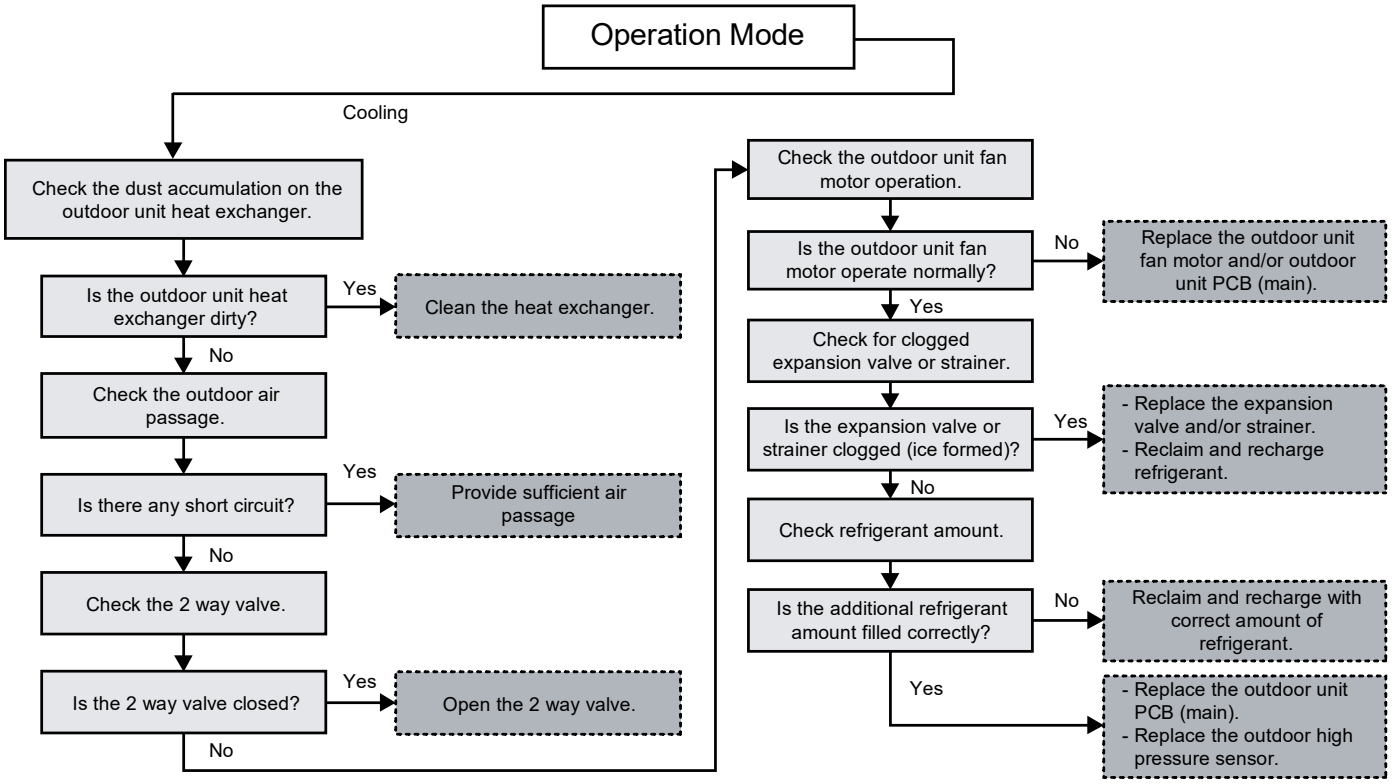
Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.26 Indoor Freeze-up Protection (H99)

Malfunction Decision Conditions:

During anti-freezing control in cooling operation, when the indoor refrigerant liquid temperature < 0°C.

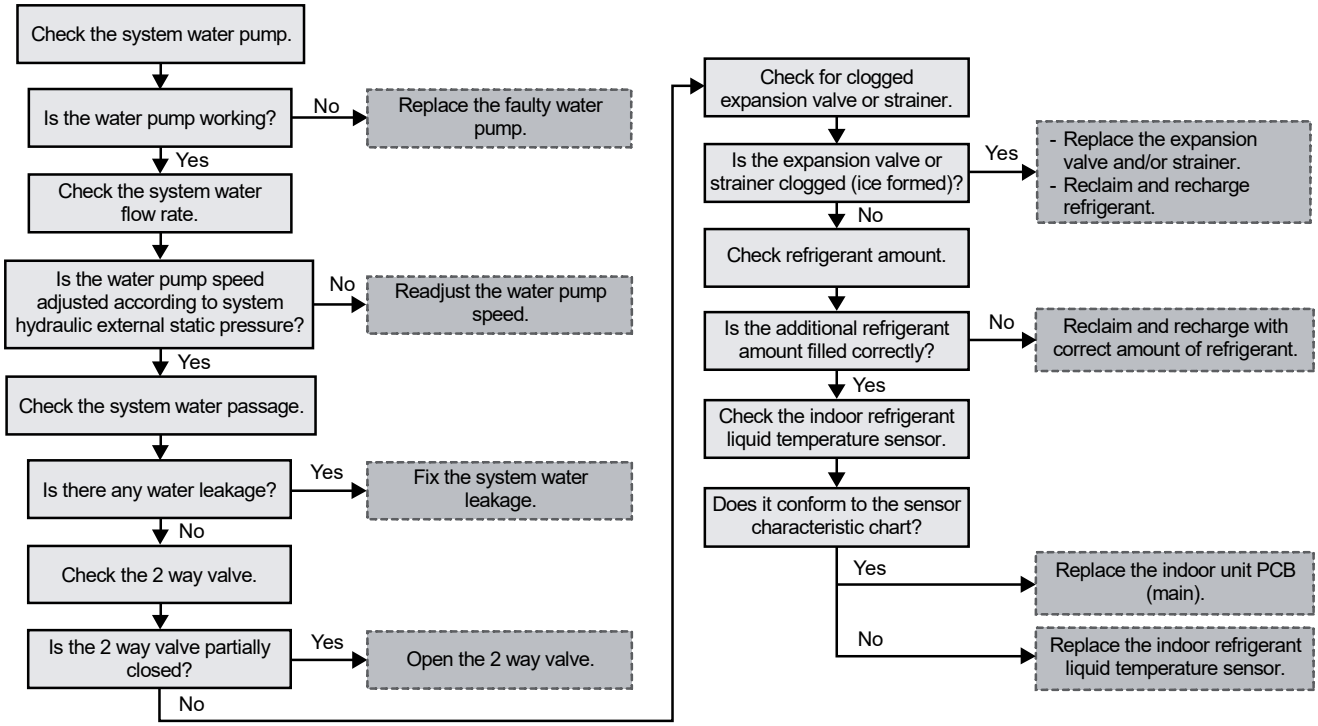
Malfunction Caused:

1 Faulty water pump.
2 Insufficient water flow rate in system.
3 Water leak in system.
4 2 way valve partially closed.
5 Clogged expansion valve or strainer.
6 Refrigerant shortage (refrigerant leakage).
7 Faulty indoor refrigerant liquid temperature sensor.
8 Faulty indoor unit PCB (main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.27 Outdoor High Pressure Switch Activate (F12)

Malfunction Decision Conditions:

During operation of cooling and heating, when pressure 4.15 MPa and above is detected by outdoor high pressure switch.

Malfunction Caused:

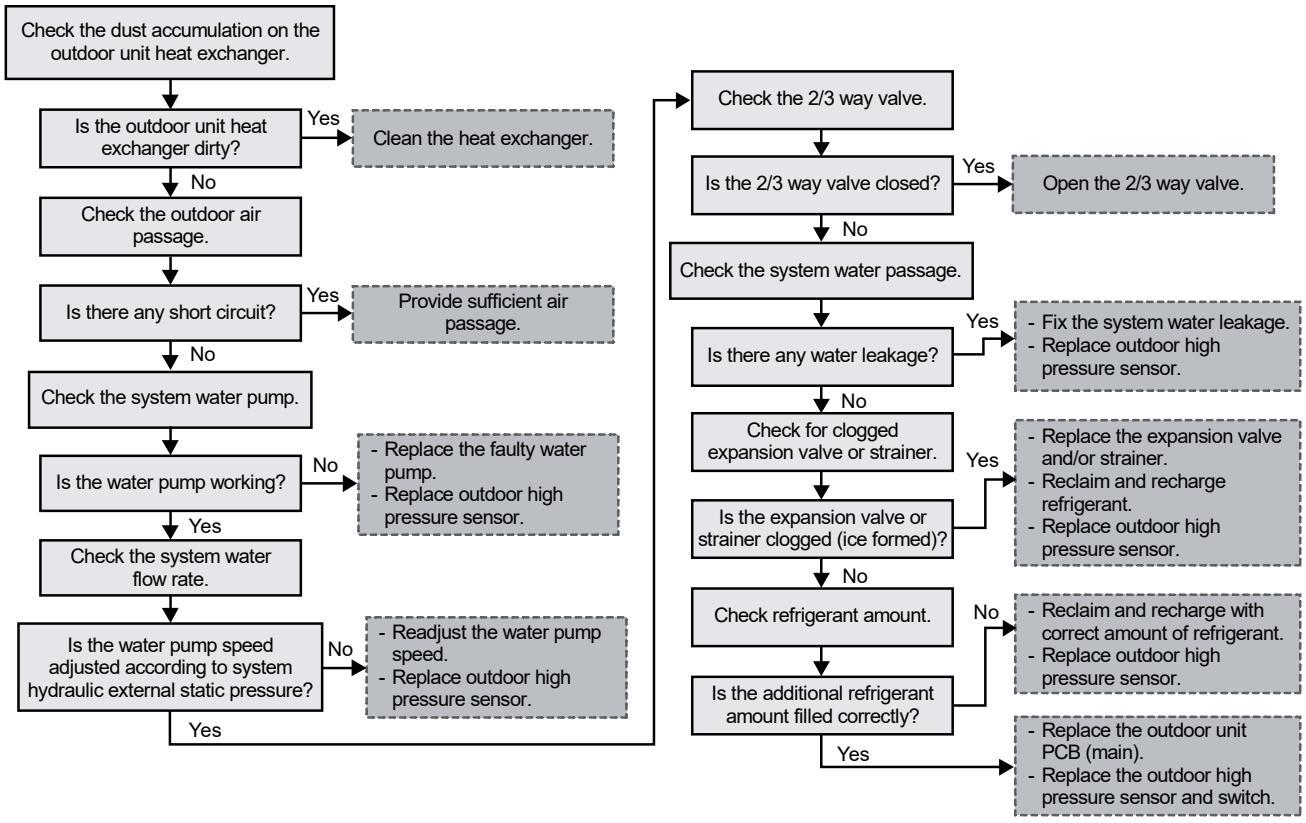
1 Dust accumulation on the outdoor unit heat exchanger.
2 Air short circuit at outdoor unit.
3 Faulty water pump.
4 Insufficient water flow rate in system.
5 Water leak in system.
6 2/3 way valve closed.
7 Clogged expansion valve or strainer.
8 Excessive refrigerant.
9 Faulty outdoor high pressure sensor and switch.
10 Faulty outdoor unit PCB.

Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.28 Compressor Rotation Failure (F14)

Malfunction Decision Conditions:

A compressor rotation failure is detected by checking the compressor running condition through the position detection circuit.

Malfunction Caused:

1 Compressor terminal disconnect.
2 Faulty outdoor unit PCB (main).
3 Faulty compressor.

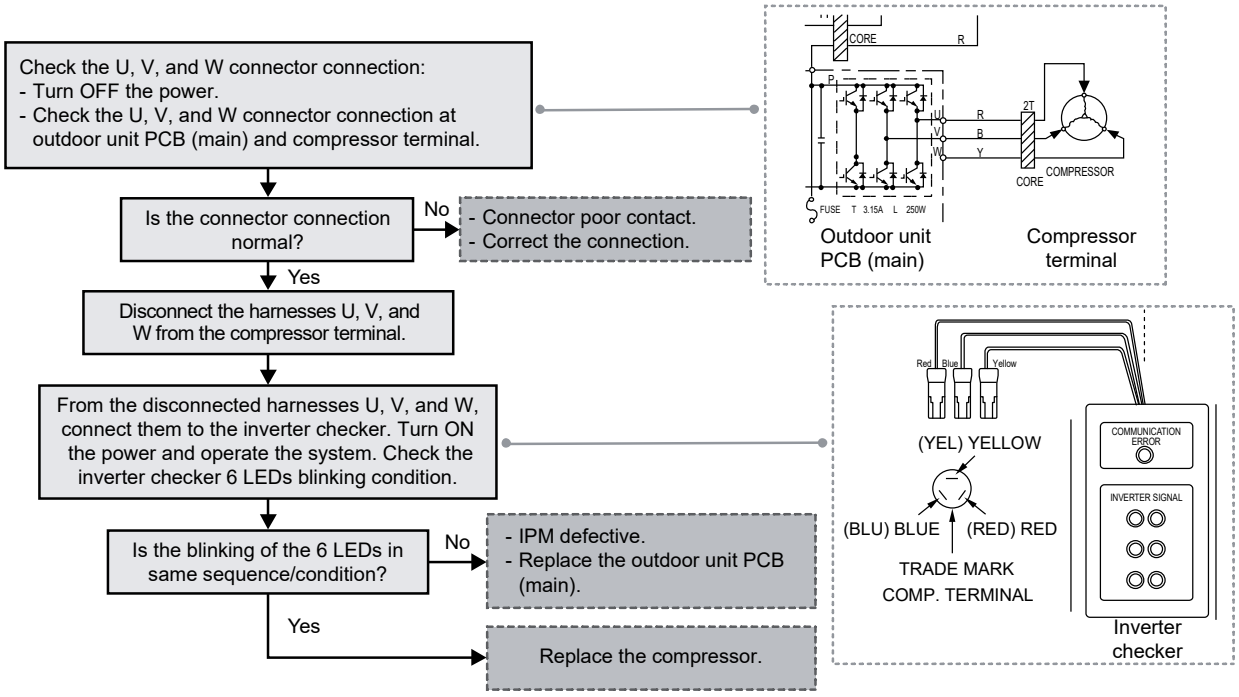
Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.29 Outdoor Fan Motor (DC Motor) Mechanism Locked (F15)

Malfunction Decision Conditions:

The rotation speed detected by the Hall IC of the fan motor during fan motor operation is used to determine abnormal fan motor (feedback of rotation > 2550 rpm or < 50 rpm).

Malfunction Caused:

1 Operation stop due to short circuit inside the fan motor winding.
2 Operation stop due to breaking of wire inside the fan motor.
3 Operation stop due to breaking of fan motor lead wires.
4 Operation stop due to fan motor Hall IC malfunction.
5 Operation error due to faulty outdoor unit PCB.

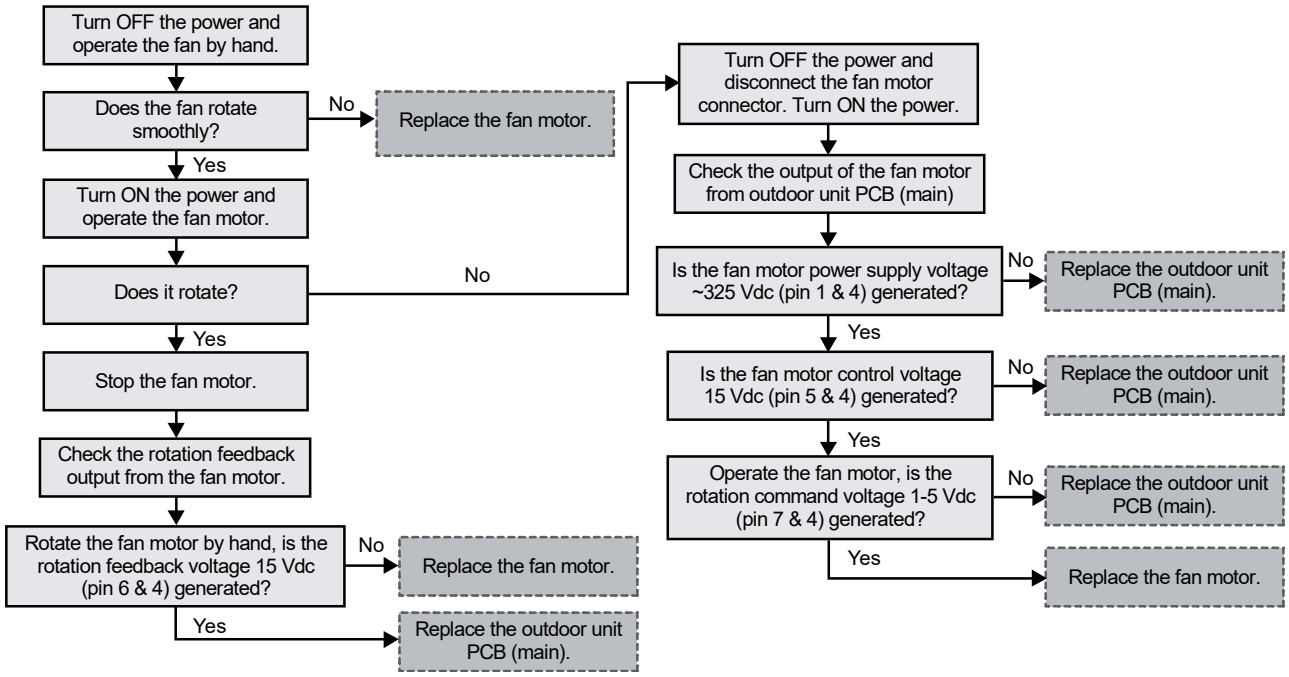
Abnormality Judgment:

Continue 2 times in 30 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.30 Input Over Current Detection (F16)

Malfunction Decision Conditions:

During operation of cooling and heating, when outdoor current above WDG05LE5 15.0A & WDG07 / 09LE5 18.0A is detected by the current transformer (CT) in the outdoor unit PCB.

Malfunction Caused:

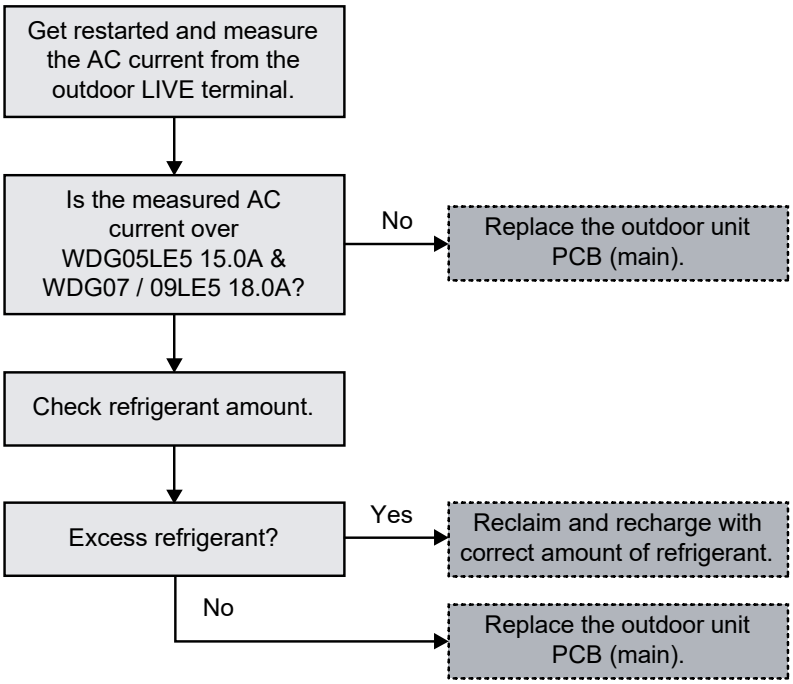
1 Excessive refrigerant.
2 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 3 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.31 Compressor Overheating (F20)

Malfunction Decision Conditions:

During operation of cooling and heating, when temperature above 112°C is detected by the compressor tank temperature sensor.

Malfunction Caused:

1 Faulty compressor tank temperature sensor.
2 2/3 way valve closed.
3 Refrigerant shortage (refrigerant leakage).
4 Clogged expansion valve or strainer.
5 Faulty outdoor unit PCB (main).
6 Faulty compressor.

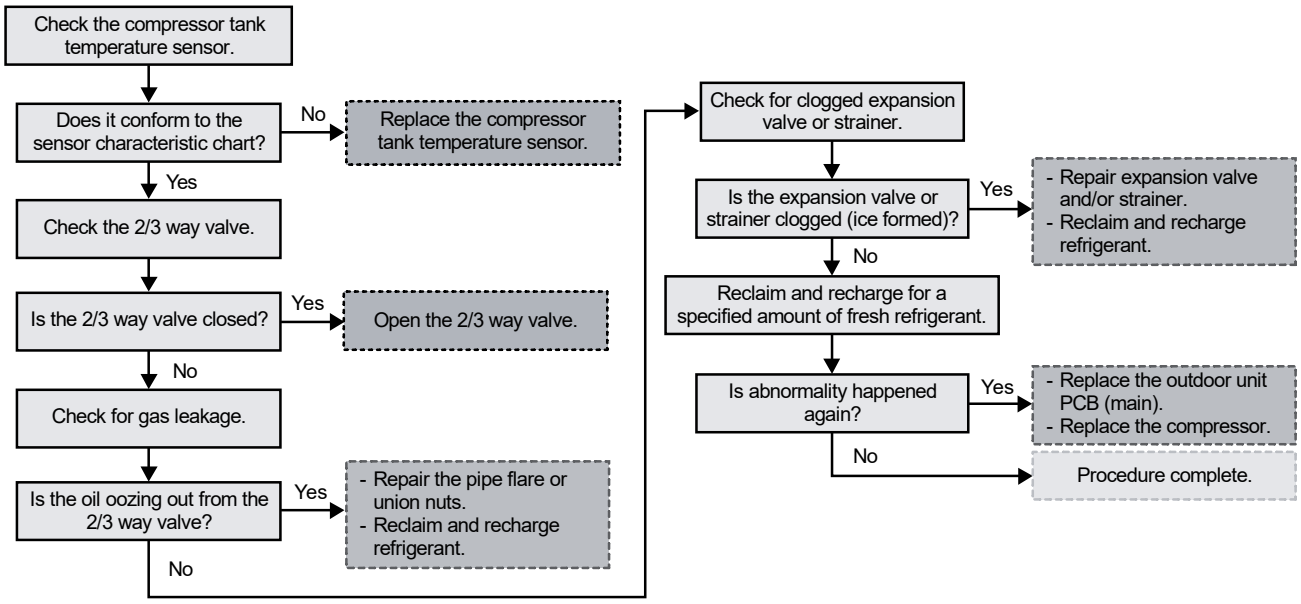
Abnormality Judgment:

Continue 4 times in 30 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.32 IPM Overheating (F22)

Malfunction Decision Conditions:

During operation of cooling and heating, when temperature 95°C is detected by the outdoor IPM temperature sensor.

Malfunction Caused:

1 Faulty outdoor unit fan motor.
2 Faulty outdoor unit PCB (main).

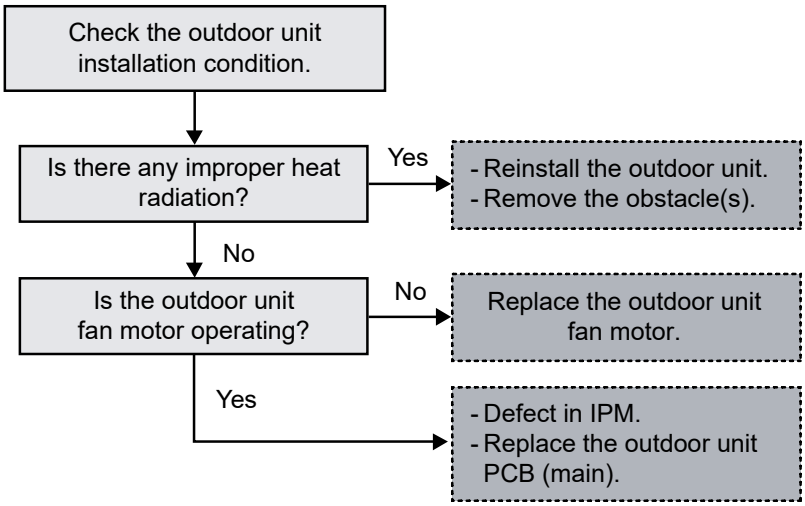
Abnormality Judgment:

Continue 3 times in 30 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.33 Output Over Current Detection (F23)

Malfunction Decision Conditions:

During operation of cooling and heating, when outdoor DC current is above set value is detected by the IPM DC Peak sensing circuitry in the outdoor unit PCB (main).

Malfunction Caused:

1 Faulty outdoor unit PCB (main).
2 Faulty compressor.

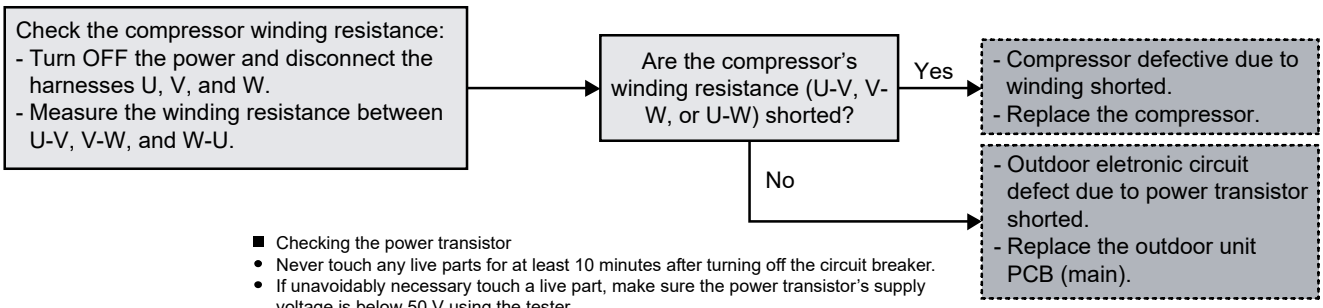
Abnormality Judgment:

Continue for 7 times.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



■ Checking the power transistor

- Never touch any live parts for at least 10 minutes after turning off the circuit breaker.
- If unavoidably necessary touch a live part, make sure the power transistor's supply voltage is below 50 V using the tester.
- For the UVW, make measurements at the Faston terminal on the board or the relay connector.

Tester's negative terminal	Power transistor (+)	UVW	Power transistor (-)	UVW
Tester's positive terminal	UVW	Power transistor (+)	UVW	Power transistor (-)
Normal resistance	Several kohms to several Mohms			
Abnormal resistance	0 or ∞			

18.5.34 Refrigeration Cycle Abnormality (F24)

Malfunction Decision Conditions:

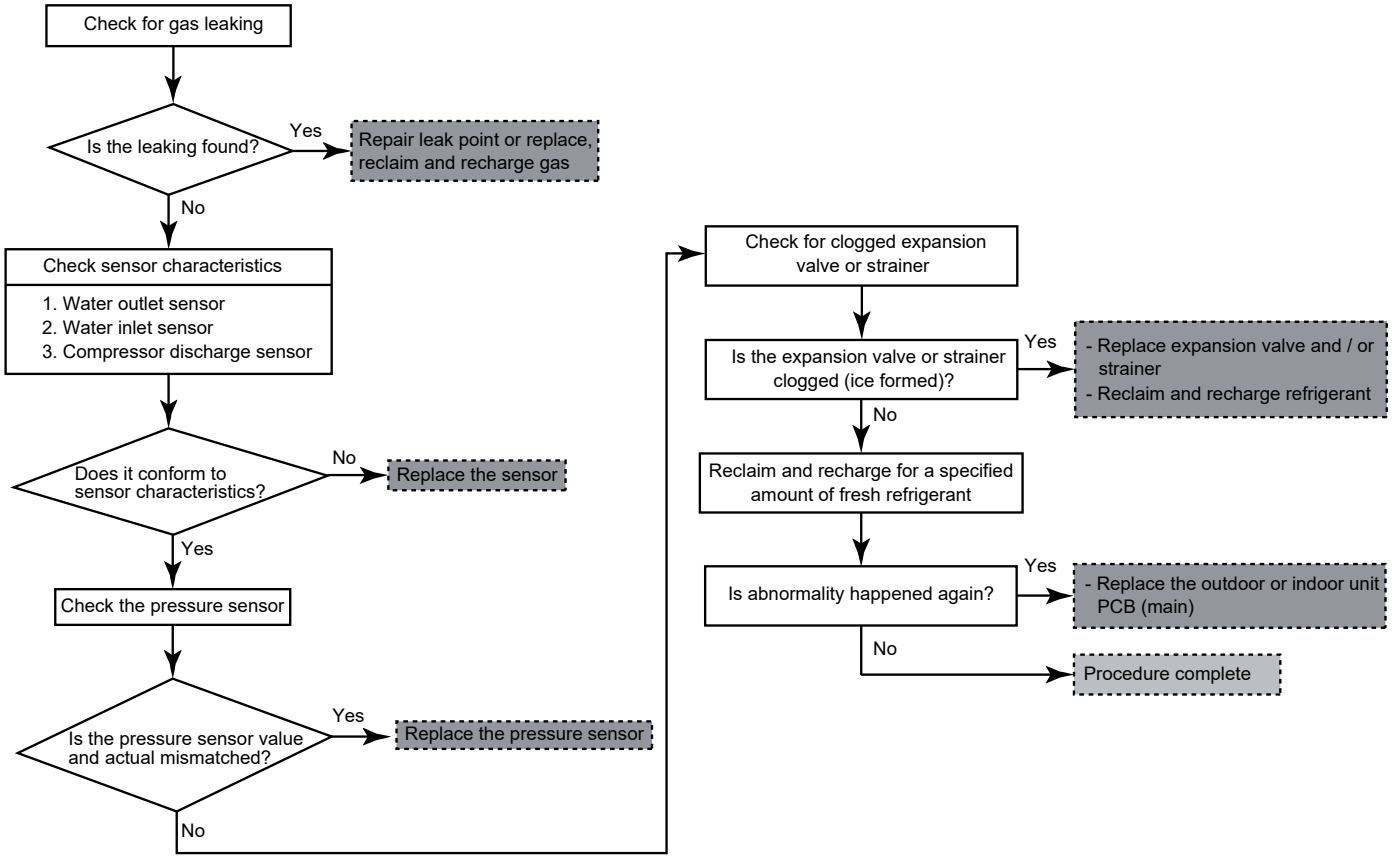
1 During compressor running (heating / cooling) for more than 10 minutes except deice, pumpdown and test mode.
2 During heating / cooling, water outlet and water inlet difference is less than 1°C.
3 During heating / cooling, high pressure < 1MPa (143 Psi) for more than 10 minutes or < 0.2 MPa (28 Psi) for more than 5 minutes.
4 During heating / cooling, discharge temperature - saturation temperature of high pressure ≥ 60°C.

Malfunction Caused:

1 Refrigerant shortage (refrigerant leakage).
2 Faulty indoor water inlet, indoor water outlet, compressor discharge temp sensor or high pressure sensor.
3 2/3 way valve closed.
4 Clogged expansion valve or strainer.
5 Faulty indoor or outdoor PCB (main).

Abnormality Judgment:

Continue 2 times in 30 minutes.



18.5.35 Four Way Valve Abnormality (F25)

Malfunction Decision Conditions:

1 During heating operation, when the indoor pipe temperature of thermostat ON indoor unit < 0°C.
2 During cooling operation, when the indoor pipe temperature of thermostat ON indoor unit > 45°C.

Malfunction Caused:

1 Faulty sensor.
2 Faulty connector connection.
3 Faulty outdoor unit PCB (noise filter/main).
4 Faulty four way valve.

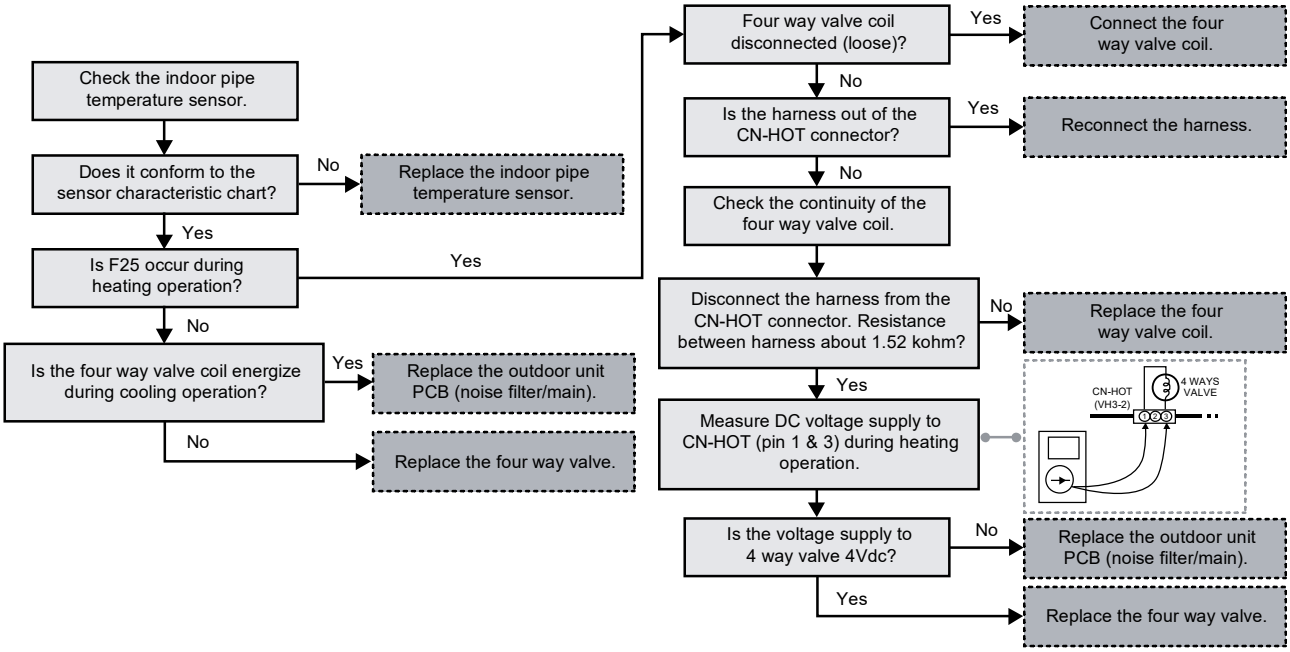
Abnormality Judgment:

Continue 4 times in 30 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.36 Outdoor High Pressure Switch Abnormal (F27)

Malfunction Decision Conditions:

During compressor stop, and outdoor high pressure switch is remain opened.

Malfunction Caused:

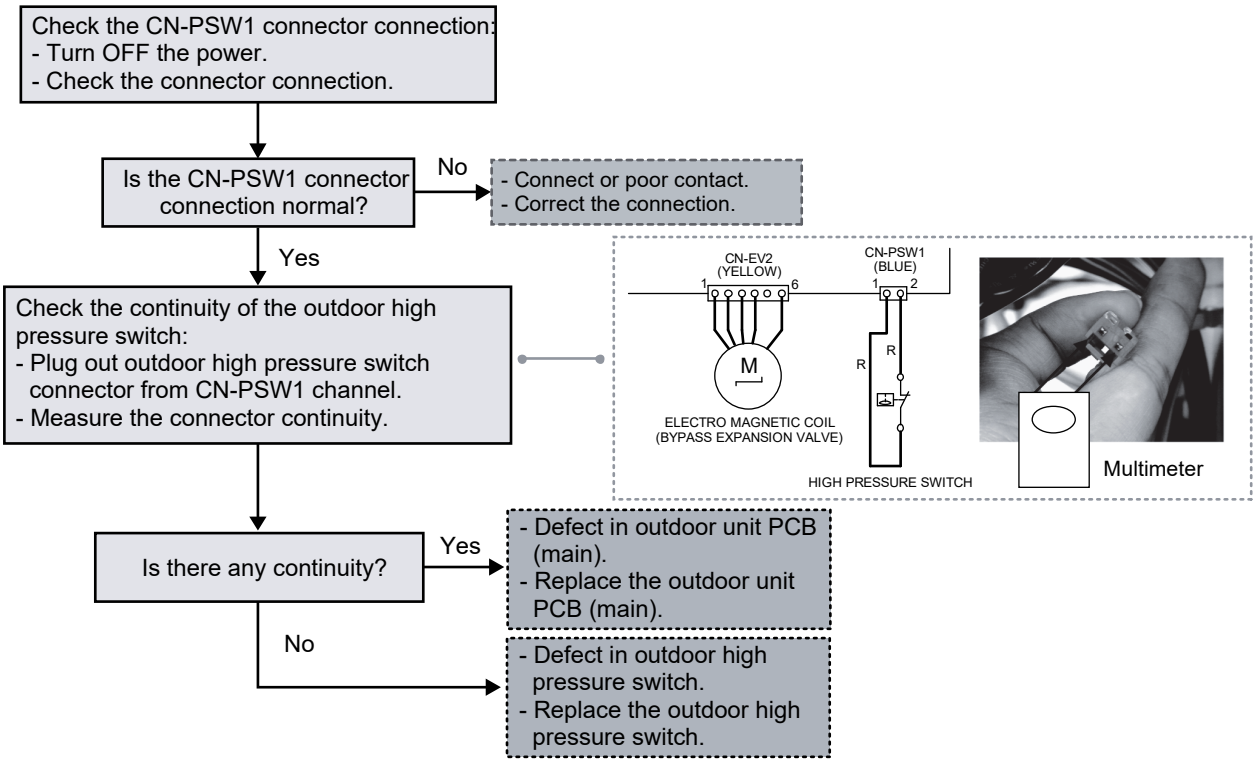
1 Faulty connector connection.
2 Faulty switch.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 1 minute.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component



18.5.37 Low Discharge Superheat (F29)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).
4 Faulty High Pressure Switch
5 Refrigerant shortage (refrigerant leakage).

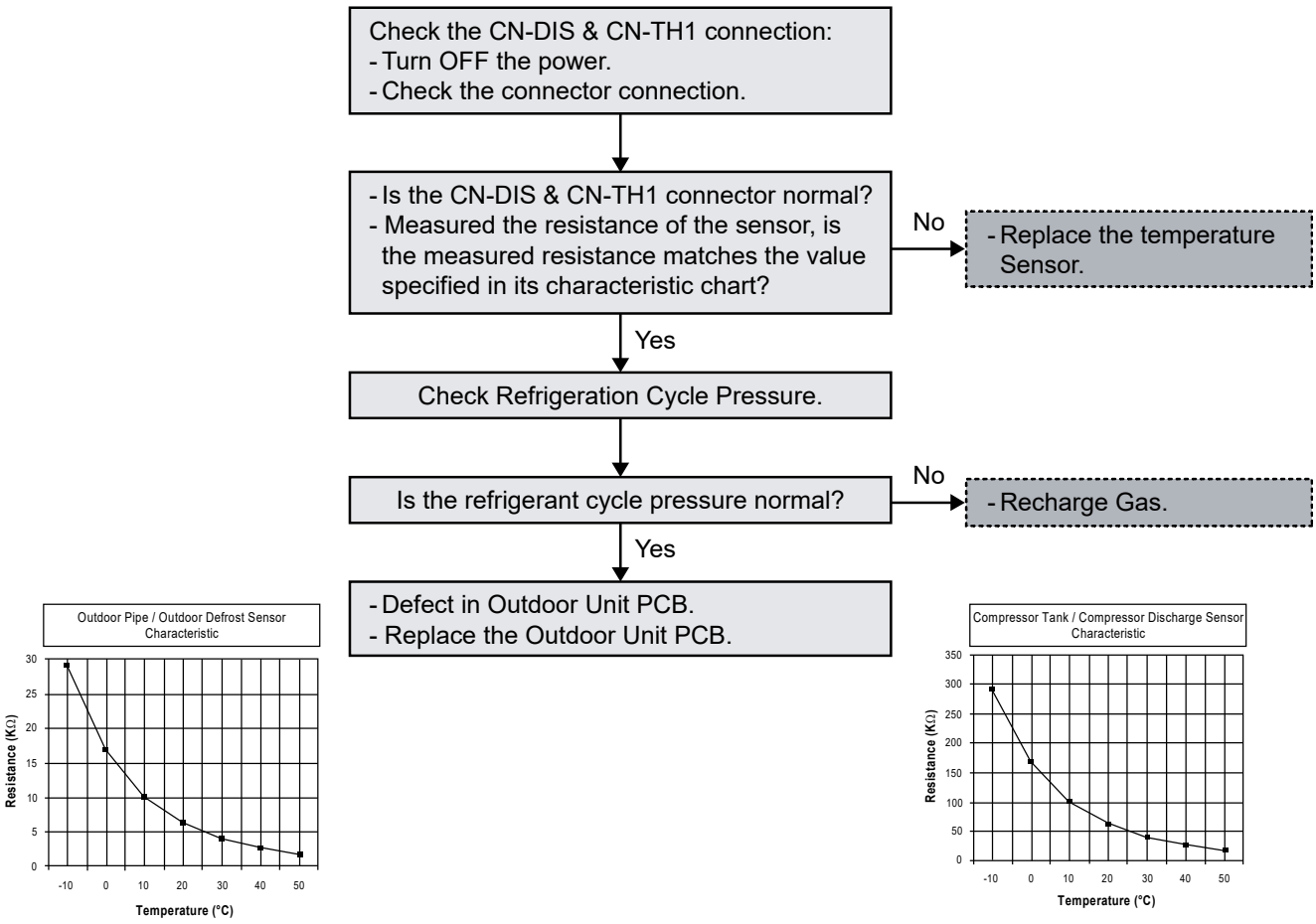
Abnormality Judgment:

1 times occurrence within 2550 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.38 Indoor Water Outlet Temperature Sensor 2 Abnormality (F30)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water outlet temperature sensor 2 are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB.

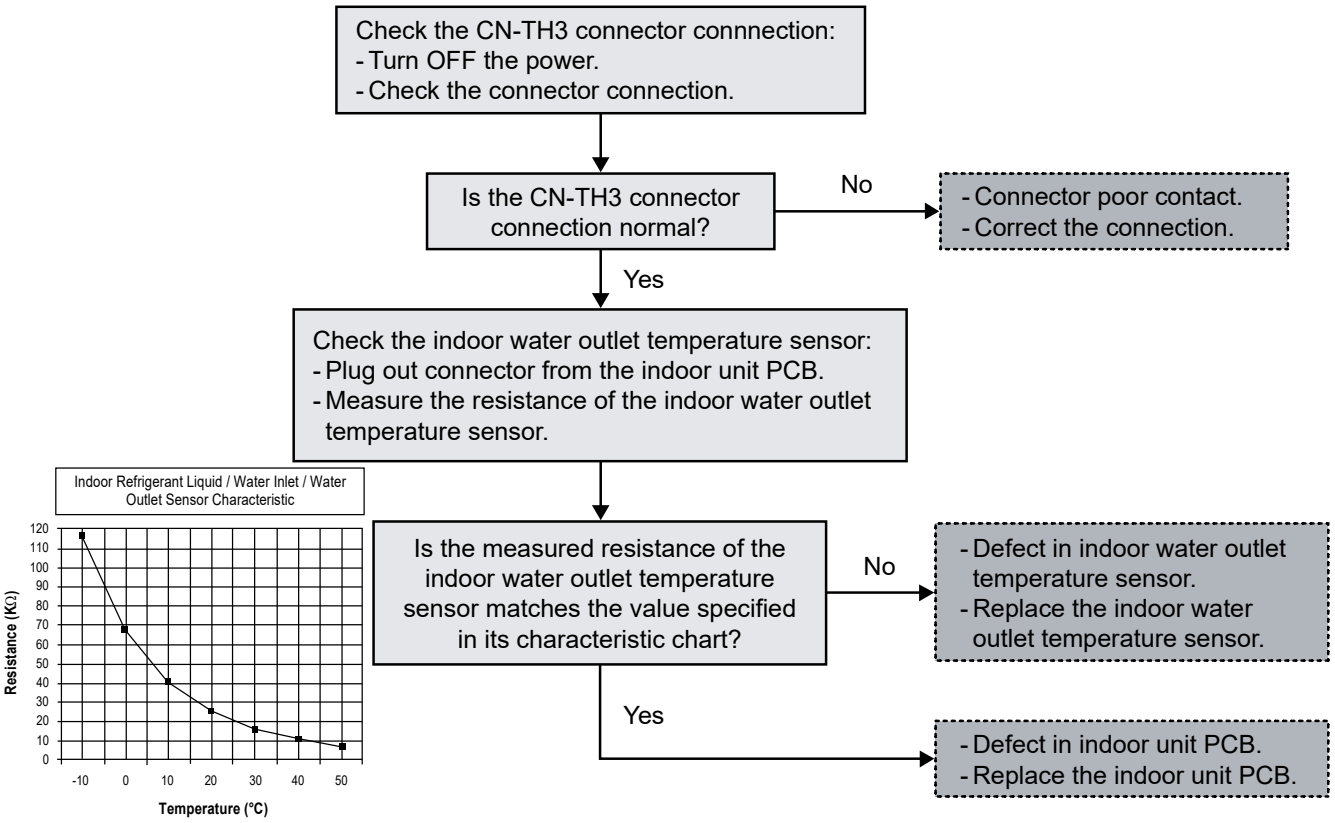
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.39 Outdoor Air Temperature Sensor Abnormality (F36)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor air temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

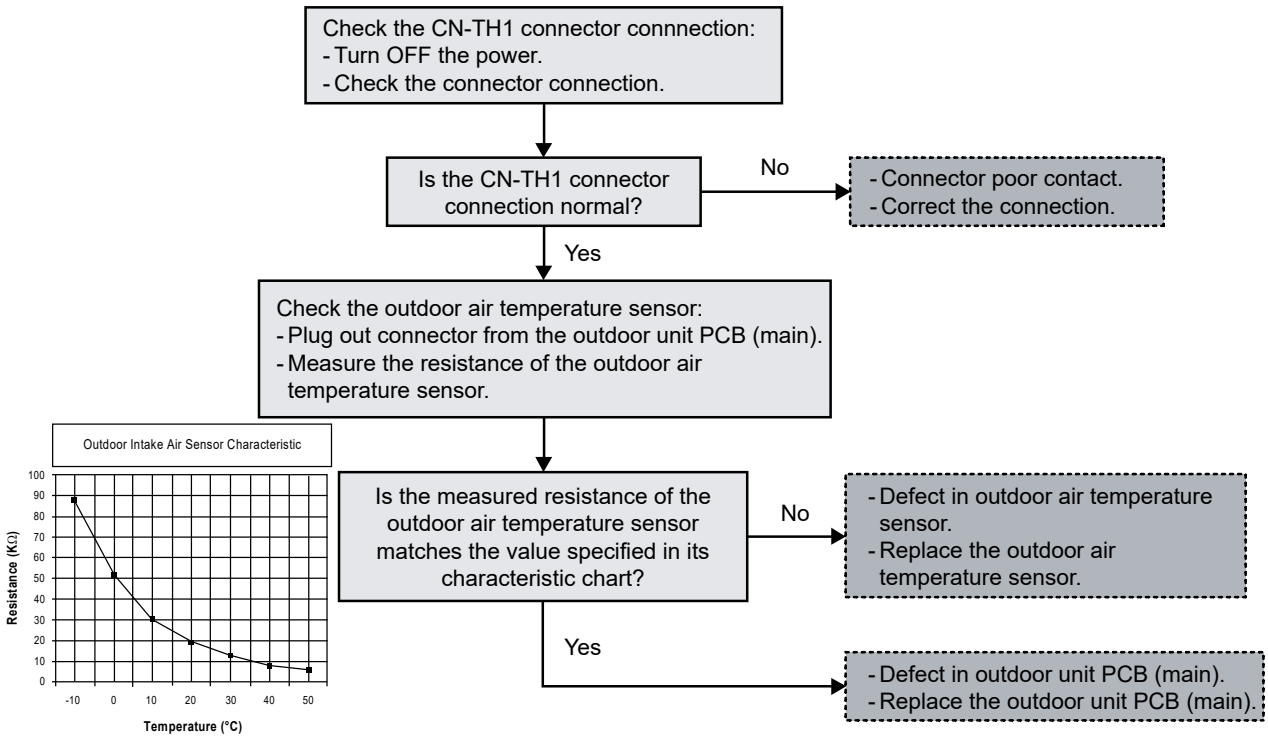
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.40 Indoor Water Inlet Temperature Sensor Abnormality (F37)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water inlet temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

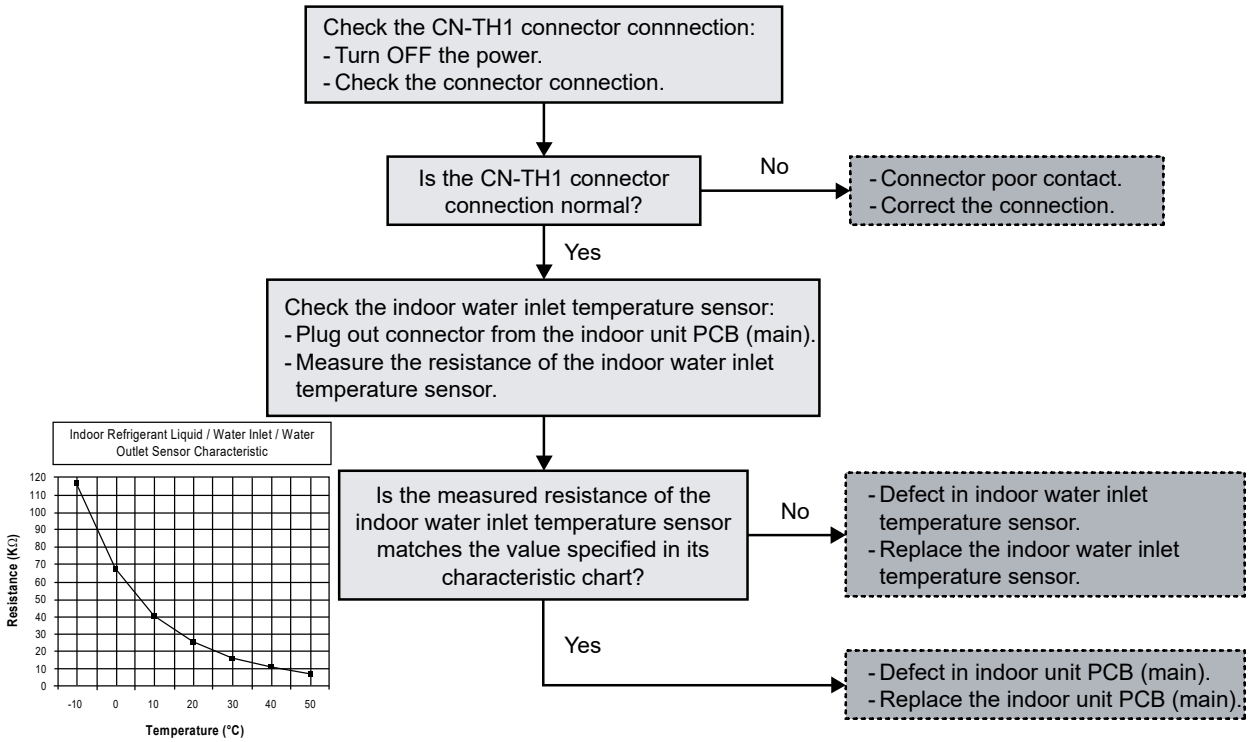
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.41 Outdoor Discharge Pipe Temperature Sensor Abnormality (F40)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor discharge pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

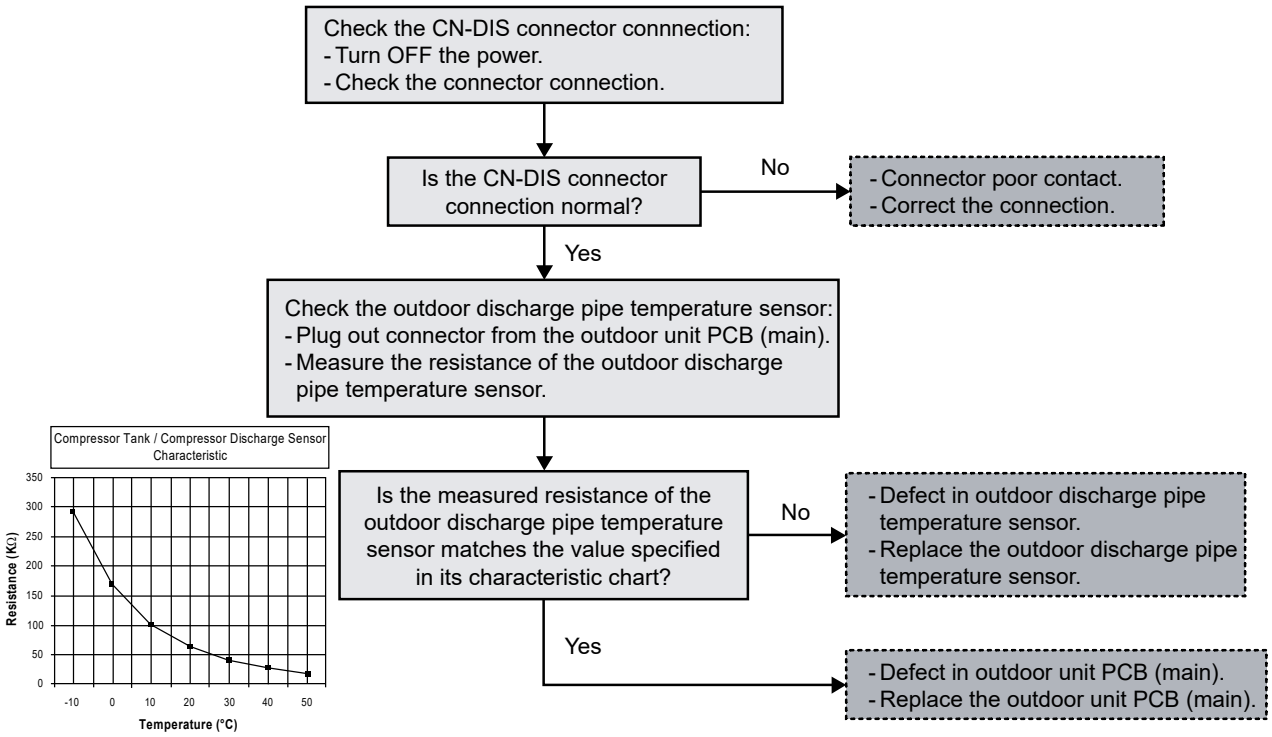
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.42 Power Factor Correction (PFC) Abnormality (F41)

Malfunction Decision Conditions:

During operation of cooling and heating, when the PFC protection circuitry in the outdoor unit PCB (main) senses abnormal high DC voltage level.

Malfunction Caused:

1 Power supply surge.
2 Compressor windings not uniform.
3 Faulty outdoor unit PCB (main).

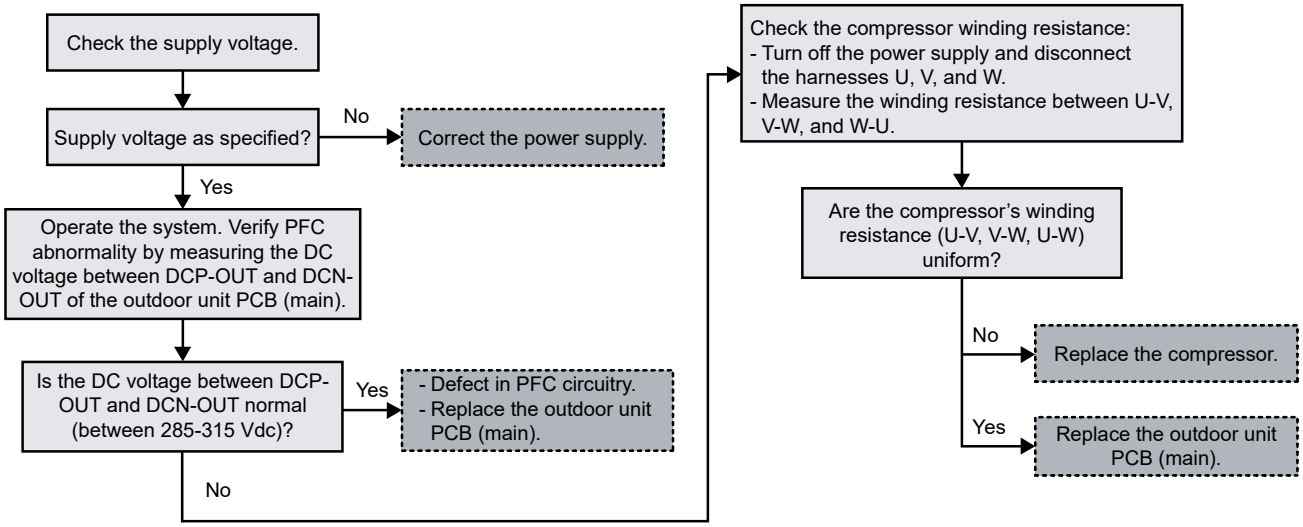
Abnormality Judgment:

Continue 4 times in 10 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.43 Outdoor Pipe Temperature Sensor Abnormality (F42)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

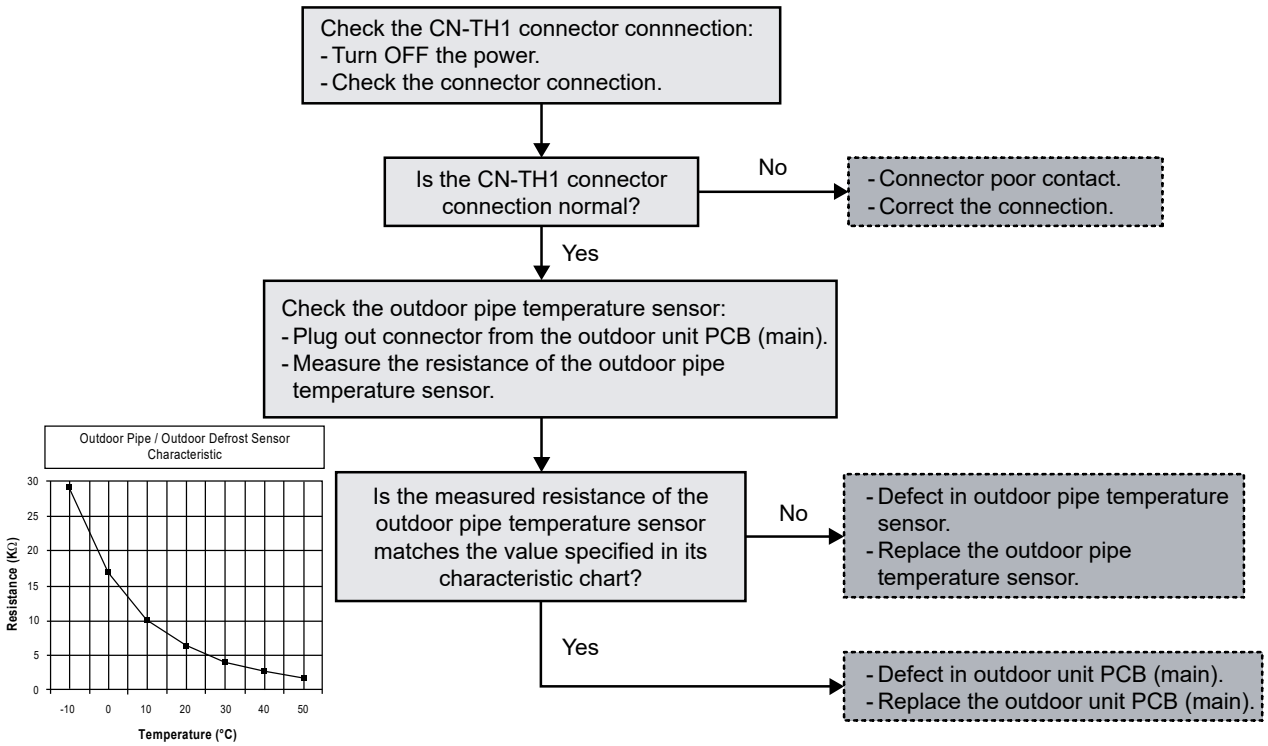
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.44 Outdoor Defrost Temperature Sensor Abnormality (F43)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor defrost temperature sensor are used to determine sensor error.

Malfunction Caused:

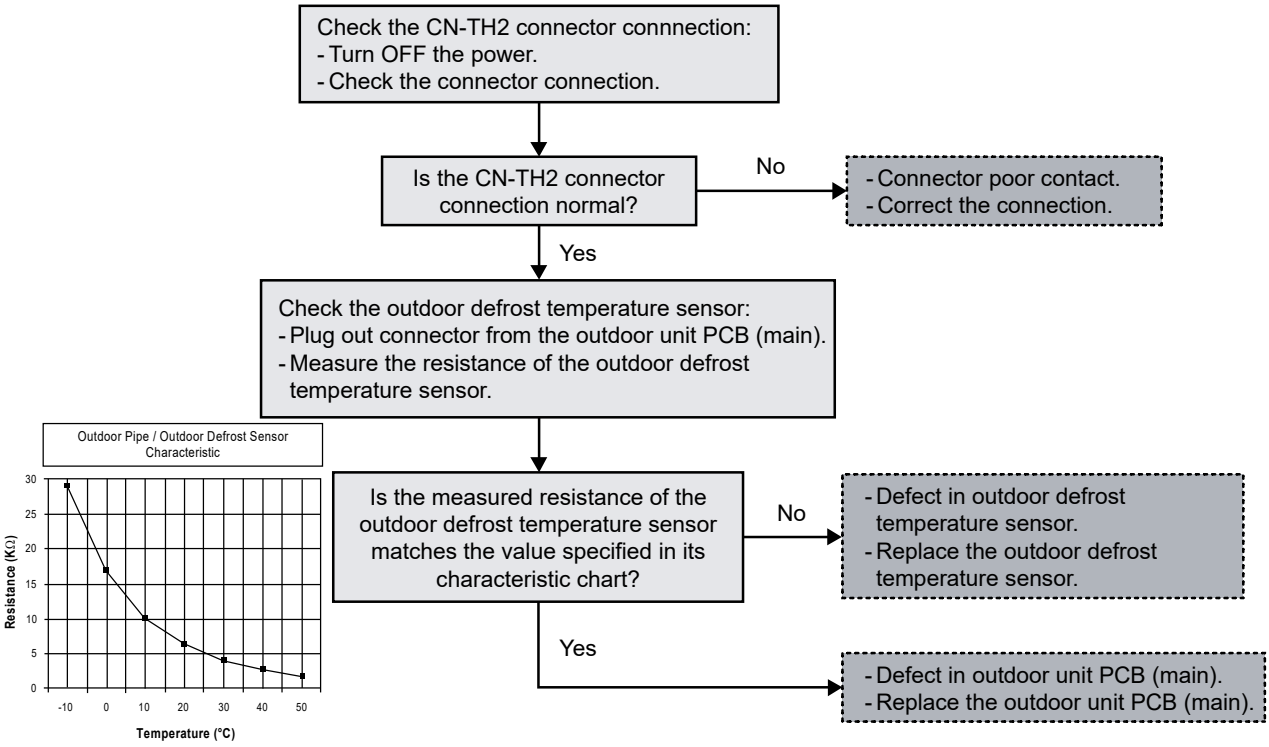
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.45 Indoor Water Outlet Temperature Sensor Abnormality (F45)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water outlet temperature sensor are used to determine sensor errors.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

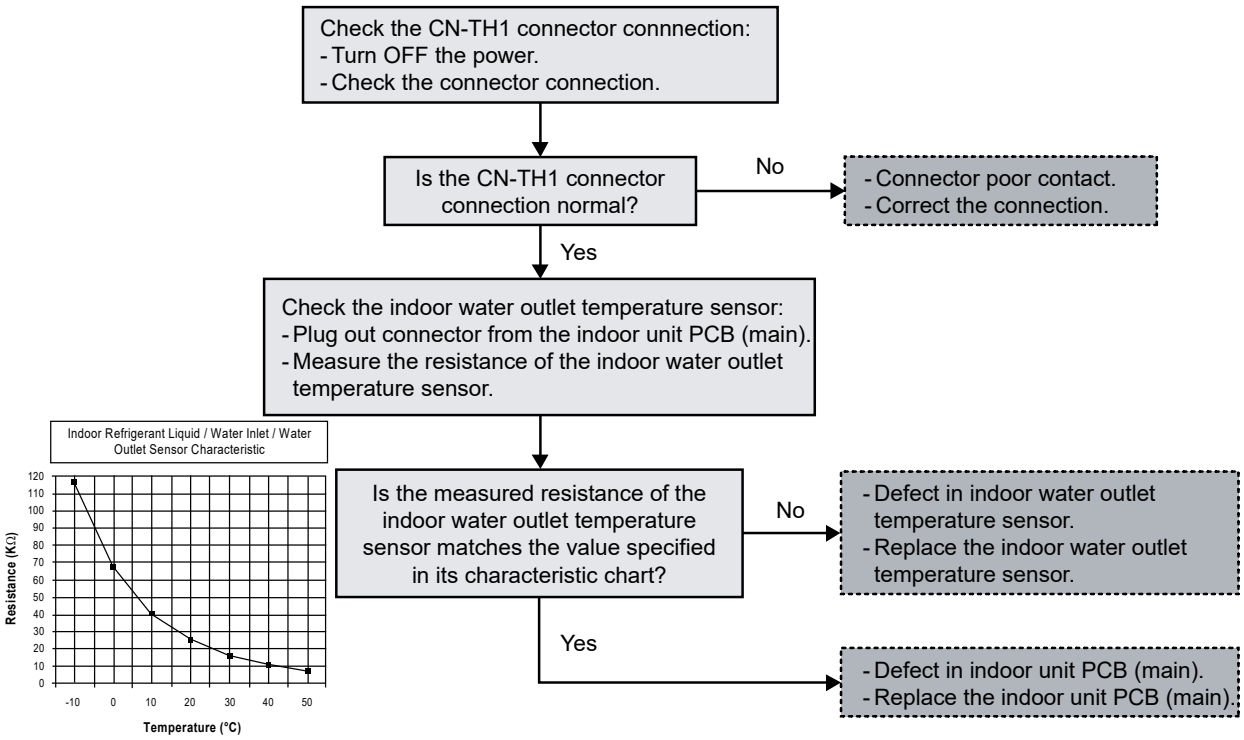
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.46 Outdoor Current Transformer Open Circuit (F46)

Malfunction Decision Conditions:

A current transformer (CT) open circuit is detected by checking the compressor running frequency (⩾ rated frequency) and CT detected input current (< 0.65 A) for continuously 20 seconds.

Malfunction Caused:

1 CT defective.
2 Faulty outdoor unit PCB (main).
3 Compressor defective (low compression).

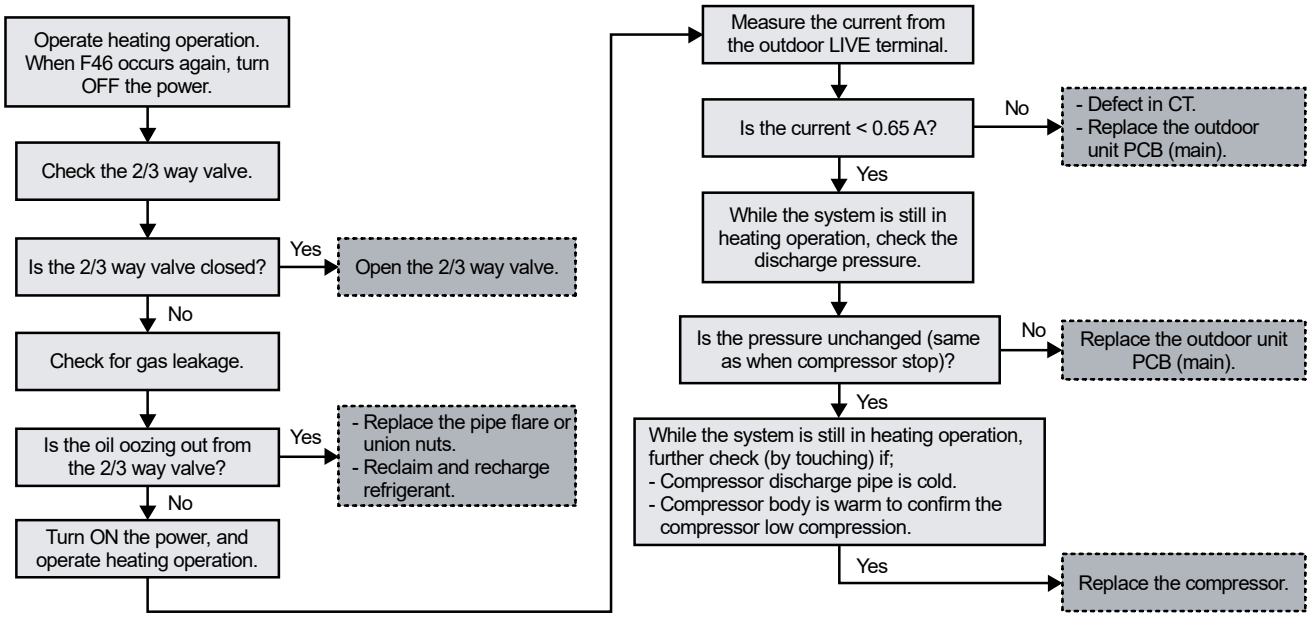
Abnormality Judgment:

Continue 3 times in 20 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.47 Outdoor Bypass Outlet Temperature Sensor (F49)

Malfunction Decision Conditions:

During start up and operation of cooling and heating, the temperature detected by outdoor bypass sensor is used to determine sensor error.

Malfunction Caused:

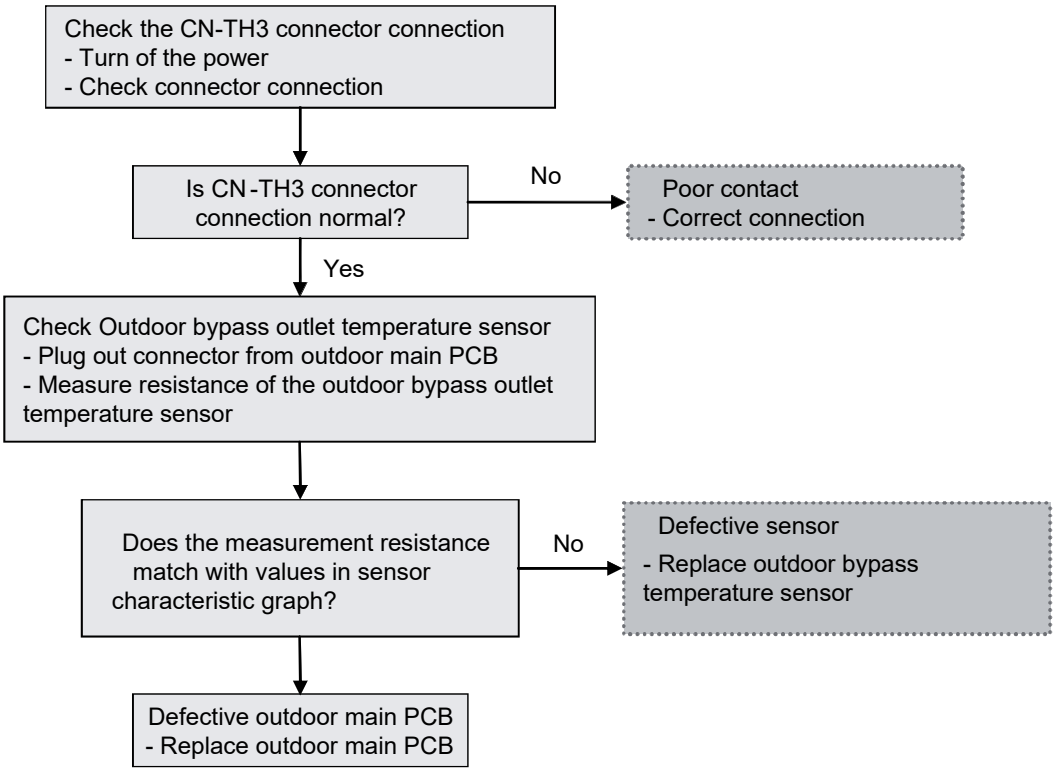
1 Faulty connector connection
2 Faulty sensor
3 Faulty outdoor unit PCB (main)

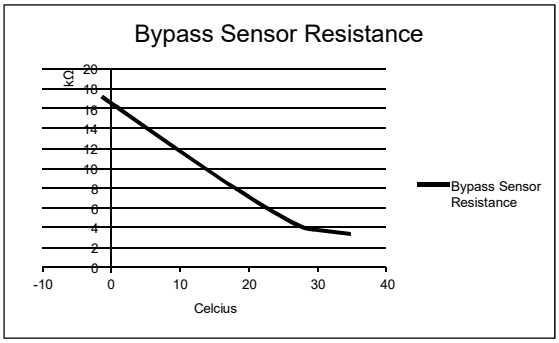
Abnormality Judgment:

Continuous for 5 seconds

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.





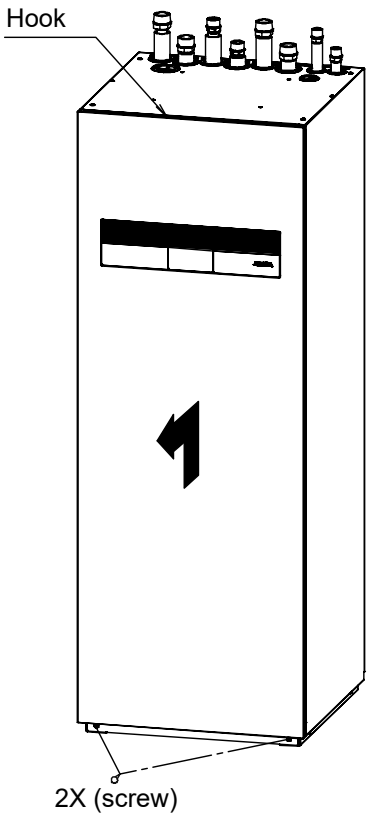
19. Disassembly and Assembly Instructions

⚠ WARNING

High Voltage are generated in the electrical parts area by the capacitor. Ensure that the capacitor has discharged sufficiently before proceeding with repair work. Failure to heed this caution may result in electric shocks.

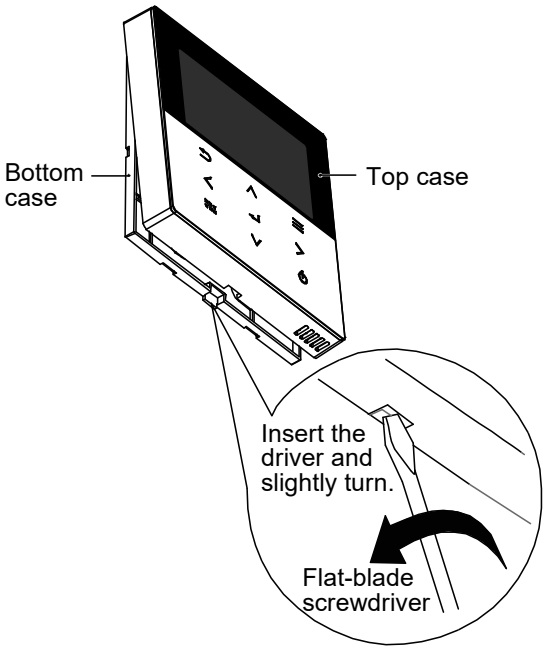
19.1 Indoor Unit

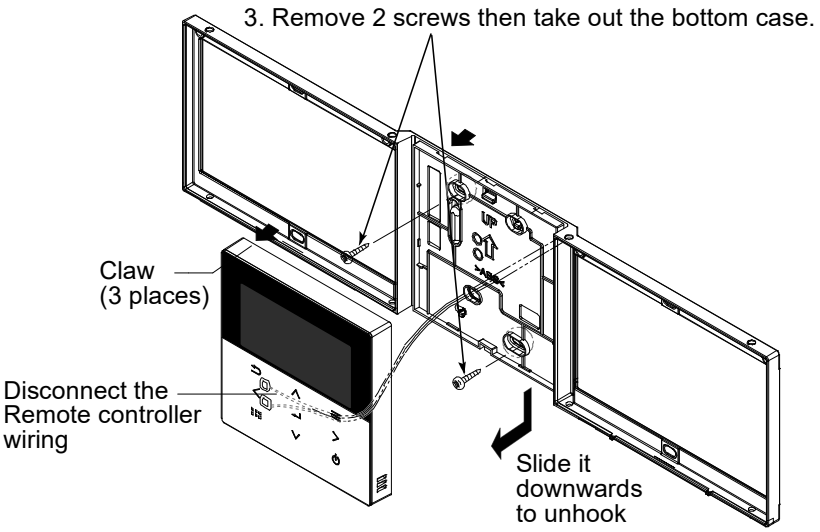
19.1.1 To Remove Front Plate and Top Plate



1 Remove 2 screw at the bottom to remove the Front Plate
2 Remove 8 screw at the top to remove the Top Plate

19.1.2 To Remove Remote Control

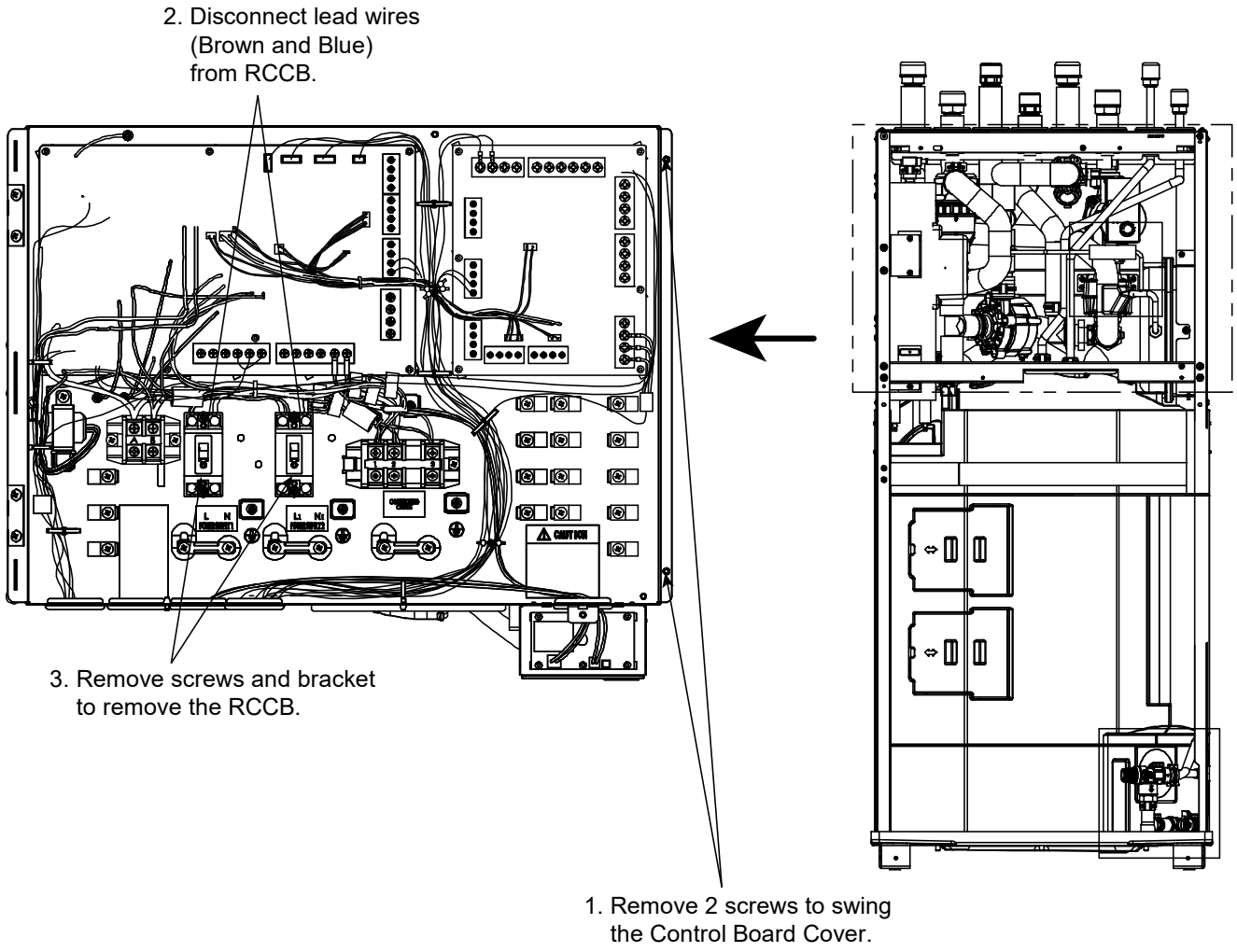




1. Remove the top case from the bottom case.

2. Disconnect the Remote Controller wiring.

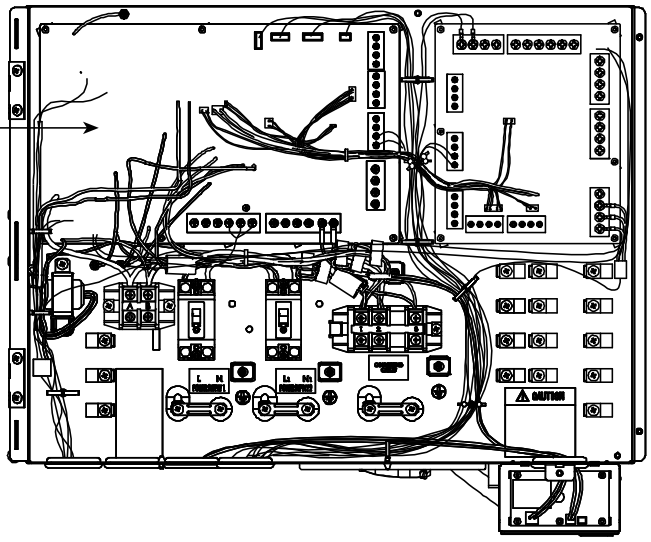
19.1.3 To Remove RCCB



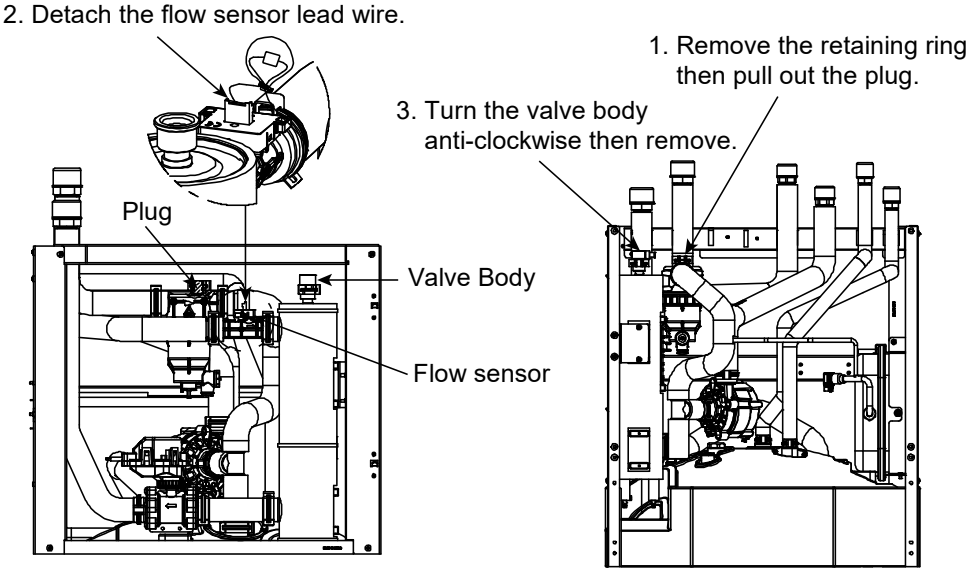
19.1.4 To Remove Electronic Controller

1. Disconnect all connectors from main PCB. (CN-FLWSEN, CN-PUMP1, CN-TH1, CN-TH2, CN-OLP1, CN1, CN2, CN3, CN4, CN5, CN6, CN-CNT2, CN-DPS, CN-CNT, CN-PWR, CN-PWR2, CN-PWR3, CN-PWR4).

2. Detach remote controller, 3-way valve and all wires from terminal board (AC1-N, G01, G05, DATA, G02, L2, L1, G03, HT1-L3, AC2-L3, AC2-N2, AC2-N3.)

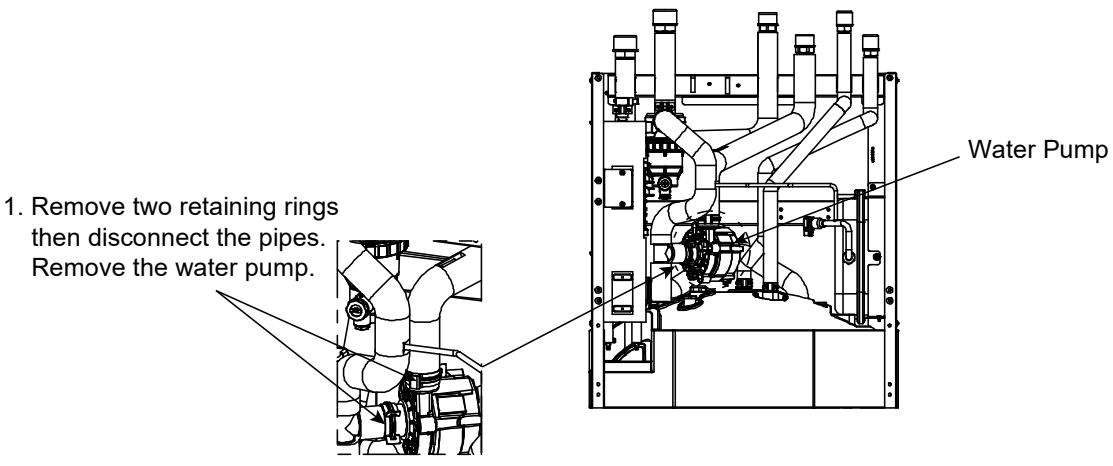


19.1.5 To Remove Flow Sensor, Plug and Valve Body



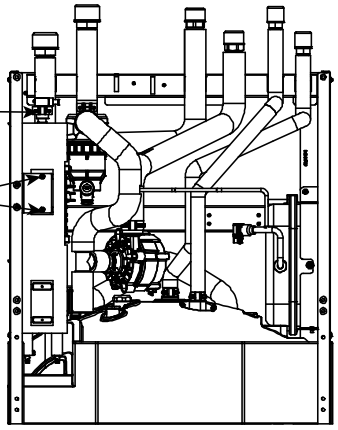
⚠ When reinstall Flow Switch, ensure the arrow on the flow switch is parallel with the pipe shaft and is facing in the direction of flow.

19.1.6 To Remove Water Pump



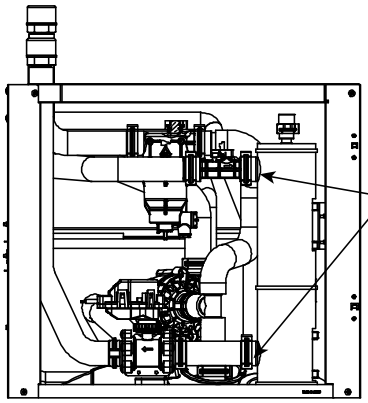
19.1.7 To Remove Heater Complete

1. Disconnect the connector CN-OLP1 from the Electronic Controller and detach the lead wire HT1-L3 (Red) and AC2-L3 (Black).



2. Remove Valve Body.

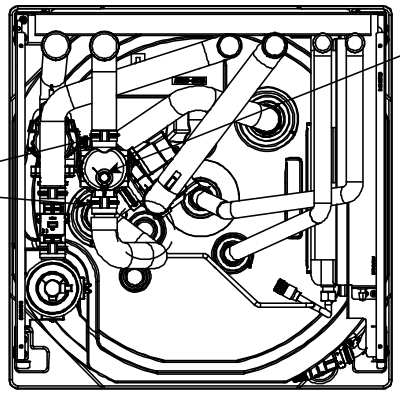
4. Remove 2 screws from the bracket.



3. Disconnect 2 pipes connection.

Side View

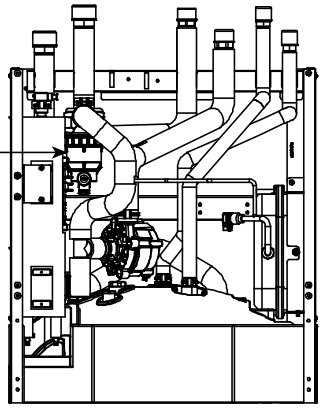
19.1.8 To Remove Water Filter



1. Remove the retaining ring then pull out the plug.

2. Remove 2 retaining rings then pull out the Magnetic Water Filter set.

Top View



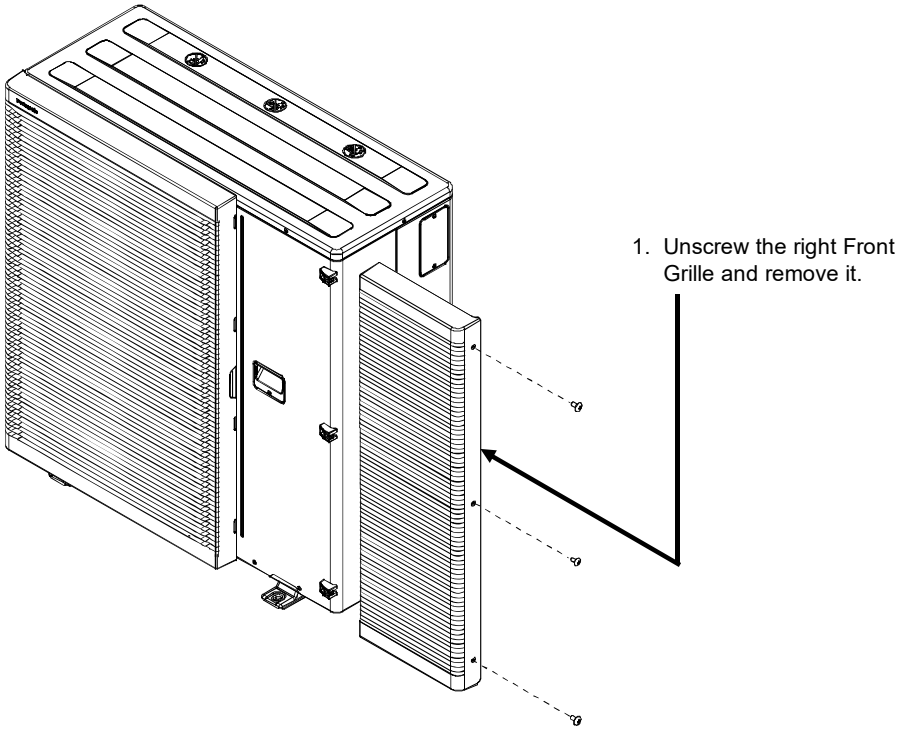
Magnetic Water Filter

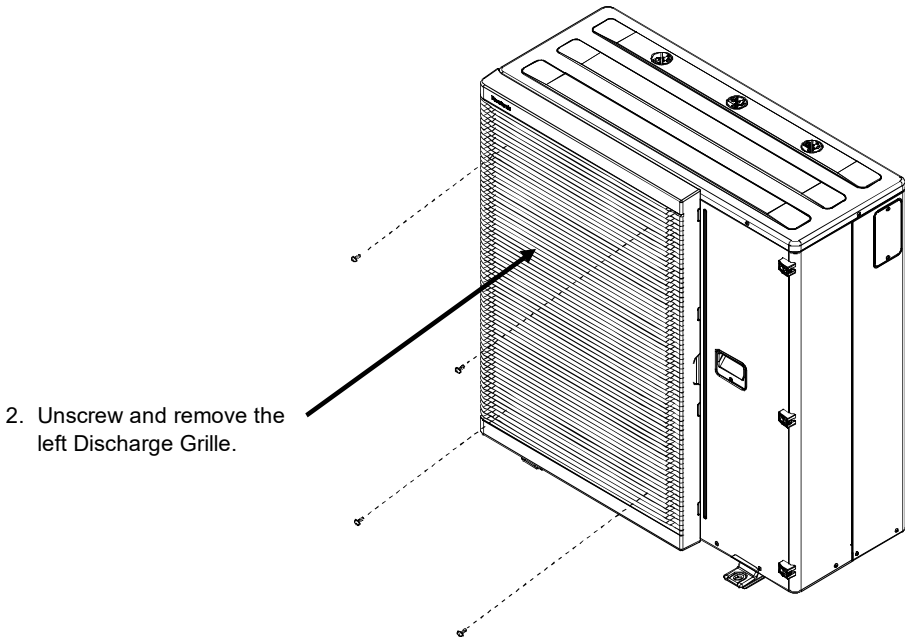
WARNING	This symbol shows that this equipment uses a flammable refrigerant with safety A3 group per ISO 817. If the refrigerant is leaked, together with an external ignition source, there is a possibility of fire / explosion.

- If you are working on the R290 product, before starting work and when entering the service area, always turn ON the combustible gas leak detector to ensure there is no leakage.
- Keep all ignition sources away from the product. In particular, open flames, hot surfaces, electrical devices that are not free from electrical sources, static discharges.
- Ensure the servicing area is well ventilated.
- Ensure all the serving tools and equipment complied with ATEX (Atmosphere Explosible) standard.
- Ensure the product is service by certified and authorized serviceman.
- Ensure to always have the approved fire extinguisher during servicing.
- Use a warning placard to ensure that unauthorized personnel cannot enter the protective zone.

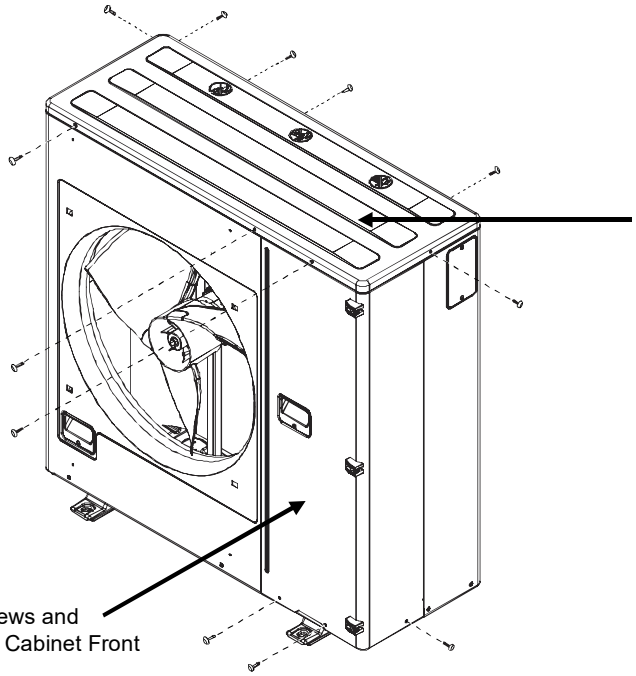
19.2 Outdoor Unit

19.2.1 Electronic Controller Removal Procedures



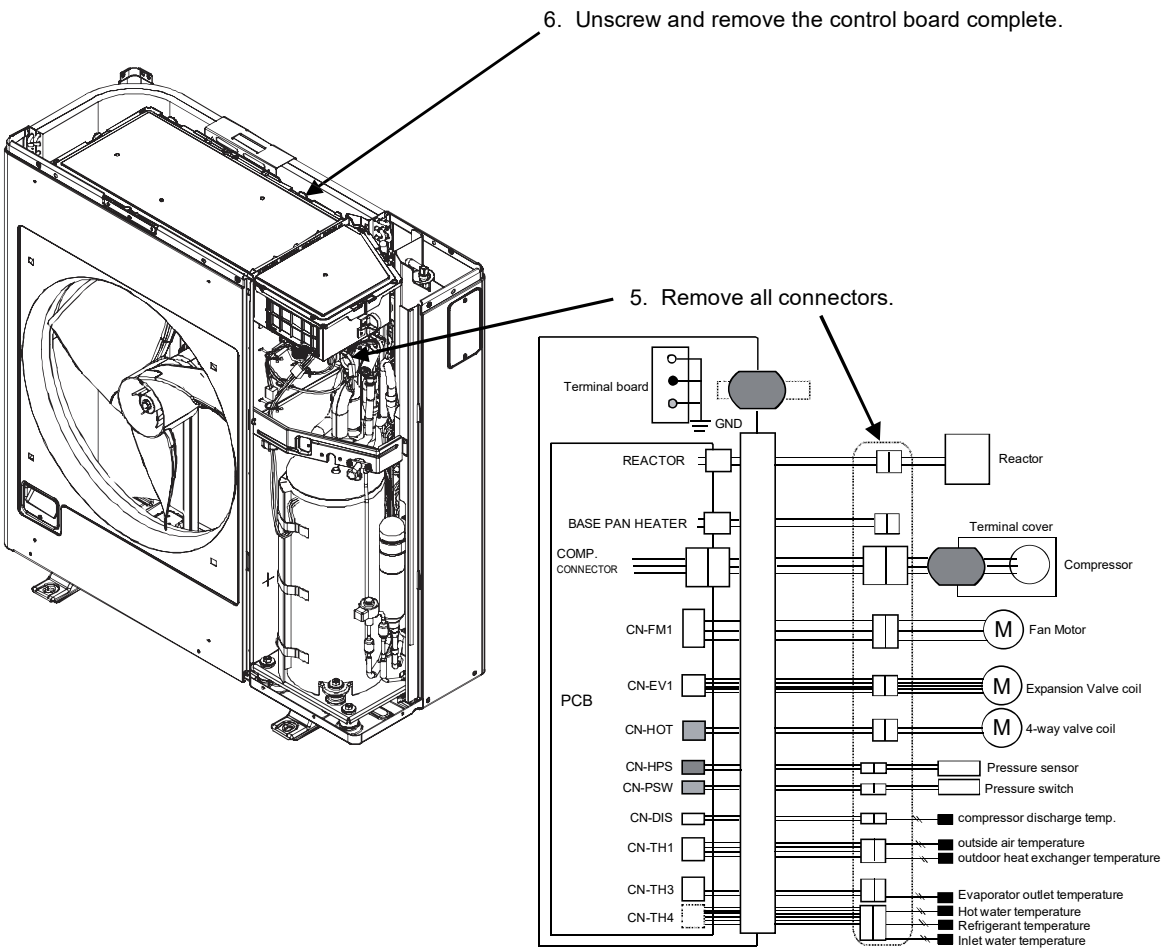


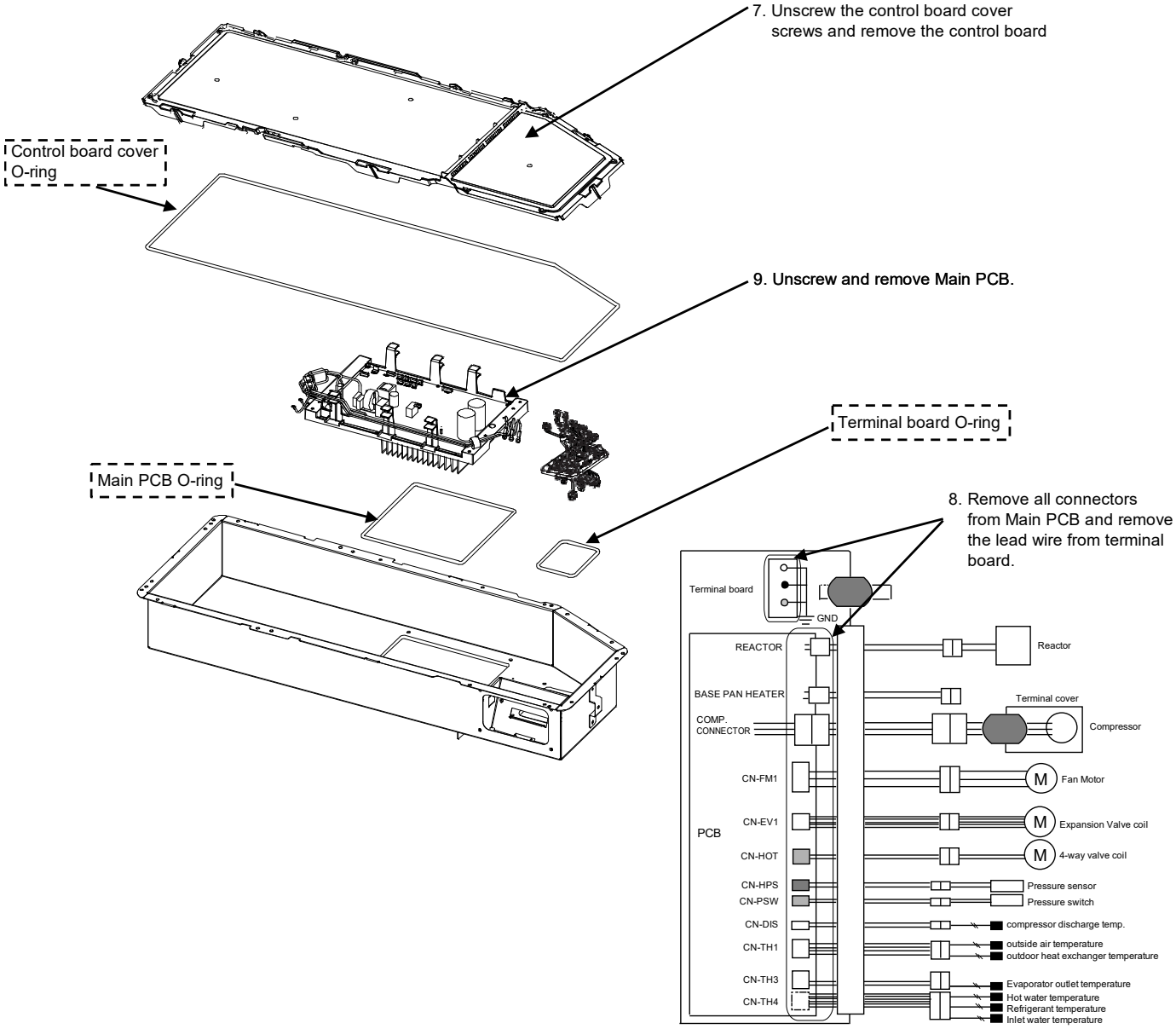
2. Unscrew and remove the left Discharge Grille.



3. Unscrew the screws and remove the Top Plate.

4. Unscrew the screws and remove the right Cabinet Front Plate.


6. Unscrew and remove the control board complete.
5. Remove all connectors.
Terminal board
GND
REACTOR
BASE PAN HEATER
COMP.
CONNECTOR
CN-FM1
CN-EV1
PCB
CN-HOT
CN-HPS
CN-PSW
CN-DIS
CN-TH1
CN-TH3
CN-TH4
Reactor
Terminal cover
Compressor
M
Fan Motor
M
Expansion Valve coil
M
4-way valve coil
Pressure sensor
Pressure switch
compressor discharge temp.
outside air temperature
outdoor heat exchanger temperature
Evaporator outlet temperature
Hot water temperature
Refrigerant temperature
Inlet water temperature



Note: During re-assemble the Main PCB, ensure to attach the O-ring properly to avoid gas leakage into the control board complete.

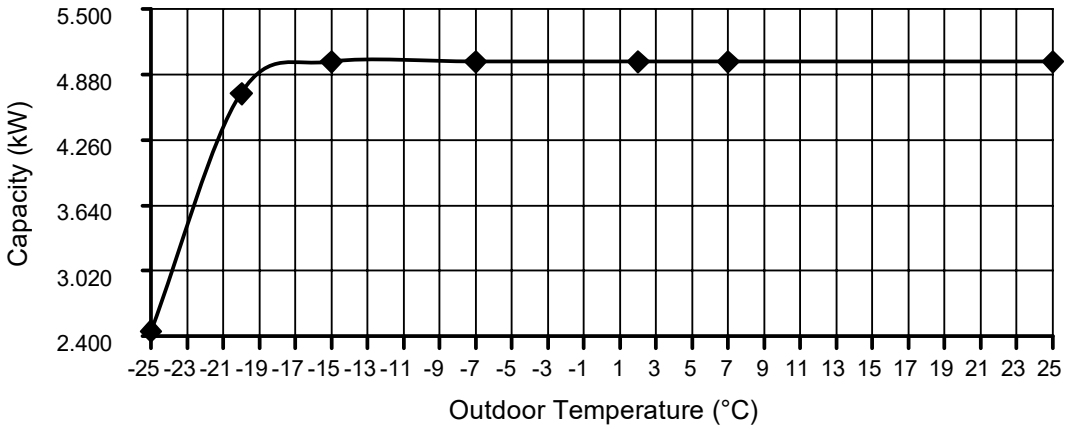
20. Technical Data

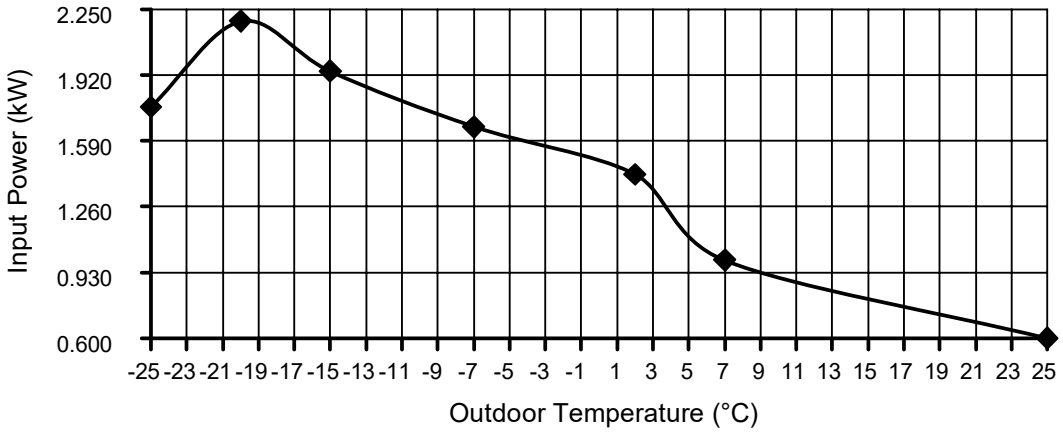
20.1 Operation Characteristics

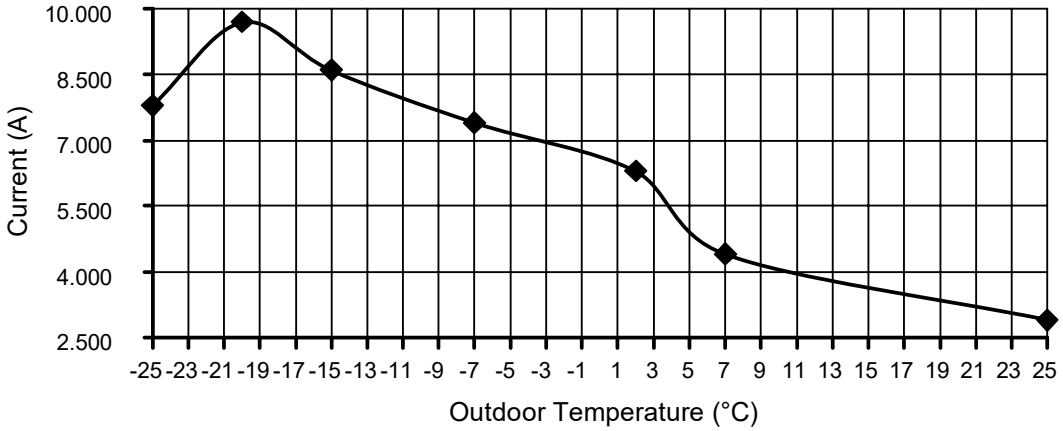
20.1.1 WH-WDG05LE5

Heating Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C
Piping length : 7 m

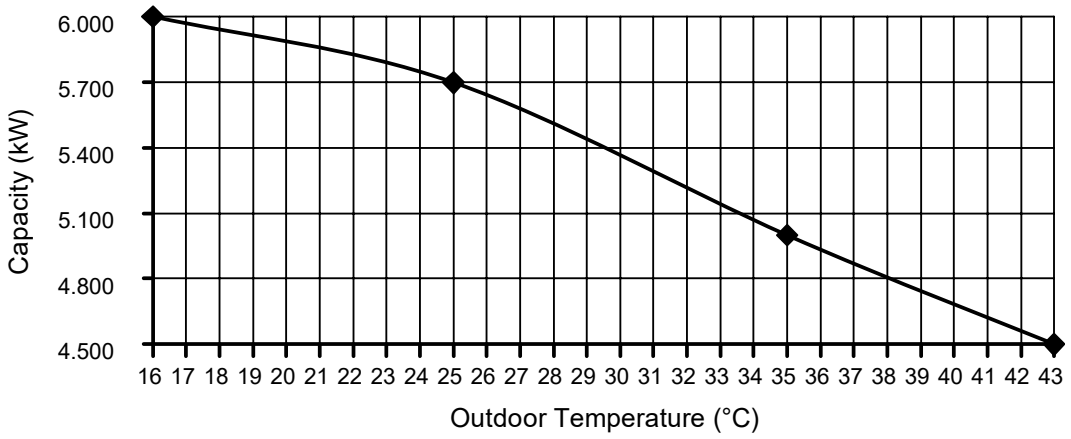


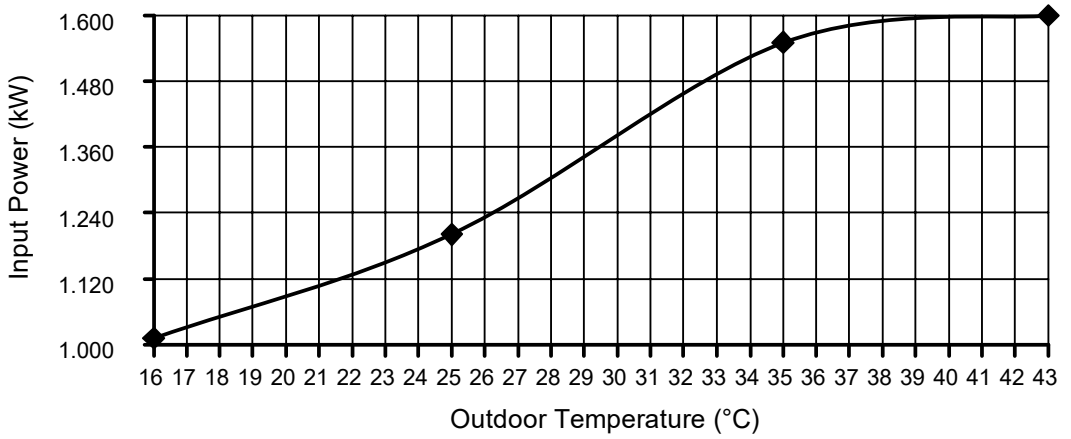


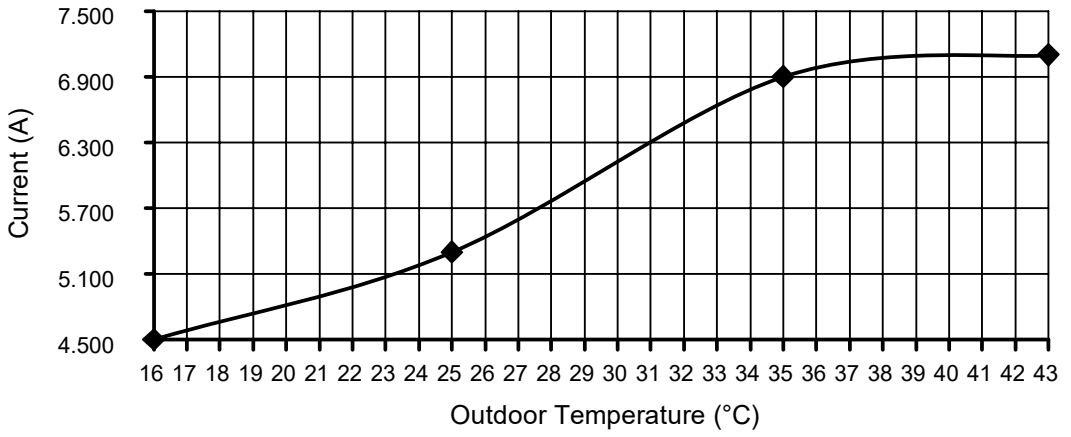


Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 12°C
Indoor water outlet temperature : 7°C
Piping length : 7 m







Cooling Characteristics at Different Outdoor Air Temperature

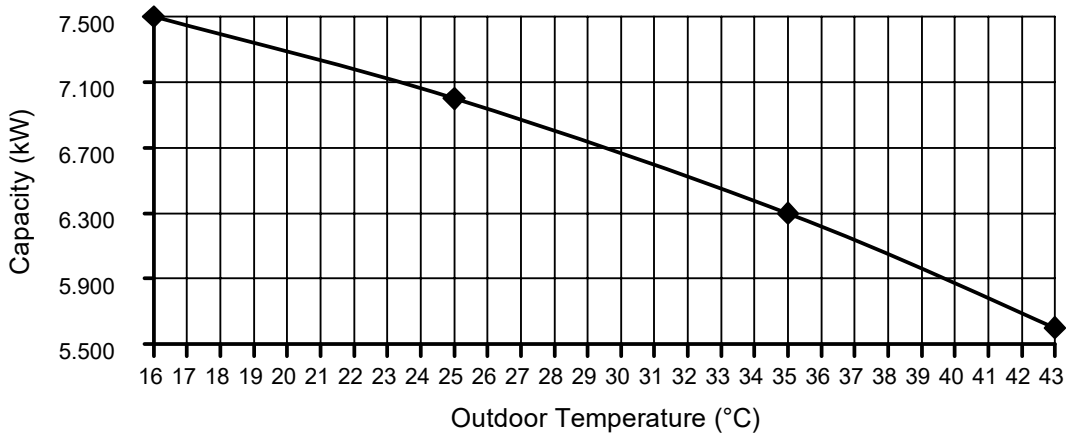
Condition

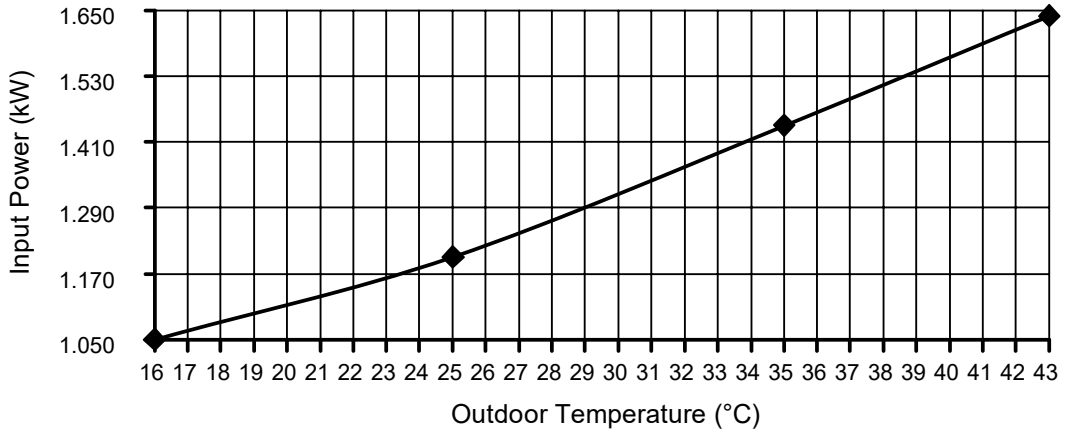
Outdoor air temperature : 35°C (DBT), -°C (WBT)

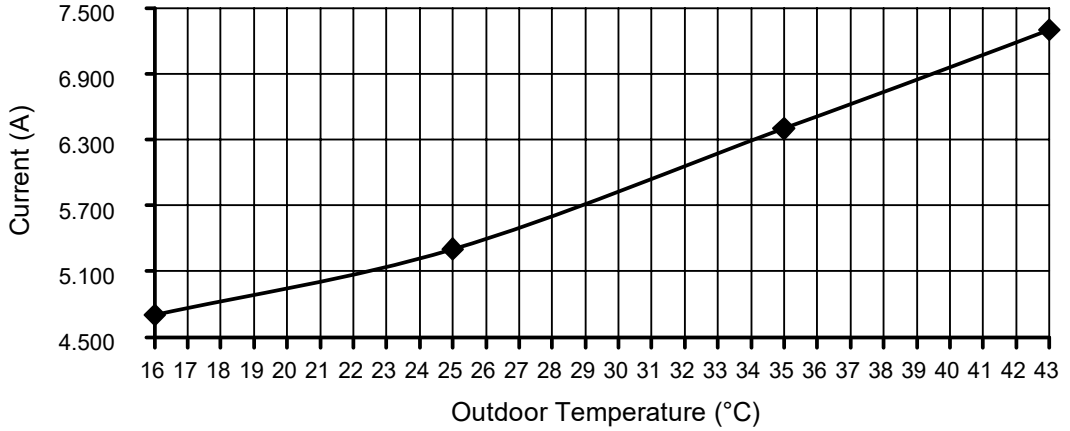
Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C

Piping length : 7 m







Cooling Characteristics at Different Outdoor Air Temperature

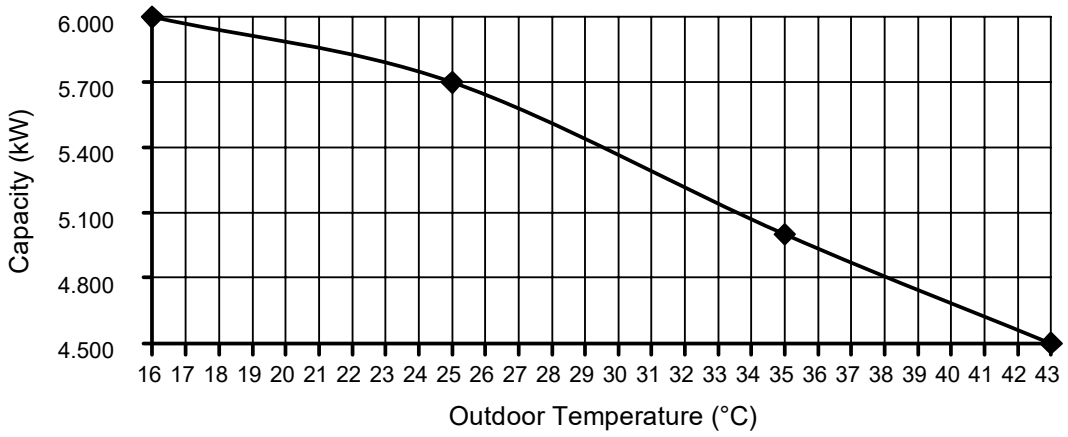
Condition

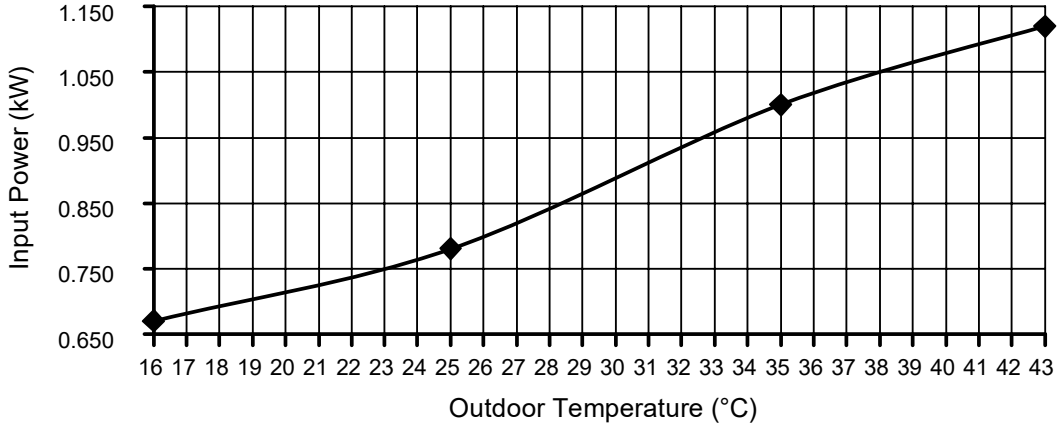
Outdoor air temperature : 35°C (DBT), -°C (WBT)

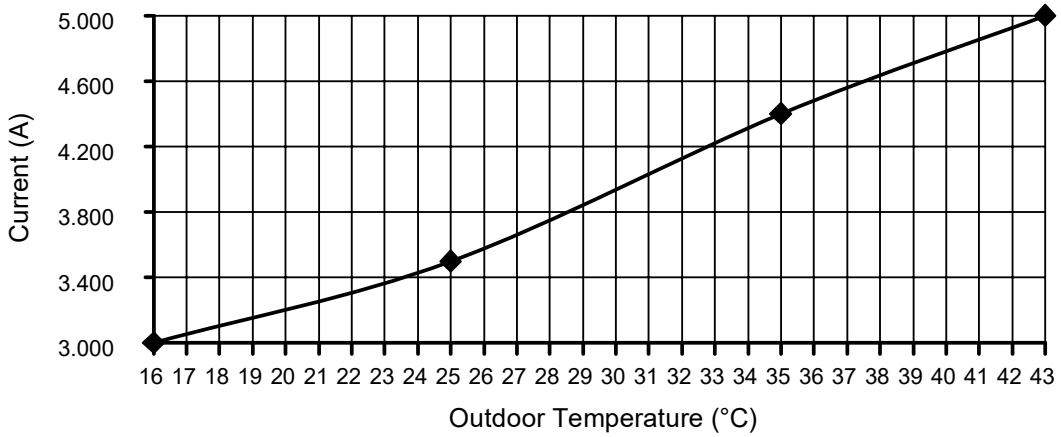
Indoor water inlet temperature : 23°C

Indoor water outlet temperature : 18°C

Piping length : 7 m



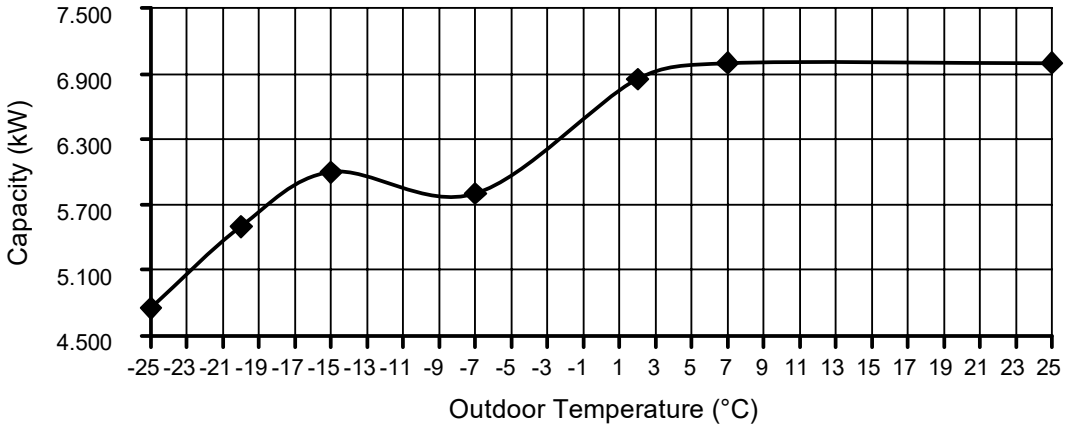


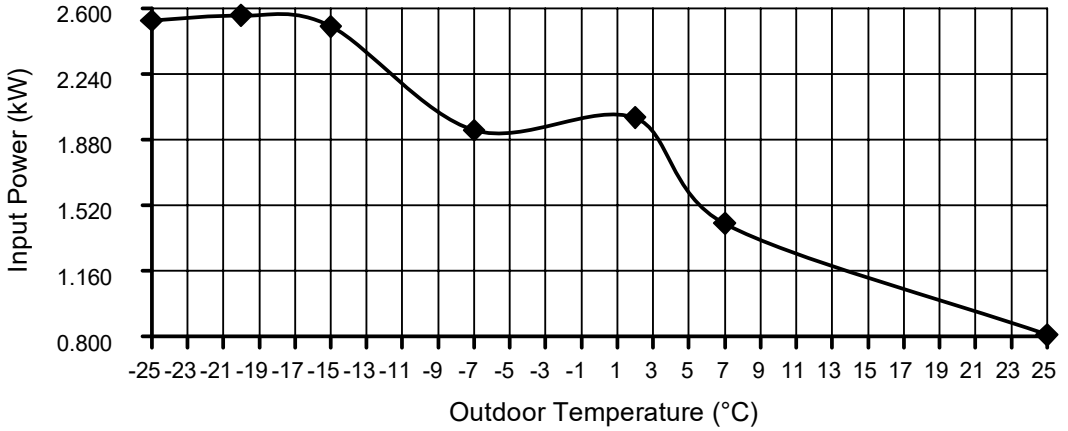


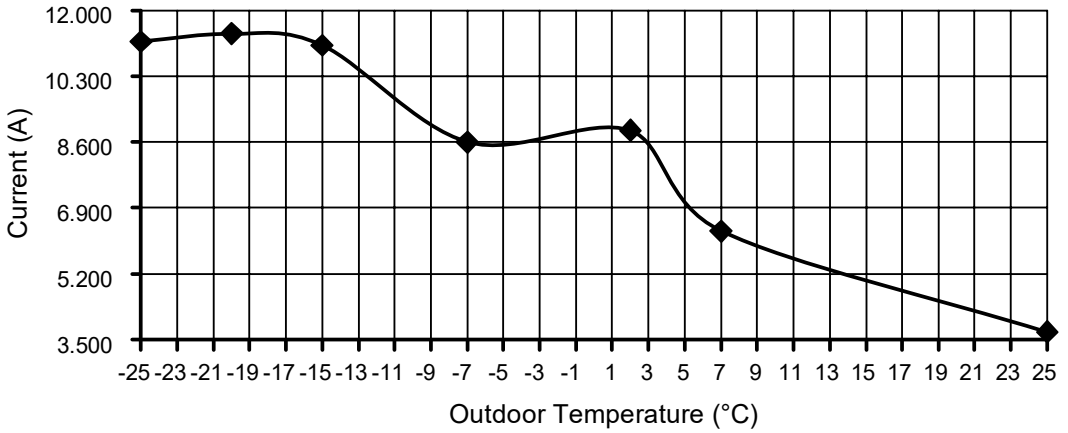
20.1.2 WH-WDG07LE5

Heating Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C
Piping length : 7 m







Cooling Characteristics at Different Outdoor Air Temperature

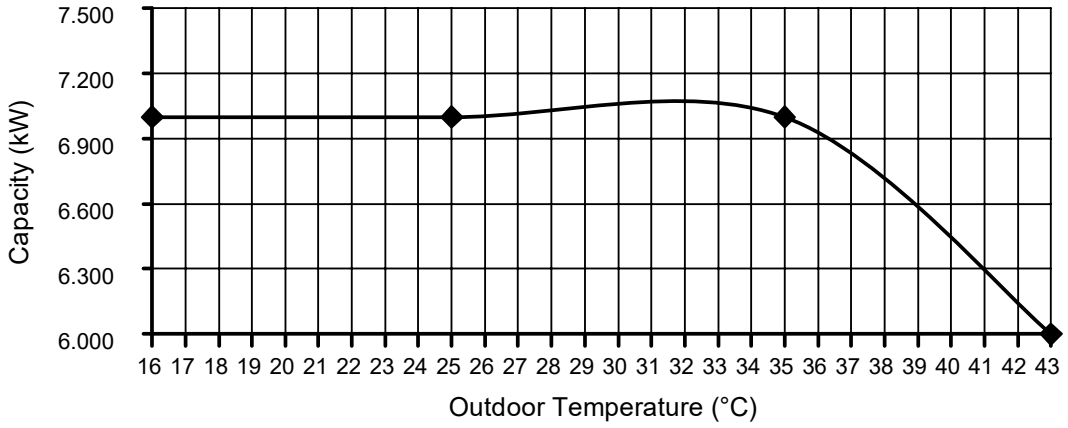
Condition

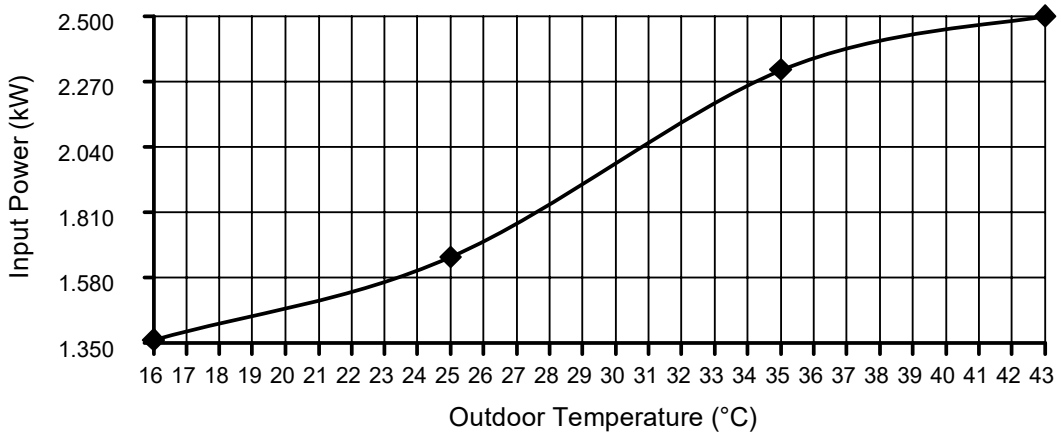
Outdoor air temperature : 35°C (DBT), -°C (WBT)

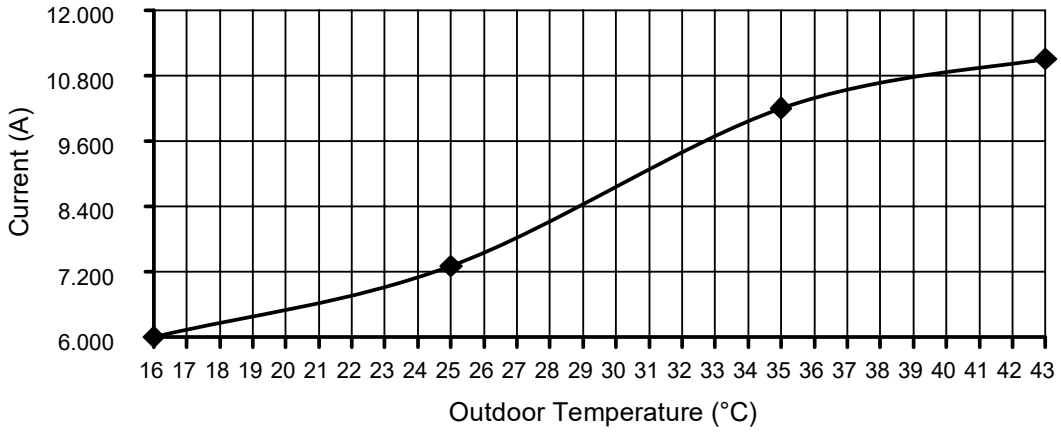
Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C

Piping length : 7 m







Cooling Characteristics at Different Outdoor Air Temperature

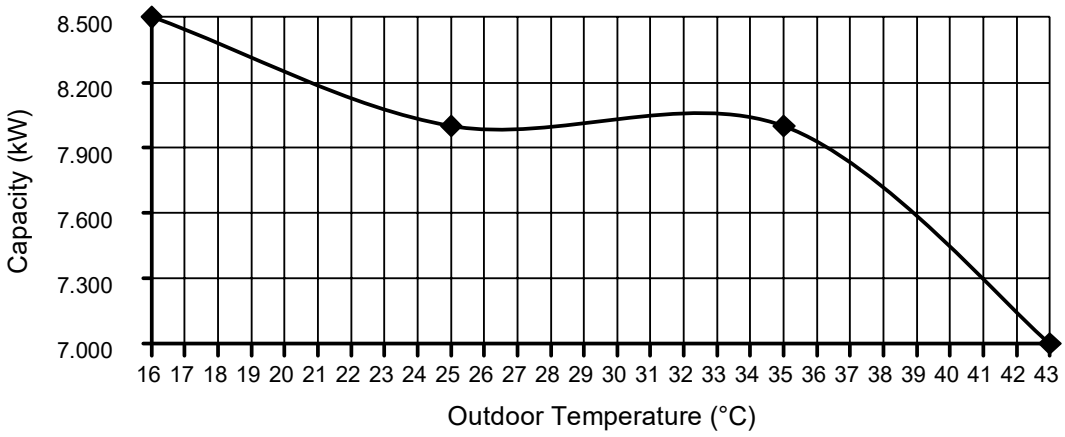
Condition

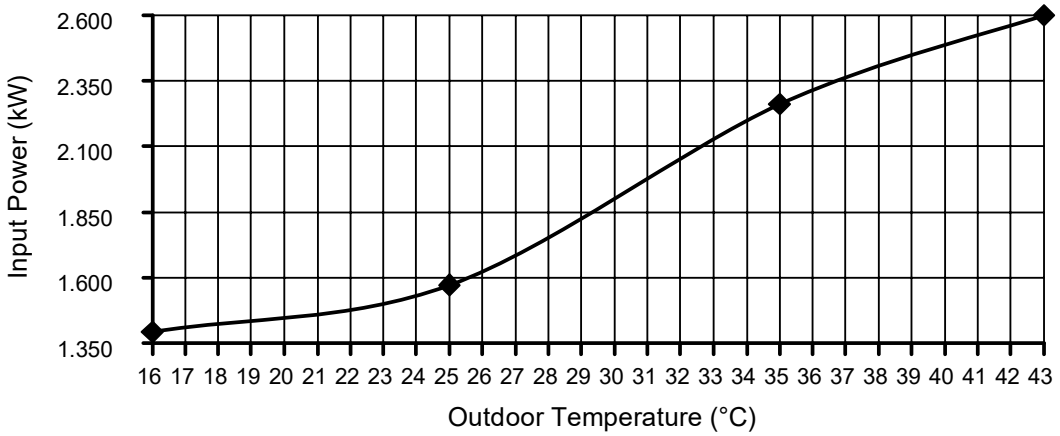
Outdoor air temperature : 35°C (DBT), -°C (WBT)

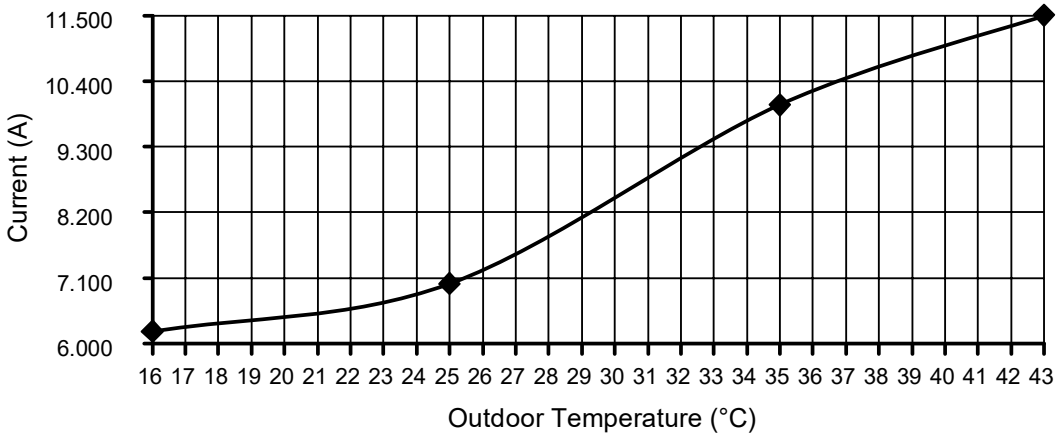
Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C

Piping length : 7 m







Cooling Characteristics at Different Outdoor Air Temperature

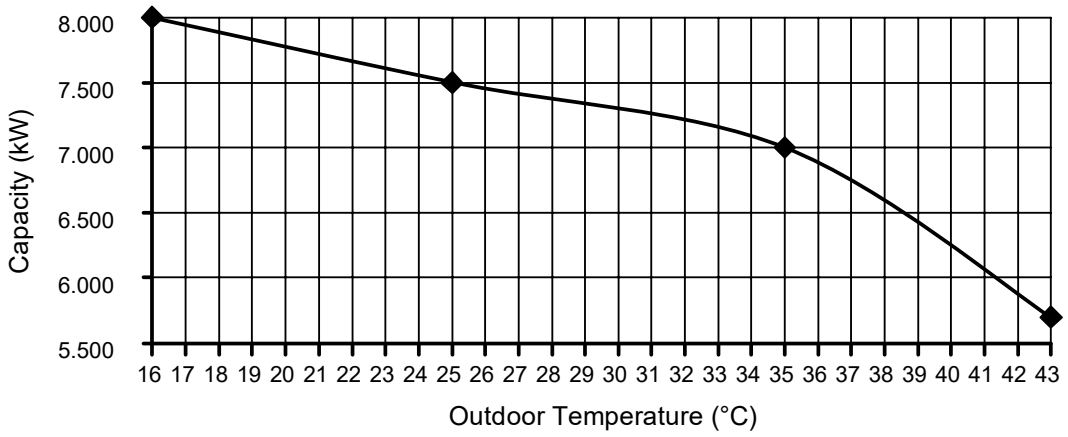
Condition

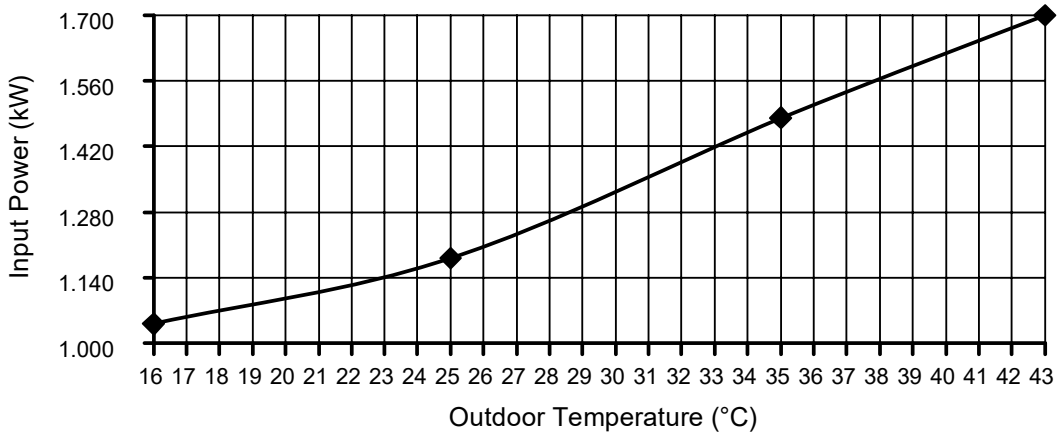
Outdoor air temperature : 35°C (DBT), -°C (WBT)

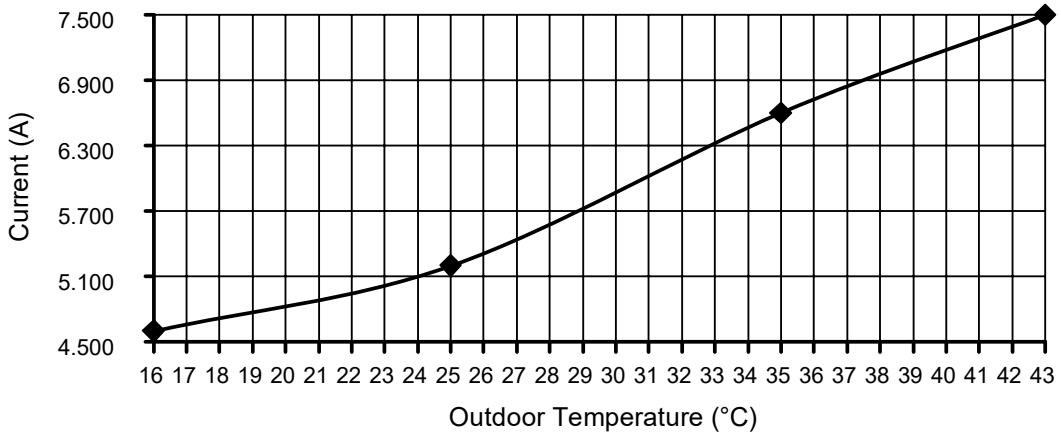
Indoor water inlet temperature : 23°C

Indoor water outlet temperature : 18°C

Piping length : 7 m



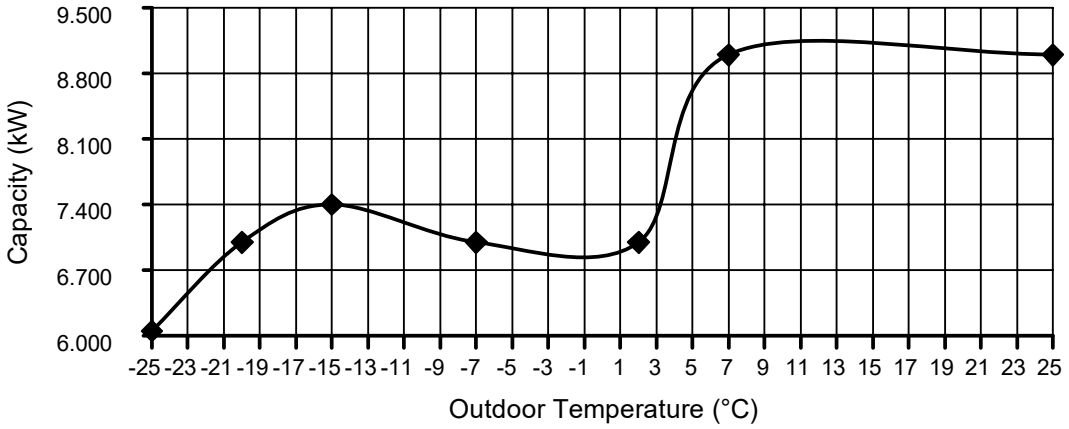


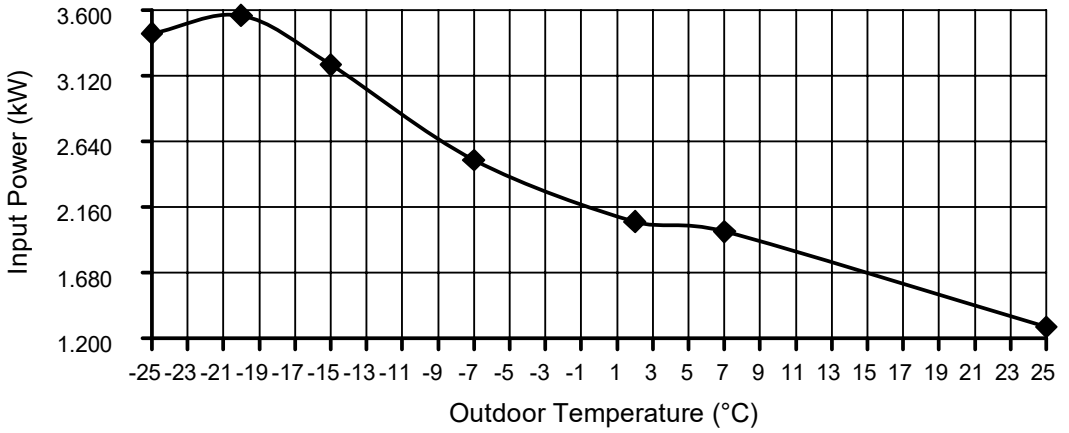


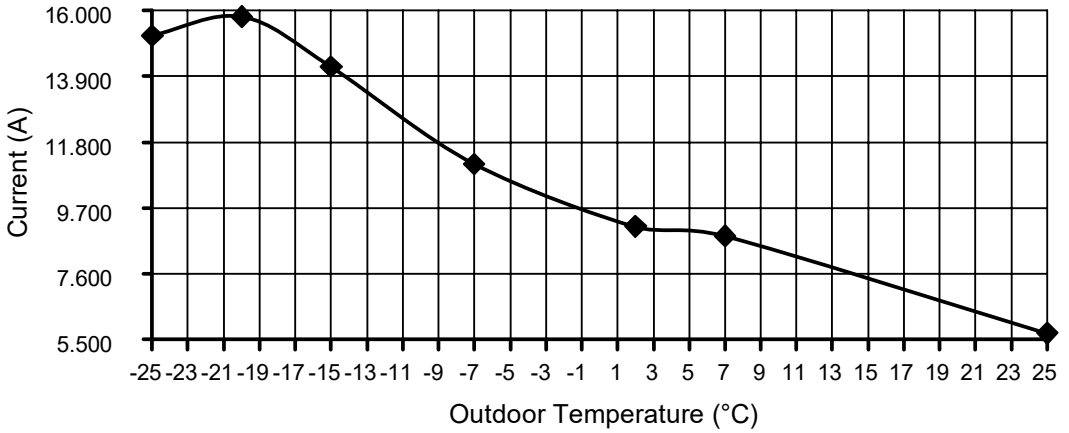
20.1.3 WH-WDG09LE5

Heating Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C
Piping length : 7 m

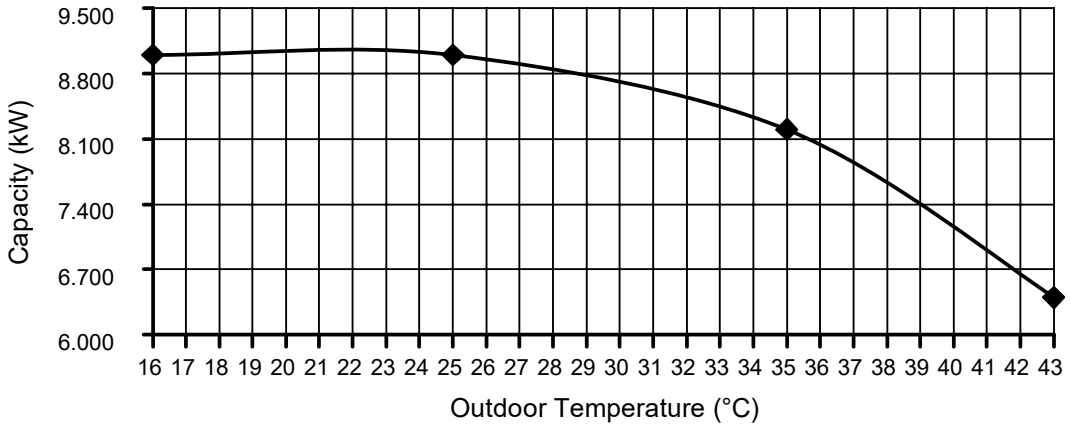


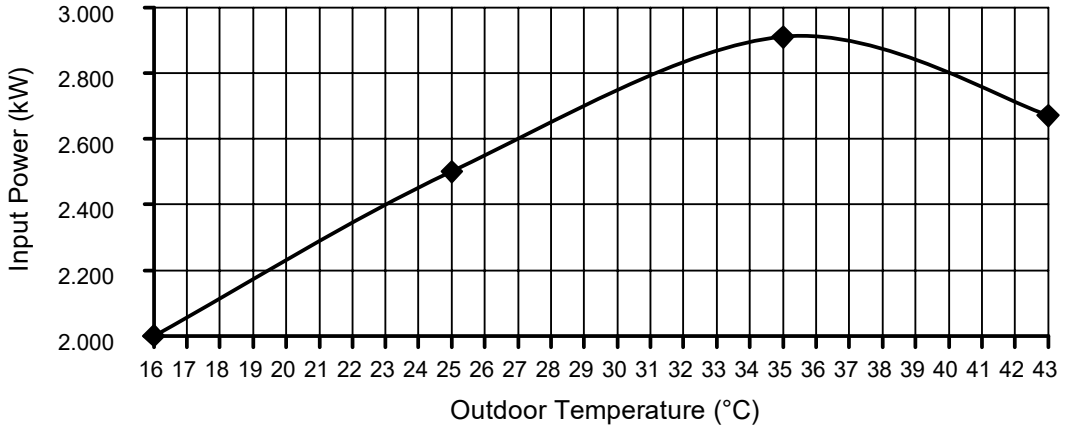


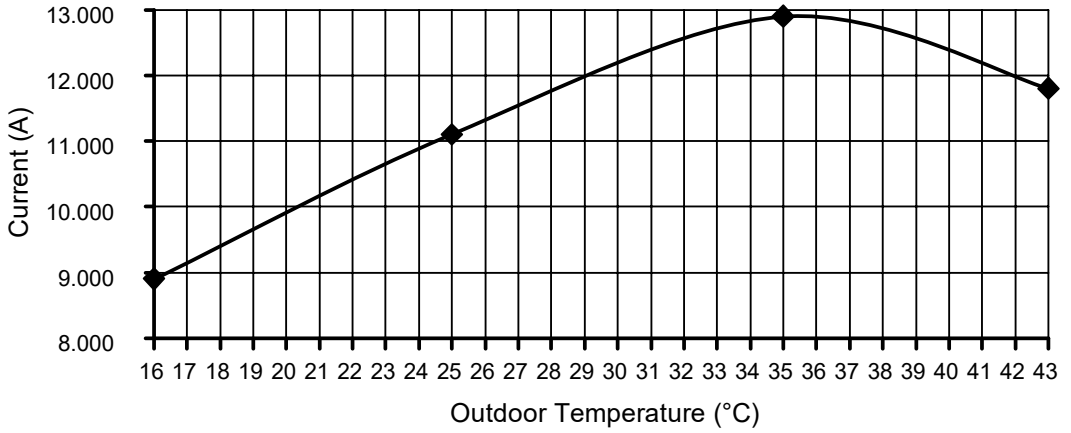


Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 12°C
Indoor water outlet temperature : 7°C
Piping length : 7 m







Cooling Characteristics at Different Outdoor Air Temperature

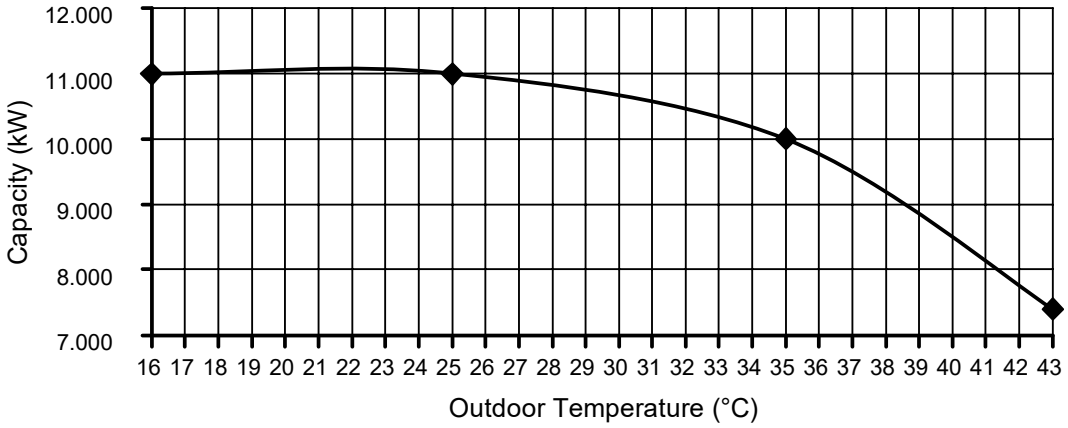
Condition

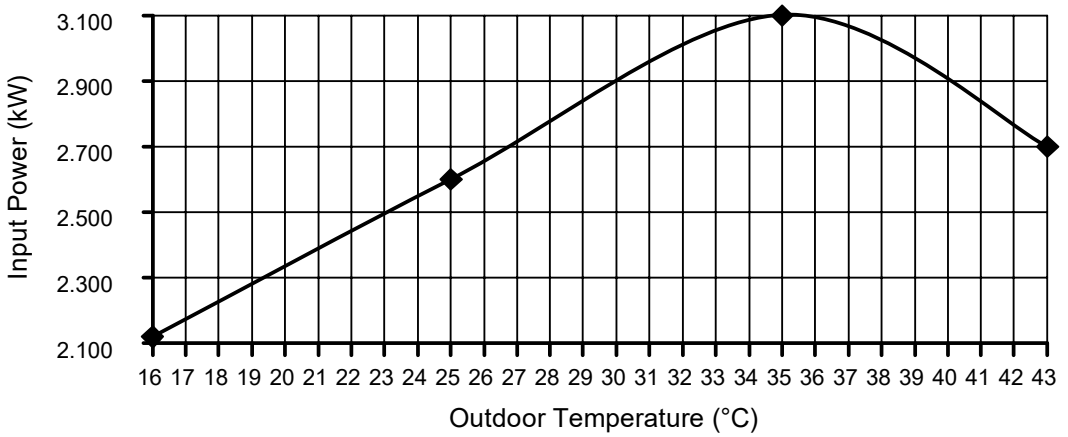
Outdoor air temperature : 35°C (DBT), -°C (WBT)

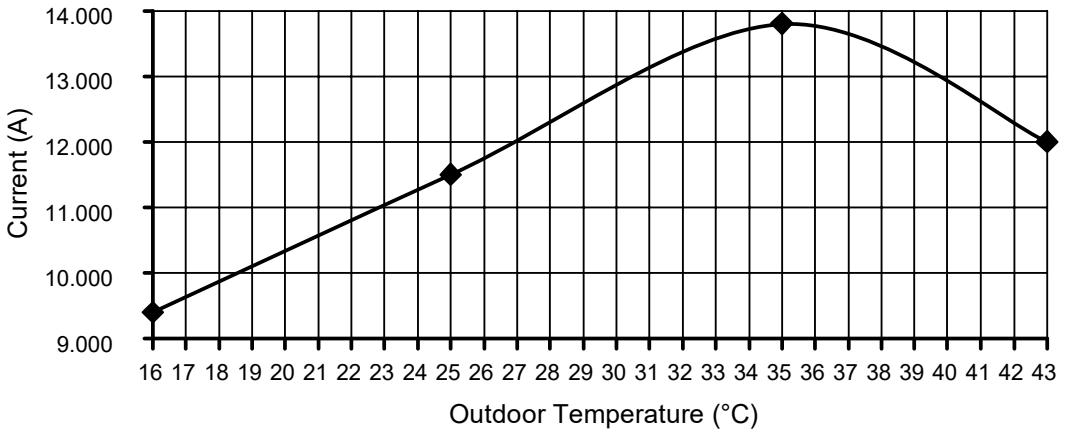
Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C

Piping length : 7 m







Cooling Characteristics at Different Outdoor Air Temperature

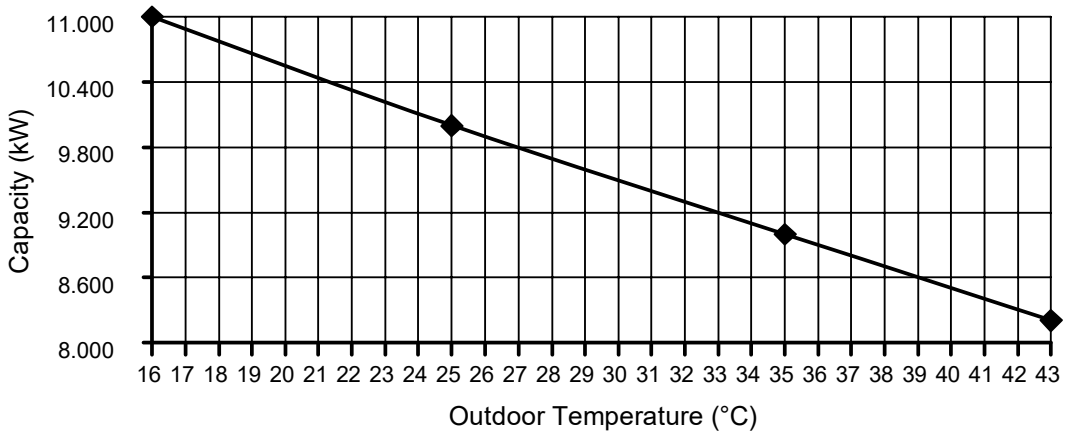
Condition

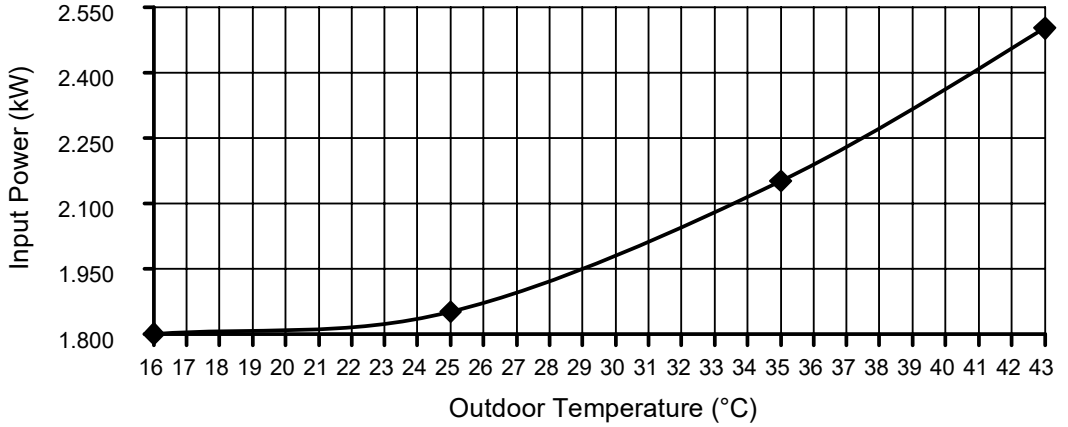
Outdoor air temperature : 35°C (DBT), -°C (WBT)

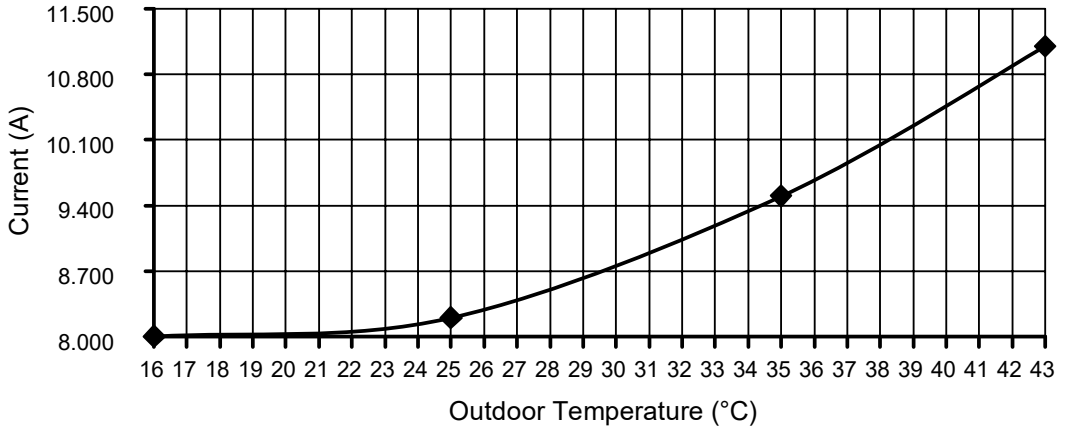
Indoor water inlet temperature : 23°C

Indoor water outlet temperature : 18°C

Piping length : 7 m







20.2 Heating Capacity Table

20.2.1 WH-WDG05LE5

Water Out (°C)	25			35			45		
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)
-25	2600	1580	7.0	2450	1760	7.8	3800	2300	10.2
-20	4900	2010	8.9	4700	2190	9.7	4500	2370	10.5
-15	5000	1750	7.8	5000	1940	8.6	5000	2310	10.2
-7	5000	1370	6.1	5000	1660	7.4	5000	1940	8.6
2	5000	1120	5.0	5000	1420	6.3	5000	1710	7.6
7	5000	760	3.4	5000	990	4.4	5000	1270	5.6
25	5000	370	1.8	5000	600	2.9	5000	860	3.8

Water Out (°C)	55			65			75		
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)
-25	3600	2460	10.9	-	-	-	-	-	-
-20	4250	2570	11.4	-	-	-	-	-	-
-15	5000	2630	11.7	4600	2880	12.8	-	-	-
-7	5000	2360	10.5	5000	2620	11.6	4300	2870	12.7
2	5000	2140	9.5	5000	2540	11.3	4600	2760	12.2
7	5000	1630	7.2	5000	2030	9.0	4700	2570	11.4
25	5000	1100	4.9	5000	1470	6.5	4700	1990	8.8

20.2.2 WH-WDG07LE5

Water Out (°C)	25			35			45		
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)
-25	3150	2020	9.0	4750	2530	11.2	4300	2660	11.8
-20	5500	2390	10.6	5500	2560	11.4	5100	2750	12.2
-15	6000	2200	9.8	6000	2500	11.1	5500	2600	11.5
-7	5600	1640	7.3	5800	1930	8.6	5800	2320	10.3
2	6850	1600	7.1	6850	2000	8.9	6600	2340	10.4
7	7000	1100	4.9	7000	1420	6.3	7000	1900	8.4
25	7000	500	2.4	7000	810	3.7	7000	1230	5.5

Water Out (°C)	55			65			75		
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)
-25	3950	2780	12.3	-	-	-	-	-	-
-20	4900	2970	13.2	-	-	-	-	-	-
-15	5200	2890	12.8	4800	3000	13.3	-	-	-
-7	5800	2740	12.2	5700	3160	14.0	4800	3560	15.8
2	6250	2670	11.8	5600	2800	12.4	5000	3130	13.9
7	7000	2350	10.4	6600	2850	12.6	6300	3400	15.1
25	7000	1650	7.3	7000	2060	9.1	7000	2800	12.4

20.2.3 WH-WDG09LE5

Water Out (°C)	25			35			45		
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)
-25	6400	3120	13.8	6050	3430	15.2	5250	3280	14.6
-20	7200	3200	14.2	7000	3560	15.8	6200	3500	15.5
-15	7600	2920	13.0	7400	3200	14.2	6800	3400	15.1
-7	7000	2150	9.5	7000	2500	11.1	7000	2980	13.2
2	7000	1650	7.3	7000	2050	9.1	7000	2500	11.1
7	9000	1580	7.0	9000	1980	8.8	9000	2580	11.4
25	9000	850	3.8	9000	1280	5.7	9000	1730	7.7

Water Out (°C)	55			65			75		
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)
-25	4650	3150	14.0	-	-	-	-	-	-
-20	5600	3430	15.2	-	-	-	-	-	-
-15	6300	3550	15.7	5600	3550	15.7	-	-	-
-7	7000	3290	14.6	6500	3530	15.7	5400	3560	15.8
2	7000	2900	12.9	6700	3350	14.9	5700	3400	15.1
7	8900	2940	13.0	8900	3560	15.8	7300	3560	15.8
25	9000	2200	9.8	9000	2700	12.0	8000	2900	12.9

20.3 Cooling Capacity Table

20.3.1 WH-WDG05LE5

Water Out (°C)	7			14			18		
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)
16	6000	1010	4.5	7500	1050	4.7	6000	670	3.0
25	5700	1200	5.3	7000	1200	5.3	5700	780	3.5
35	5000	1550	6.9	6300	1440	6.4	5000	1000	4.4
43	4500	1600	7.1	5600	1640	7.3	4500	1120	5.0

20.3.2 WH-WDG07LE5

Water Out (°C)	7			14			18		
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)
16	7000	1360	6.0	8500	1390	6.2	8000	1040	4.6
25	7000	1650	7.3	8000	1570	7.0	7500	1180	5.2
35	7000	2310	10.2	8000	2260	10.0	7000	1480	6.6
43	6000	2500	11.1	7000	2600	11.5	5700	1700	7.5

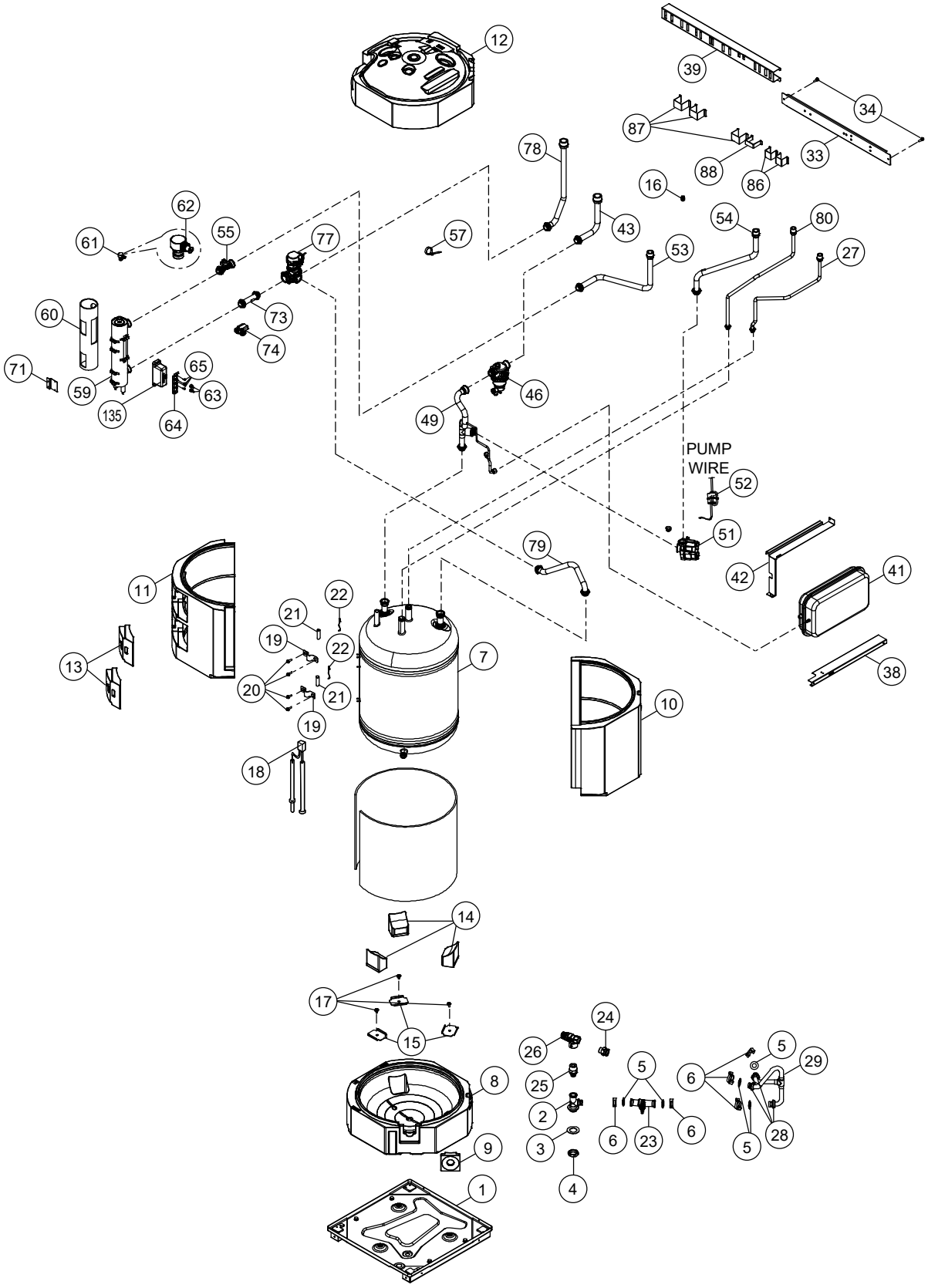
20.3.3 WH-WDG09LE5

Water Out (°C)	7			14			18		
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)	Capacity (W)	Input Power (W)	Current (A)
16	9000	2000	8.9	11000	2120	9.4	11000	1800	8.0
25	9000	2500	11.1	11000	2600	11.5	10000	1850	8.2
35	8200	2910	12.9	10000	3100	13.8	9000	2150	9.5
43	6400	2670	11.8	7400	2700	12.0	8200	2500	11.1

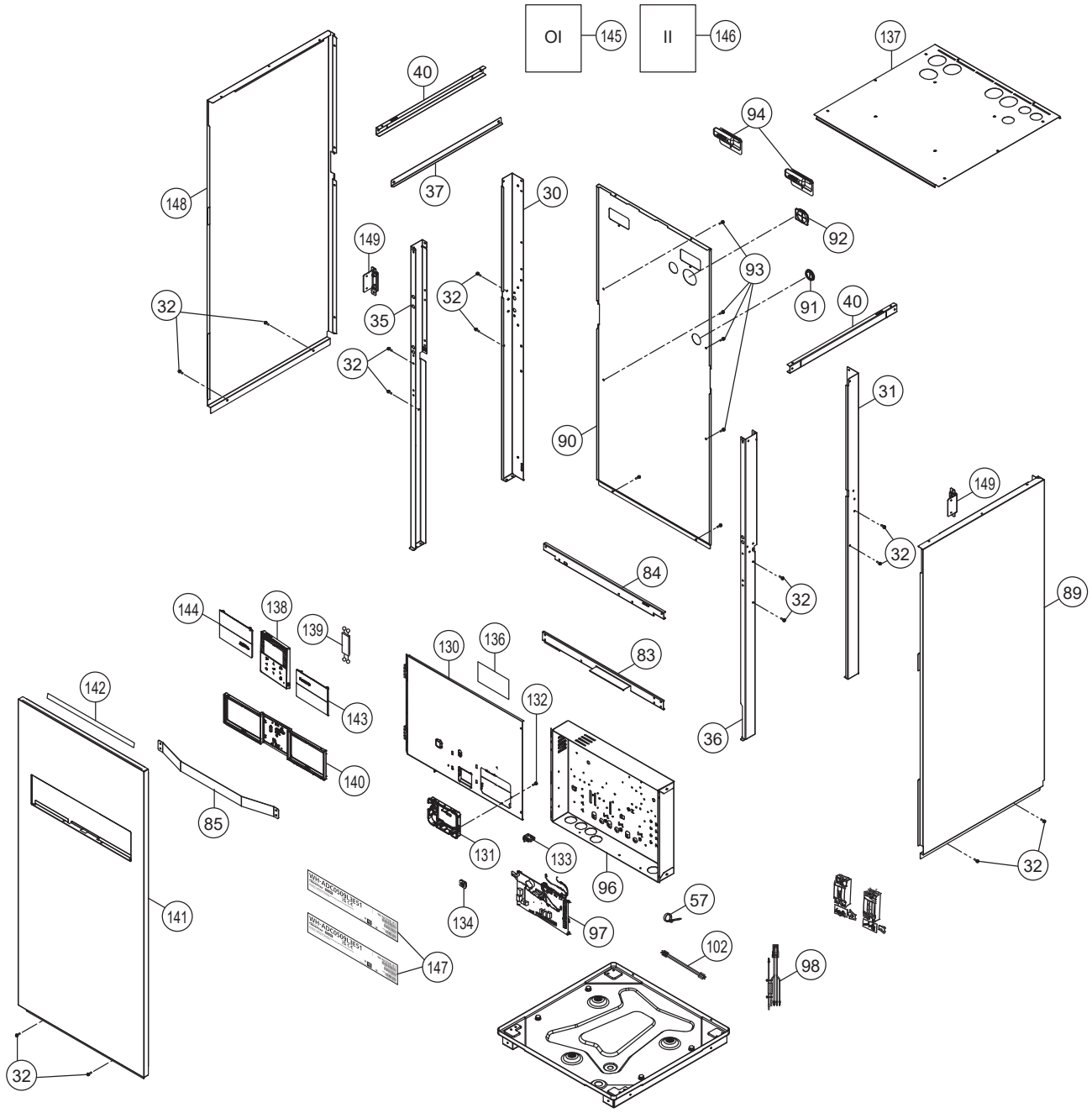
21. Exploded View and Replacement Parts List

21.1 Indoor Unit

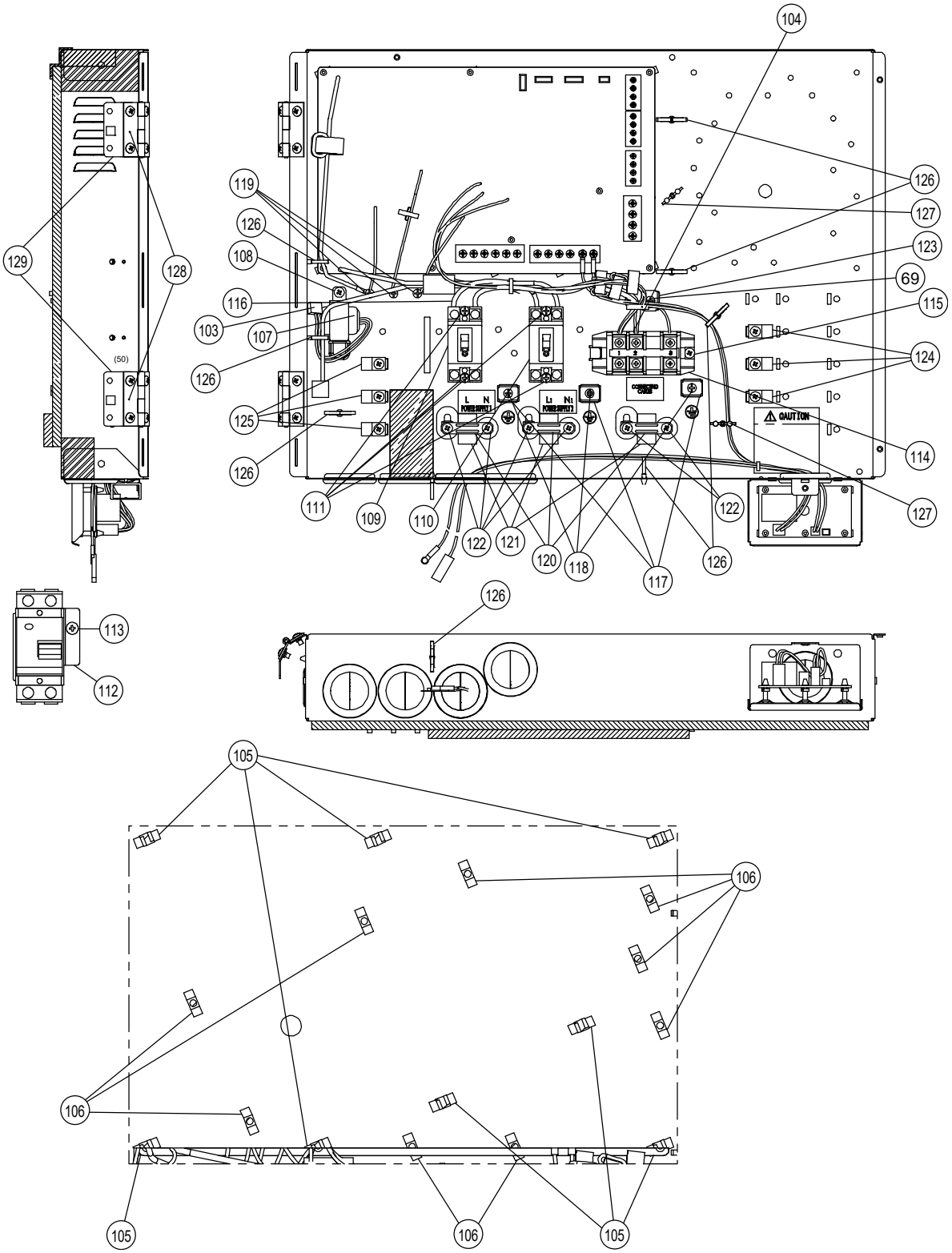
21.1.1 WH-ADC0509L3E51



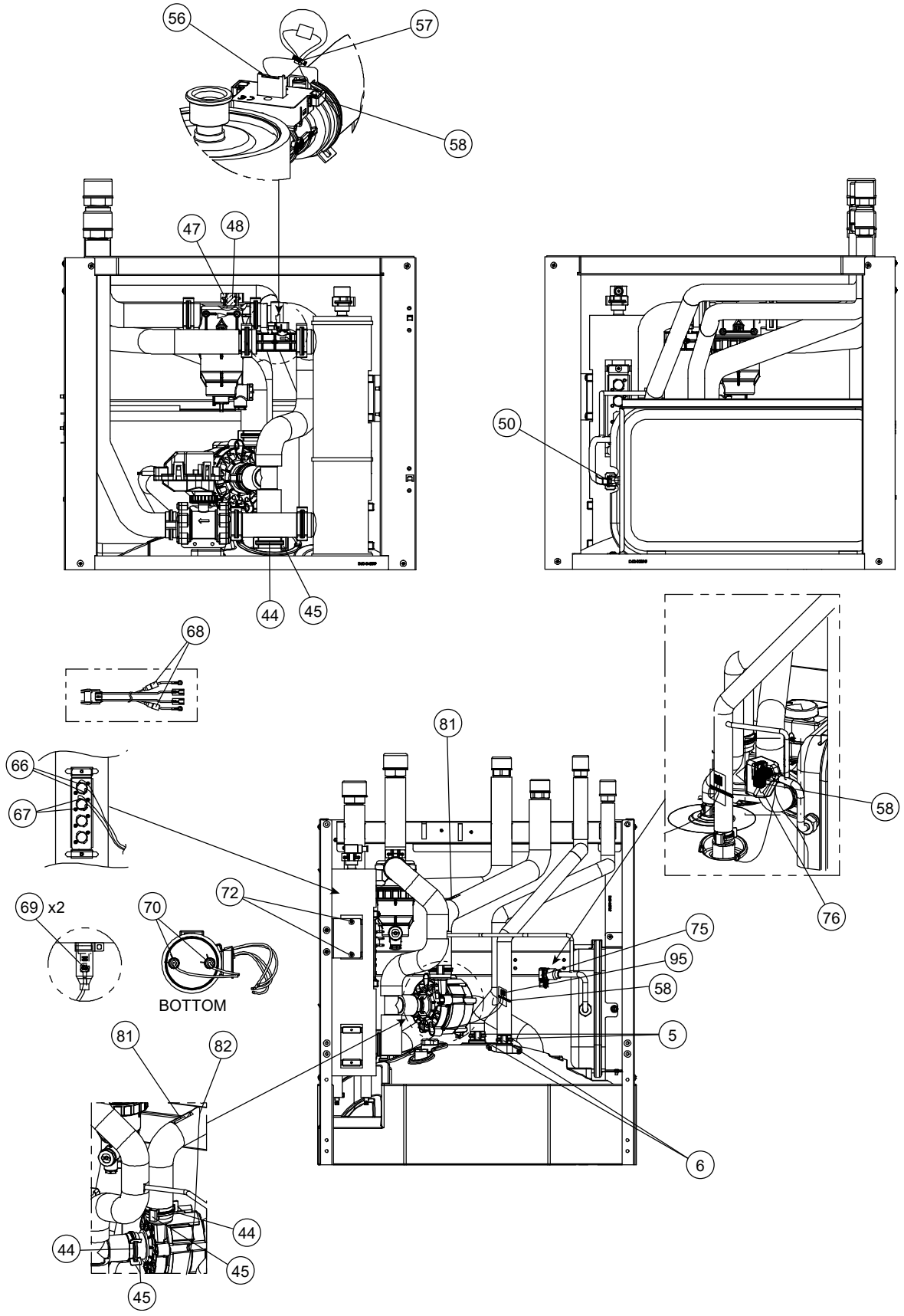
Note:
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The non-numbered parts are not kept as standard service parts.



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Note:
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SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0509L3E51	REMARK
	1	CHASSIS - COMPLETE	1	ACXD50C04840	
	2	TUBE CONNECTER	1	ACXT29-02570	
	3	PACKING	1	ACXB81-07070	
	4	NUT	1	ACXH56-01370	
⚠	5	PACKING	7	ACXB81-00010	
	6	RETAINING RING	7	H581007	
⚠	7	TANK COMPLETE	1	ACXB56C01530	
	8	FOAMED POLYSTYRENE	1	ACXG07-07690	
	9	FOAMED POLYSTYRENE	1	ACXG07-07700	
	10	FOAMED POLYSTYRENE	1	ACXG07-09260	
	11	FOAMED POLYSTYRENE	1	ACXG07-09270	
	12	FOAMED POLYSTYRENE	1	ACXG07-07731	
	13	FOAMED POLYSTYRENE	2	ACXG07-07740	
	14	FOAMED POLYSTYRENE	3	ACXG07-07750	
	15	U-SHAPED PLATE	3	ACXD62-02690	
	16	CAP	1	H521306	
	17	SCREW	3	H551029J	
⚠	18	SENSOR - COMPLETE	1	ACXA50C18700	O
	19	PARTICULAR PIECE	2	D934023	
	20	NUT	4	H561116	
	21	STRAIGHT TUBE	2	T102044	
	22	PLATE SPRING	2	H711010	
⚠	23	COCK - COMPLETE	1	ACXB65C00111	O
	24	SOCKET	1	ACXT27-00090	
	25	SOCKET	1	ACXT27-00020	
⚠	26	VALVE BODY [RBM SAFETY VALVE]	1	ACXB62-00920	O
	27	TUBE ASSY - COMPLETE	1	ACXT00C40900	
	28	SOCKET	3	T271092	
	29	TUBE ASSY	1	ACXT00-77551	
	30	L-SHAPED PLATE [LEFT REAR PILLAR]	1	ACXD60-05190	
	31	L-SHAPED PLATE [RIGHT REAR PILLAR]	1	ACXD60-05200	
	32	SCREW	14	H551040J	
	33	CONNECTING BAR	1	ACXE26-02841	
	34	SCREW	2	H551217	
	35	U-SHAPED PLATE [LEFT FRONT PILLAR]	1	ACXD62-03070	
	36	U-SHAPED PLATE [RIGHT FRONT PILLAR]	1	ACXD62-03080	
	37	L-SHAPED PLATE	1	ACXD60-04532	
	38	U-SHAPED PLATE [RIGHT MIDDLE BEAM]	1	ACXD62-02390CZ	
	39	CONNECTING BAR	1	ACXE26-02851	
	40	U-SHAPED PLATE	2	ACXD62-02720	
	41	RECEIVER	1	ACXB14-00720	O

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0509L3E51	REMARK
	42	PARTICULAR PLATE	1	ACXD90-26060CZ	
	43	TUBE ASSY - COMPLETE	1	ACXT00C45260	
⚠	44	PACKING	3	ACXB81-06910	
	45	RETAINING RING	3	ACXH58-00370	
	46	FILTER COMPLETE	1	ACXB51C00170	O
	47	RETAINING RING	1	H581038	
⚠	48	PACKING	1	ACXB81-06810	
	49	TUBE ASSY - COMPLETE	1	ACXT00C45481	
	50	PACKING	1	B811179	
⚠	51	PUMP	1	ACXB53-00850	O
	52	V-COIL [FERRITE CORE]	1	A431122	O
	53	TUBE ASSY - COMPLETE	1	ACXT00C45380	
	54	TUBE ASSY - COMPLETE	1	ACXT00C45390	
⚠	55	VALVE BODY [FLOW SENSOR]	1	ACXB62-00932	O
	56	LEAD WIRE - COMPLETE (FLWSEN)	1	ACXA60C82910	O
	57	BAND	3	H88000	
	58	BAND	3	4605004	
⚠	59	HEATER ASSY	1	ACXA34K00480	O
	60	SOUND PROOF MATERIAL	1	ACXG30-10530CZ	
	61	VALVE BODY - COMPLETE [AIR PURGE]	1	ACXB62C01130	O
⚠	62	PACKING	1	ACXB81-00020	
⚠	63	THERMOSTAT	2	ACXA15-00260	O
	64	U-SHAPED PIECE	1	D721031	
	65	MACHINE SCREW & WASHER ASSY	4	XYN3+C5FJ	
	66	LEAD WIRE - COMPLETE (HEAT-OLP1)	1	ACXA60C93670	O
⚠	67	LEAD WIRE - COMPLETE (HEAT-CON)	1	ACXA60C93680	O
	68	CAP	2	H521215	
	69	MACHINE SCREW & WASHER ASSY	3	XYN4DC8FJ	
	70	BAND	2	H881137	
	71	HEATER ATTACHMENT PLATE	1	ACXD56-00240	
	72	SCREW	2	H55293J	
	73	TUBE ASSY - COMPLETE	1	ACXT00C37270	
⚠	74	VALVE BODY [PRESSURE TRANSDUCER]	1	ACXB62-01110	O
	75	WIRE SPRING	1	ACXH72-00341	
	76	LEAD WIRE - COMPLETE (SENS-CN)	1	ACXA60C93661	O
⚠	77	VALVE BODY [3-WAY VALVE]	1	ACXB62-00092	O
	78	TUBE ASSY - COMPLETE	1	ACXT00C45370	
	79	TUBE ASSY - COMPLETE	1	ACXT00C45320	
	80	TUBE ASSY - COMPLETE	1	ACXT00C41030	
	81	PLATE SPRING	2	H711019	
⚠	82	SENSOR - COMPLETE	1	ACXA50C19160	O

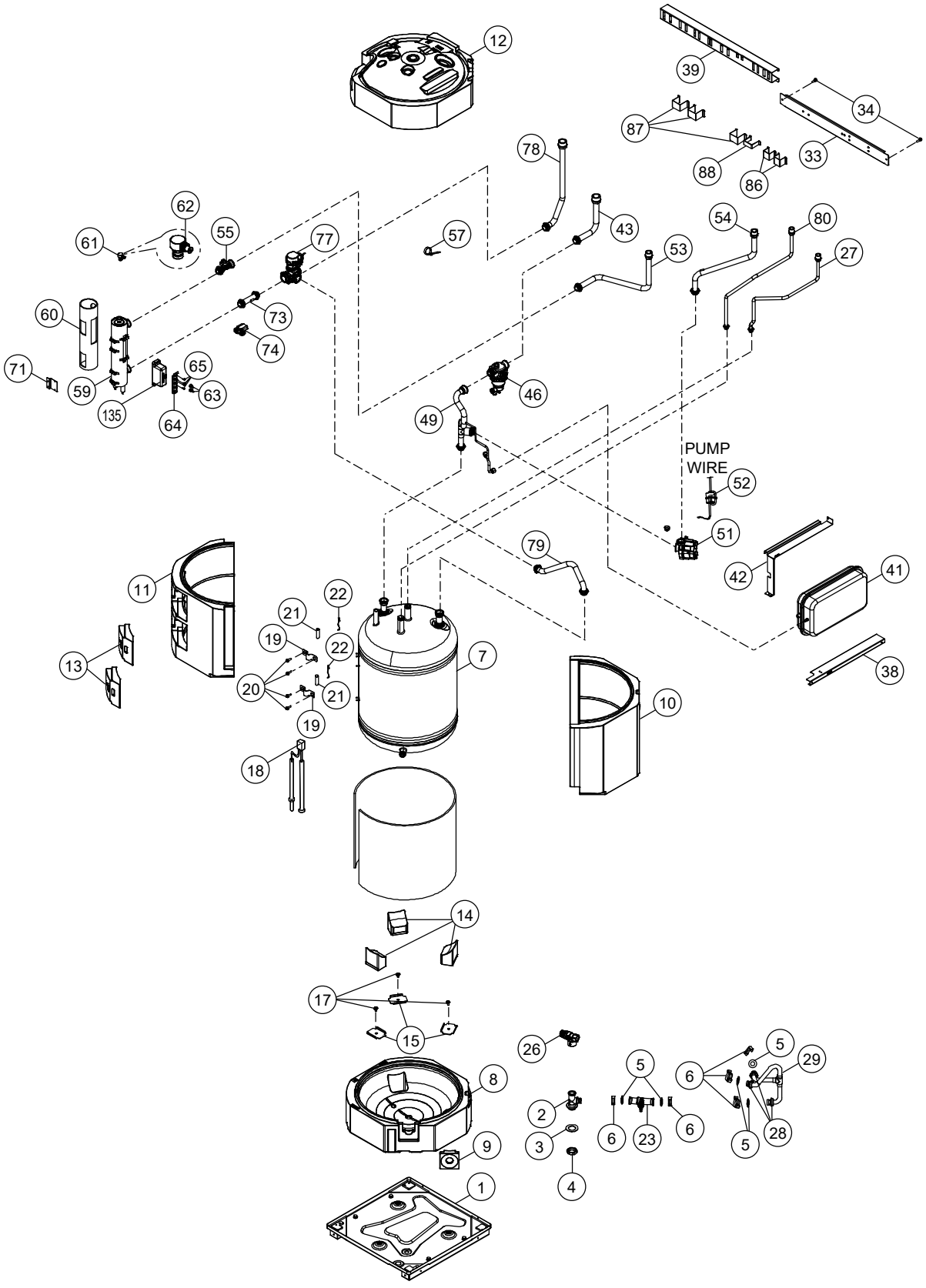
SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0509L3E51	REMARK
	83	CONNECTING BAR	1	ACXE26-02860	
	84	CONNECTING BAR	1	ACXE26-02870	
	85	CONNECTING BAR	1	ACXE26-02880	
	86	PARTICULAR PIECE	2	ACXD93-25020	
	87	PARTICULAR PIECE	3	ACXD93-25030	
	88	PARTICULAR PIECE	1	ACXD93-25040	
	89	CABINET SIDE PLATE [RIGHT]	1	ACXE04-14380A	
	90	CABINET REAR PLATE	1	ACXE02-03790	
	91	PACKING	1	ACXB81-00030	
	92	PACKING	1	ACXB81-00040	
	93	SCREW	4	ACXH55-07220	
	94	HANDLE	2	ACXE16-00300	
	95	LEAD WIRE - COMPLETE (CON-PUMP)	1	ACXA60C73630	O
	96	CONTROL BOARD	1	ACXH10-09860	O
⚠	97	ELECTRONIC CONTROLLER	1	ACXA74C20720	O
⚠	98	LEAD WIRE - COMPLETE (AC2-L3)	1	ACXA60C94310	O
	102	LEAD WIRE - COMPLETE (CN8-CN9)	1	ACXA60C96530	O
⚠	103	LEAD WIRE - COMPLETE (L1-L2)	1	A68C1161	O
⚠	104	LEAD WIRE - COMPLETE (AC1-N)	1	ACXA61C09090	O
	105	SPACER	8	H54294	
	106	SPACER	9	H54295	
⚠	107	REACTOR	1	G0C103Z00003	O
	108	SELF TAPPING SCREW	1	XTB4+8FFJ	
⚠	109	CIRCUIT BREAKER	1	K5KYYAY00003	O
⚠	110	CIRCUIT BREAKER	1	ACXA18-00011	O
	111	SELF TAPPING SCREW	4	XTN4+25CFJ	
	112	U-SHAPED PIECE	1	D721014	
	113	SELF TAPPING SCREW	1	XTT4+8CFJ	
⚠	114	TERMINAL BOARD ASSY	1	A28K1217	O
	115	SELF TAPPING SCREW	1	XTT4+12CFJ	
⚠	116	LEAD WIRE - COMPLETE (WIRE1)	1	ACXA60C96540	O
	117	WASHER	3	H571044	
	118	MACHINE SCREW & WASHER ASSY	3	XYN5DC10FJ	
	119	SELF TAPPING SCREW	3	XTT4+8FFJ	
	120	HOLDER - P.S. CORD	3	H31103	
	121	SHEET	3	H86450	
	122	SELF TAPPING SCREW	6	XTT4+25DFJ	
	123	WASHER	1	H57094	
	124	HOLDER - P.S. CORD	3	H31042	
	125	A-PIECE	3	D77001	
	126	BAND	8	4605011	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0509L3E51	REMARK
	127	BAND	2	4605009	
	128	HINGE	2	H611006	
	129	L-PIECE	2	ACXD70-02300	
	130	CONTROL BOARD COVER	1	ACXH13-09290	
	131	BOX SHAPED PLATE	1	ACXD66-03961	
	132	SELF TAPPING SCREW	1	XTB4+8CFJ	
	133	HOLDER - P.S. CORD	1	ACXH31-00210	
	134	PACKING	1	B811177	
	135	TERMINAL COVER	1	H171051	
⚠	136	WIRING DIAGRAM	1	ACXF29-00880	
	137	CABINET TOP PLATE	1	ACXE03-04930	
⚠	138	REMOTE CONTROL SWITCH - COMPLETE	1	ACXA75C23211	O
	139	LEAD WIRE - COMPLETE (REMOTE)	1	ACXA60C93691	O
	140	BOX SHAPED PLATE	1	ACXD66-03970	
	141	CABINET FRONT PLATE	1	ACXE06-05780A	
	142	PACKING	1	ACXB81-07400	
	143	DECORATION BASE ASSY	1	ACXE35K03630	
	144	DECORATION BASE ASSY	1	ACXE35K03640	
⚠	145	OPERATING INSTRUCTION - COMPLETE	1	ACXF55C32270	
⚠	146	INSTALLATION INSTRUCTION COMPLETE	1	ACXF60C21470	
⚠	147	MODEL LABEL	2	ACXF87-52080	
	148	CABINET SIDE PLATE [LEFT]	1	ACXE04-14370A	
	149	BRACKET - UNIT MOUNTING COMPLETE	2	ACXD57C00450	

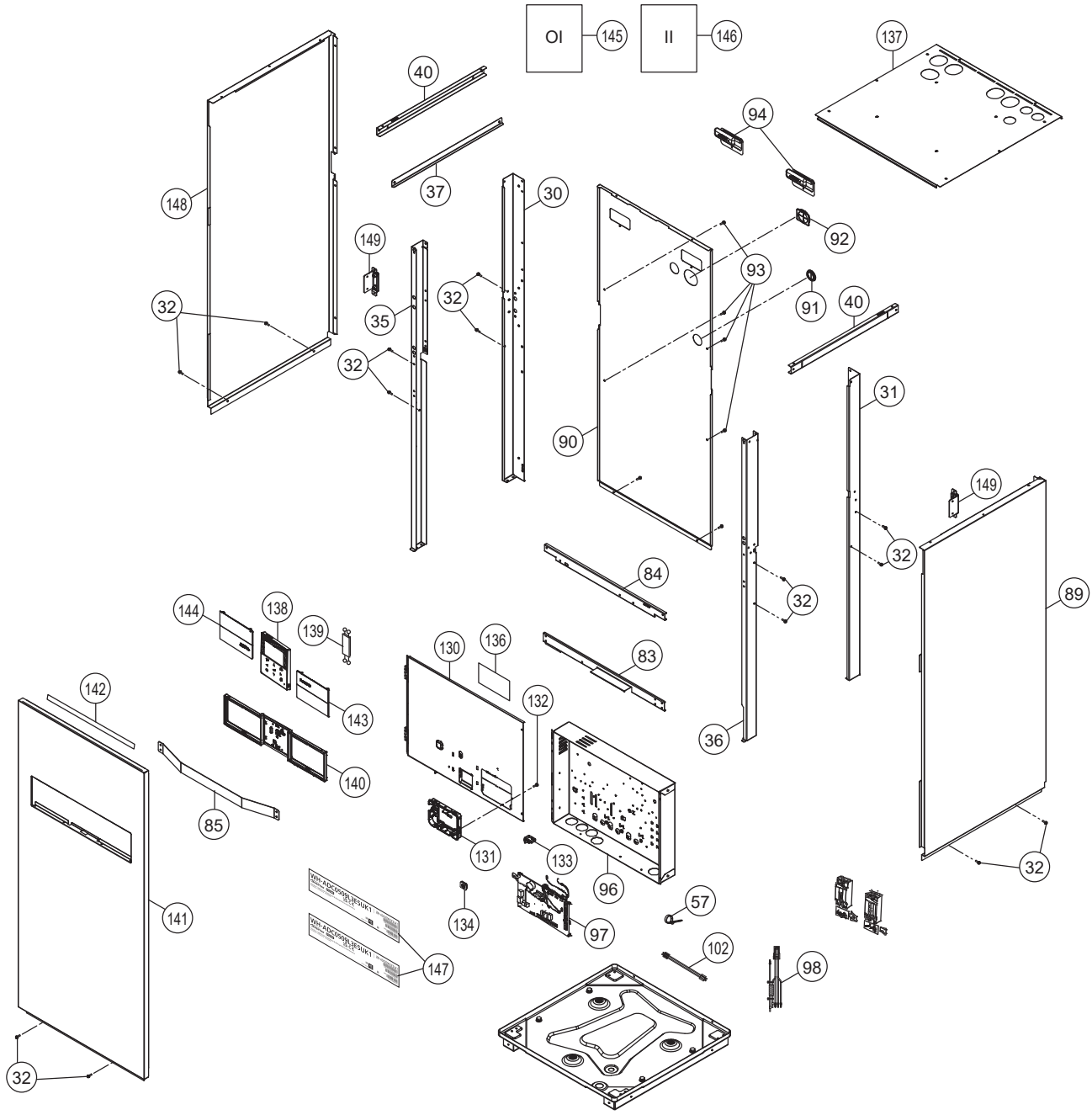
Note:

- All parts are supplied from PHVACCZ, Czech (Vendor Code: 00029407).
- “O” marked parts are recommended to be kept in stock.

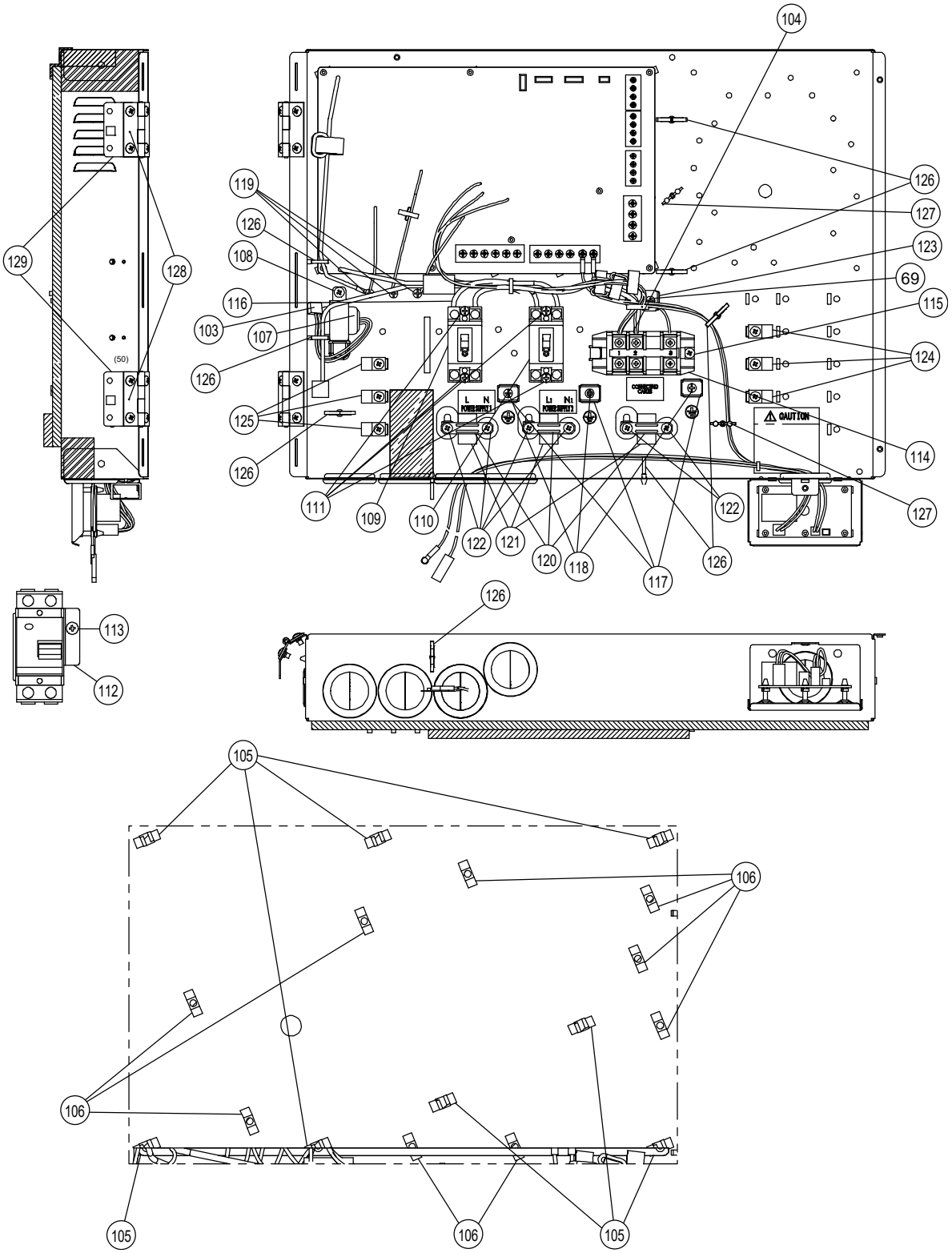
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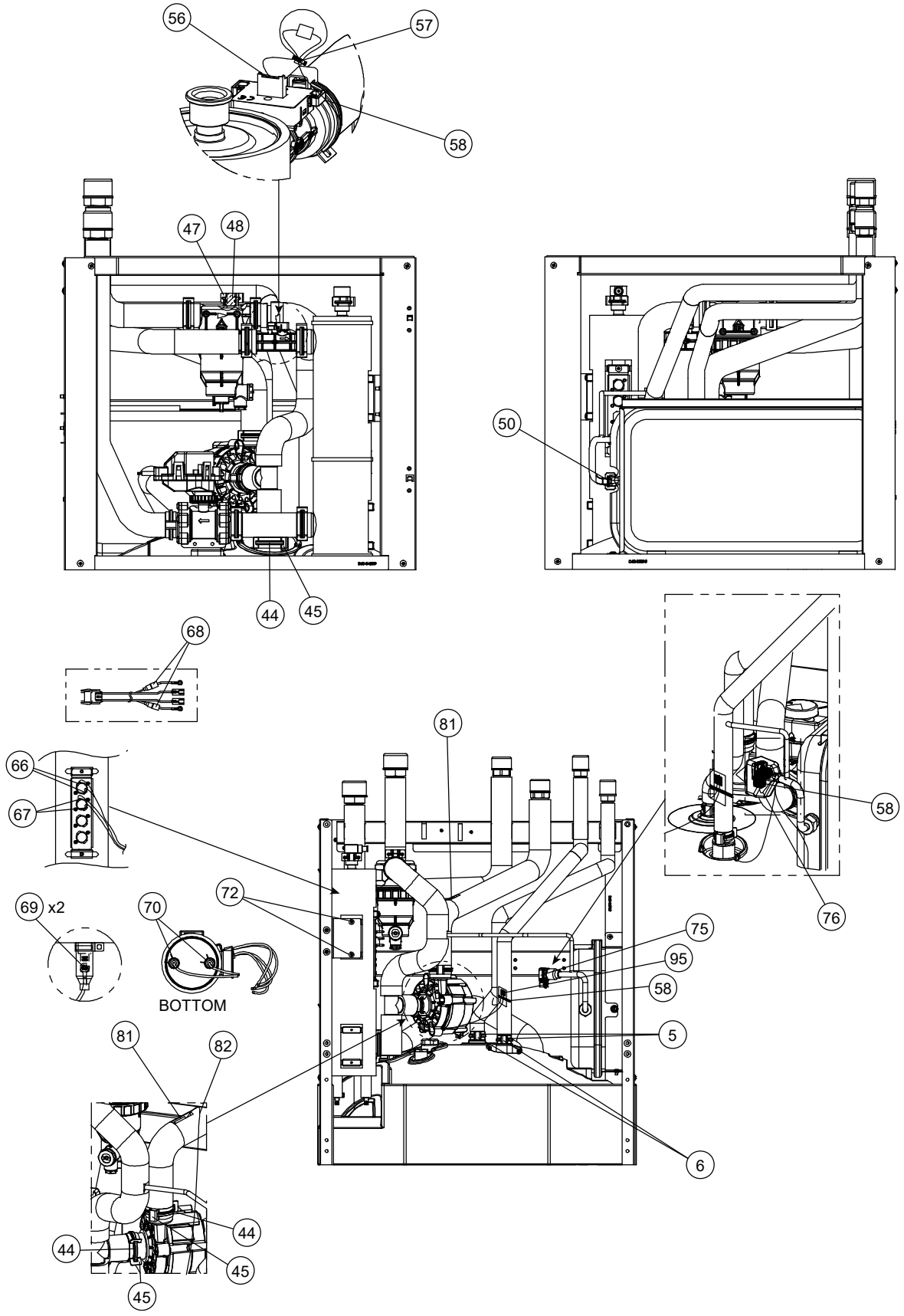
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SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0509L3E5UK1	REMARK
	1	CHASSIS - COMPLETE	1	ACXD50C04840	
	2	TUBE CONNECTER	1	ACXT29-02600	
	3	PACKING	1	ACXB81-07070	
	4	NUT	1	ACXH56-01370	
⚠	5	PACKING	7	ACXB81-00010	
	6	RETAINING RING	7	H581007	
⚠	7	TANK COMPLETE	1	ACXB56C01530	
	8	FOAMED POLYSTYRENE	1	ACXG07-07690	
	9	FOAMED POLYSTYRENE	1	ACXG07-07700	
	10	FOAMED POLYSTYRENE	1	ACXG07-09260	
	11	FOAMED POLYSTYRENE	1	ACXG07-09270	
	12	FOAMED POLYSTYRENE	1	ACXG07-07731	
	13	FOAMED POLYSTYRENE	2	ACXG07-07740	
	14	FOAMED POLYSTYRENE	3	ACXG07-07750	
	15	U-SHAPED PLATE	3	ACXD62-02690	
	17	SCREW	3	H551029J	
⚠	18	SENSOR - COMPLETE	1	ACXA50C18700	O
	19	PARTICULAR PIECE	2	D934023	
	20	NUT	4	H561116	
	21	STRAIGHT TUBE	2	T102044	
	22	PLATE SPRING	2	H711010	
⚠	23	COCK - COMPLETE	1	ACXB65C00111	O
⚠	26	VALVE BODY [RBM SAFETY VALVE]	1	B621135	O
	27	TUBE ASSY - COMPLETE	1	ACXT00C40900	
	28	SOCKET	3	T271092	
	29	TUBE ASSY	1	ACXT00-77930	
	30	L-SHAPED PLATE [LEFT REAR PILLAR]	1	ACXD60-05190	
	31	L-SHAPED PLATE [RIGHT REAR PILLAR]	1	ACXD60-05200	
	32	SCREW	14	H551040J	
	33	CONNECTING BAR	1	ACXE26-02841	
	34	SCREW	2	H551217	
	35	U-SHAPED PLATE [LEFT FRONT PILLAR]	1	ACXD62-03070	
	36	U-SHAPED PLATE [RIGHT FRONT PILLAR]	1	ACXD62-03080	
	37	L-SHAPED PLATE	1	ACXD60-04532	
	38	U-SHAPED PLATE [RIGHT MIDDLE BEAM]	1	ACXD62-02390CZ	
	39	CONNECTING BAR	1	ACXE26-02851	
	40	U-SHAPED PLATE	2	ACXD62-02720	
	41	RECEIVER	1	ACXB14-00720	O
	42	PARTICULAR PLATE	1	ACXD90-26060CZ	
	43	TUBE ASSY - COMPLETE	1	ACXT00C45260	
⚠	44	PACKING	3	ACXB81-06910	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0509L3E5UK1	REMARK
	45	RETAINING RING	3	ACXH58-00370	
	46	FILTER COMPLETE	1	ACXB51C00170	O
	47	RETAINING RING	1	H581038	
⚠	48	PACKING	1	ACXB81-06810	
	49	TUBE ASSY - COMPLETE	1	ACXT00C45481	
	50	PACKING	1	B811179	
⚠	51	PUMP	1	ACXB53-00850	O
	52	V-COIL [FERRITE CORE]	1	A431122	O
	53	TUBE ASSY - COMPLETE	1	ACXT00C45380	
	54	TUBE ASSY - COMPLETE	1	ACXT00C45390	
⚠	55	VALVE BODY [FLOW SENSOR]	1	ACXB62-00932	O
	56	LEAD WIRE - COMPLETE (FLWSEN)	1	ACXA60C82910	O
	57	BAND	3	H88000	
	58	BAND	3	4605004	
⚠	59	HEATER ASSY	1	ACXA34K00480	O
	60	SOUND PROOF MATERIAL	1	ACXG30-10530CZ	
	61	VALVE BODY - COMPLETE [AIR PURGE]	1	ACXB62C01130	O
⚠	62	PACKING	1	ACXB81-00020	
⚠	63	THERMOSTAT	2	ACXA15-00260	O
	64	U-SHAPED PIECE	1	D721031	
	65	MACHINE SCREW & WASHER ASSY	4	XYN3+C5FJ	
	66	LEAD WIRE - COMPLETE (HEAT-OLP1)	1	ACXA60C93670	O
⚠	67	LEAD WIRE - COMPLETE (HEAT-CON)	1	ACXA60C93680	O
	68	CAP	2	H521215	
	69	MACHINE SCREW & WASHER ASSY	3	XYN4DC8FJ	
	70	BAND	2	H881137	
	71	HEATER ATTACHMENT PLATE	1	ACXD56-00240	
	72	SCREW	2	H55293J	
	73	TUBE ASSY - COMPLETE	1	ACXT00C37270	
⚠	74	VALVE BODY [PRESSURE TRANSDUCER]	1	ACXB62-01110	O
	75	WIRE SPRING	1	ACXH72-00341	
	76	LEAD WIRE - COMPLETE (SENS-CN)	1	ACXA60C93661	O
⚠	77	VALVE BODY [3-WAY VALVE]	1	ACXB62-00092	O
	78	TUBE ASSY - COMPLETE	1	ACXT00C45370	
	79	TUBE ASSY - COMPLETE	1	ACXT00C45320	
	80	TUBE ASSY - COMPLETE	1	ACXT00C41030	
	81	PLATE SPRING	2	H711019	
⚠	82	SENSOR - COMPLETE	1	ACXA50C19160	O
	83	CONNECTING BAR	1	ACXE26-02860	
	84	CONNECTING BAR	1	ACXE26-02870	
	85	CONNECTING BAR	1	ACXE26-02880	

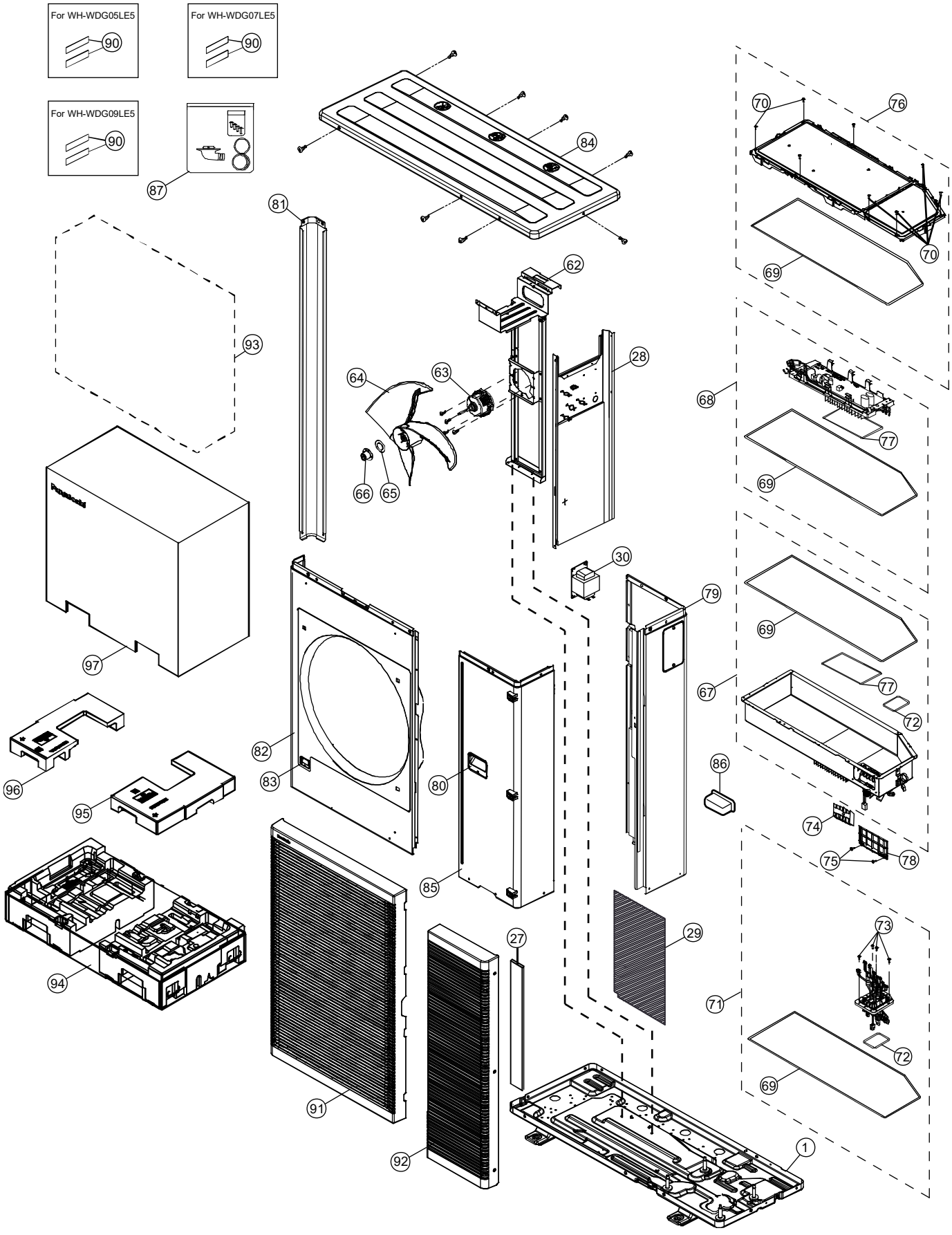
SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0509L3E5UK1	REMARK
	86	PARTICULAR PIECE	2	ACXD93-25020	
	87	PARTICULAR PIECE	3	ACXD93-25030	
	88	PARTICULAR PIECE	1	ACXD93-25040	
	89	CABINET SIDE PLATE [RIGHT]	1	ACXE04-14380A	
	90	CABINET REAR PLATE	1	ACXE02-03980	
	91	PACKING	1	ACXB81-00030	
	92	PACKING	1	ACXB81-00040	
	93	SCREW	4	ACXH55-07220	
	94	HANDLE	2	ACXE16-00300	
	95	LEAD WIRE - COMPLETE (CON-PUMP)	1	ACXA60C73630	O
	96	CONTROL BOARD	1	ACXH10-09860	O
⚠	97	ELECTRONIC CONTROLLER	1	ACXA74C20730	O
⚠	98	LEAD WIRE - COMPLETE (AC2-L3)	1	ACXA60C94310	O
	102	LEAD WIRE - COMPLETE (CN8-CN9)	1	ACXA60C96530	O
⚠	103	LEAD WIRE - COMPLETE (L1-L2)	1	A68C1161	O
⚠	104	LEAD WIRE - COMPLETE (AC1-N)	1	ACXA61C09090	O
	105	SPACER	8	H54294	
	106	SPACER	9	H54295	
⚠	107	REACTOR	1	G0C103Z00003	O
	108	SELF TAPPING SCREW	1	XTB4+8FFJ	
⚠	109	CIRCUIT BREAKER	1	K5KYYAY00003	O
⚠	110	CIRCUIT BREAKER	1	ACXA18-00011	O
	111	SELF TAPPING SCREW	4	XTN4+25CFJ	
	112	U-SHAPED PIECE	1	D721014	
	113	SELF TAPPING SCREW	1	XTT4+8CFJ	
⚠	114	TERMINAL BOARD ASSY	1	A28K1217	O
	115	SELF TAPPING SCREW	1	XTT4+12CFJ	
⚠	116	LEAD WIRE - COMPLETE (WIRE1)	1	ACXA60C96540	O
	117	WASHER	3	H571044	
	118	MACHINE SCREW & WASHER ASSY	3	XYN5DC10FJ	
	119	SELF TAPPING SCREW	3	XTT4+8FFJ	
	120	HOLDER - P.S. CORD	3	H31103	
	121	SHEET	3	H86450	
	122	SELF TAPPING SCREW	6	XTT4+25DFJ	
	123	WASHER	1	H57094	
	124	HOLDER - P.S. CORD	3	H31042	
	125	A-PIECE	3	D77001	
	126	BAND	8	4605011	
	127	BAND	2	4605009	
	128	HINGE	2	H611006	
	129	L-PIECE	2	ACXD70-02300	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0509L3E5UK1	REMARK
	130	CONTROL BOARD COVER	1	ACXH13-09290	
	131	BOX SHAPED PLATE	1	ACXD66-03961	
	132	SELF TAPPING SCREW	1	XTB4+8CFJ	
	133	HOLDER - P.S. CORD	1	ACXH31-00210	
	134	PACKING	1	B811177	
	135	TERMINAL COVER	1	H171051	
⚠	136	WIRING DIAGRAM	1	ACXF29-00880	
	137	CABINET TOP PLATE	1	ACXE03-04930	
⚠	138	REMOTE CONTROL SWITCH - COMPLETE	1	ACXA75C23211	O
	139	LEAD WIRE - COMPLETE (REMOTE)	1	ACXA60C93691	O
	140	BOX SHAPED PLATE	1	ACXD66-03970	
	141	CABINET FRONT PLATE	1	ACXE06-05780A	
	142	PACKING	1	ACXB81-07400	
	143	DECORATION BASE ASSY	1	ACXE35K03630	
	144	DECORATION BASE ASSY	1	ACXE35K03640	
⚠	145	OPERATING INSTRUCTION - COMPLETE	1	ACXF55C32280	
⚠	146	INSTALLATION INSTRUCTION COMPLETE	1	ACXF60C21480	
⚠	147	MODEL LABEL	2	ACXF87-52100	
	148	CABINET SIDE PLATE [LEFT]	1	ACXE04-14370A	
	149	BRACKET - UNIT MOUNTING COMPLETE	2	ACXD57C00450	

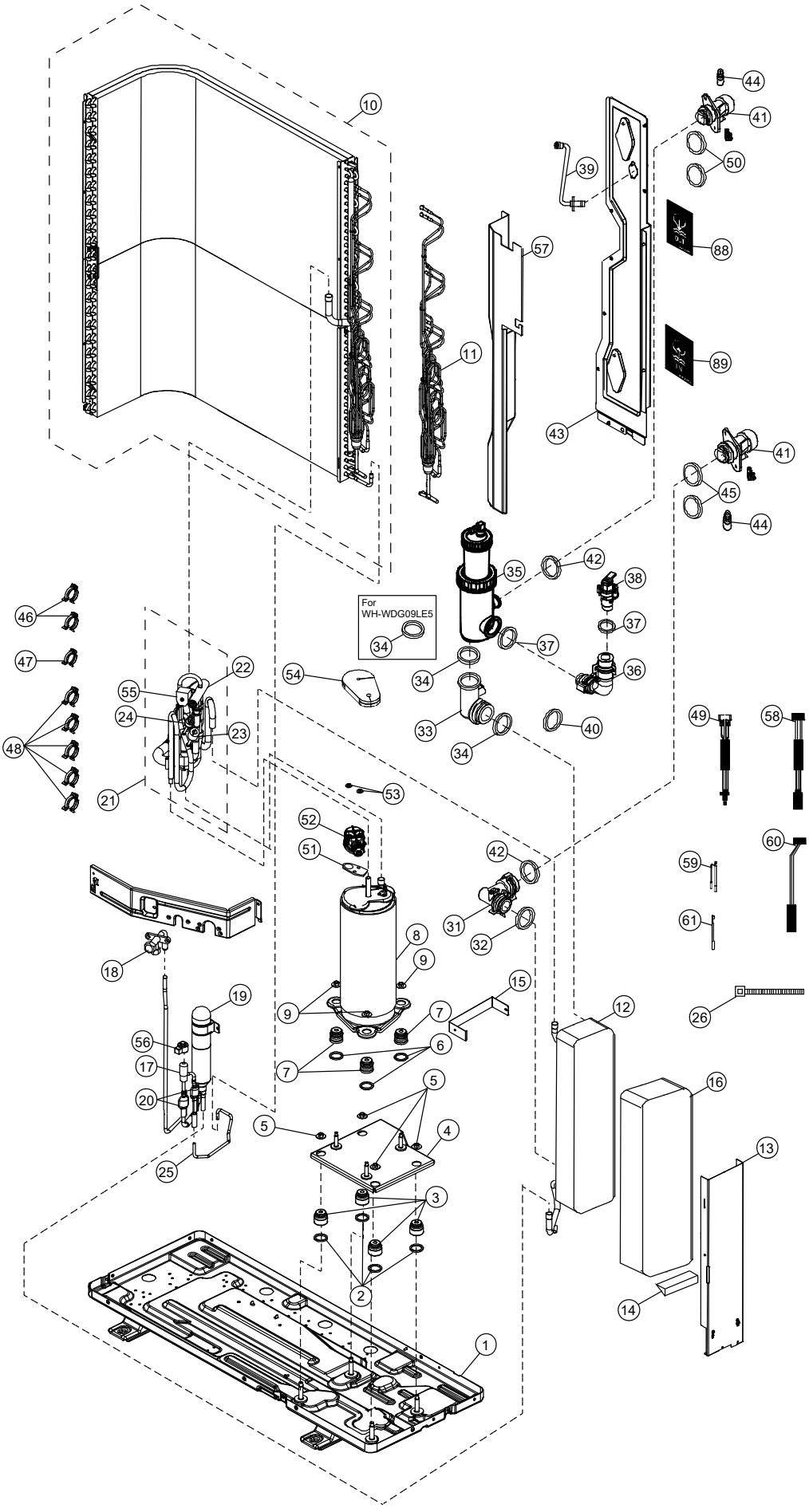
Note:

- All parts are supplied from PHVACCZ, Czech (Vendor Code: 00029407).
- “O” marked parts are recommended to be kept in stock.

21.2 Outdoor Unit



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-WDG05LE5	WH-WDG07LE5	REMARK
	1	BASE PAN ASSY	1	ACXD52K05240	←	O
	2	O-RING (COMPRESSOR)	4	B811017	←	
	3	ANTI - VIBRATION BUSHING	4	ACXH50-00480	←	
	4	FLAT PLATE ASSY	1	ACXD64K00031	←	O
	5	NUT	4	H561049	←	
	6	O-RING (COMPRESSOR)	3	B811017	←	
	7	ANTI - VIBRATION BUSHING	3	H50055	←	
⚠	8	COMPRESSOR	1	ACXB09-09340	←	O
	9	NUT	3	H561049	←	
	10	CONDENSER COMPLETE	1	ACXB32C27490	←	O
	11	MANIFOLD TUBE ASSY	1	ACXT07K10850	←	O
	12	HOT WATER COIL - COMPLETE	1	ACXB90C01860CZ	←	O
	13	PARTICULAR PLATE	1	ACXD90-28980	←	O
	14	PARTICULAR PLATE	1	ACXD90-28990	←	O
	15	PARTICULAR PLATE	1	ACXD90-29010	←	O
	16	FOAMED POLYSTYRENE	1	ACXG07-07800	←	O
	17	EXPANTION VALVE	1	B051029	←	
	18	2-WAYS VALVE	1	ACXB02-03960	←	O
	19	RECEIVER	1	ACXB14-00760	←	O
	20	STRAINER	2	B111024	←	
	21	4-WAYS VALVE COMPLETE	1	ACXB00C03121	←	O
	22	4-WAYS VALVE	1	ACXB00-01520	←	O
⚠	23	HIGH PRESSURE SENSOR	1	ACXA50-06450	←	O
⚠	24	HIGH PRESSURE SWITCH	1	ACXA10-00710	←	O
	25	MULTIBENT TUBE	1	ACXT31-23920	←	O
	26	HOSE BAND	1	4090023	←	
	27	SOUND PROOF MATERIAL	1	ACXG30-13340	←	O
	28	SOUND - PROOF BOARD	1	ACXH15-03931	←	O
	29	SOUND PROOF MATERIAL	1	ACXG30-13380	←	O
⚠	30	FIXED INDUCTORS	1	G0C392J00060	←	O
	31	L-SHAPED TUBE	1	ACXT20-14160	←	O
	32	O-RING (TUBE)	1	ACXB81-06910	←	O
	33	L-SHAPED TUBE	1	ACXT20-14150	←	O
	34	O-RING (TUBE)	2	ACXB81-06910	←	O
	35	FILTER COMPLETE	1	ACXB51C00130	←	O
	36	TUBE ASSY	1	ACXT00-78420	←	O
	37	O-RING (PRESSURE RELIEF VALVE)	2	ACXB81-06820	←	O
⚠	38	RELIEF VALVE	1	ACXB62-00740	←	
	39	TUBE ASSY	1	ACXT00-78430	←	O
	40	O-RING (PRESSURE RELIEF VALVE)	1	ACXB81-06820	←	O
	41	TUBE CONNECTER	2	ACXT29-02620	←	O
	42	O-RING (TUBE)	2	ACXB81-06910	←	O

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-WDG05LE5	WH-WDG07LE5	REMARK
	43	HOLDER - COUPLING	1	ACXH35-02320	←	O
	44	PLUG	2	B821027	←	
	45	O-RING (PLUG)	2	ACXB81-06770	←	
	46	RETAINING RING	2	H581007	←	
	47	RETAINING RING	1	H581038	←	
	48	RETAINING RING	5	ACXH58-00370	←	
⚠	49	SENSOR - CO. WATER IN OUT (CN-TH4)	1	ACXA50C18860	←	O
	50	O-RING (SENSOR)	2	ACXB81-06780	←	O
	51	O-RING (TERMINAL COVER)	1	ACXB81-07090	←	O
	52	TERMINAL COVER COMPLETE	1	ACXH17C00091	←	O
	53	NUT	2	ACXH56-00120	7080300J	
	54	SOUND PROOF MATERIAL	1	ACXG30-13360	←	O
⚠	55	V-COIL CO. (4 WAY VALVE)	1	ACXA43C07580	←	O
⚠	56	V-COIL CO. (EXPENSION VALVE)	1	ACXA43C07590	←	O
	57	PARTICULAR PLATE	1	ACXD90-29000	←	O
⚠	58	SENSOR - CO. OUTDOOR DISCHARGE (CN-DIS)	1	ACXA50C18870	←	O
⚠	59	SENSOR - CO. OUTDOOR HEAT & AMBIENT (CN-TH1)	1	ACXA50C18880	←	O
⚠	60	SENSOR - CO. EVA OUTLET (CN-TH3)	1	ACXA50C18900	←	O
⚠	61	SENSOR - CO. REF TEMP (CN-TH4)	1	ACXA50C18910	←	O
	62	FAN MOTOR BRACKET	1	ACXD54-05030	←	O
⚠	63	FAN MOTOR DC 120W 3PH	1	L6CAYYYL0195	←	O
	64	FAN ASSY	1	ACXH03K01200	←	
	65	WASHER	1	H571075A	←	
	66	NUT	1	H561112A	←	
	67	CONTROL BOARD ASSY	1	ACXH10K03170	←	O
⚠	68	ELECTRONIC CONTROLLER - MAIN	1	ACXA70C00100	ACXA70C00110	O
	69	O-RING (ELECTRONIC CO. - MAIN)	1	ACXB81-07210	←	O
	70	SCREW	6	H551198	←	
⚠	71	BOX SHAPED PLATE - CO. (LEADWIRE)	1	ACXD66C00710	←	O
	72	O-RING (BOX SHAPE PLATE)	1	ACXB81-07140	←	O
	73	SCREW	4	H551198	←	
⚠	74	TERMINAL BOARD ASS'Y	1	A28K1213	←	
	75	SCREW	2	H551198	←	
	76	CONTROL BOARD COVER	1	ACXH13K00910	←	O
	77	O-RING (CONTROL BOARD COVER)	1	ACXB81-07230	←	O
	78	CONTROL BOARD COVER - COMPLETE	1	ACXH13C06430	←	O
	79	CABINET SIDE PLATE - COMPLETE	1	ACXE04C08840	←	O
	80	HANDLE	1	ACXE16-00230G	←	O
	81	CABINET SIDE PLATE	1	ACXE04-13370	←	O
	82	CABINET FRONT PLATE - COMPLETE	1	ACXE06-05250	←	O
	83	HANDLE	1	ACXE16-00230G	←	O

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-WDG05LE5	WH-WDG07LE5	REMARK
	84	CABINET TOP PLATE - COMPLETE	1	ACXE03C02370	←	O
	85	CABINET FRONT PLATE - COMPLETE	1	ACXE06C04900	←	O
	86	HANDLE	1	ACXE16-00230G	←	O
	87	ACCESSORY - COMPLETE	1	ACXH82C24921	←	O
	88	INDICATION LABEL	1	ACXF71-17601	←	O
	89	INDICATION LABEL	1	ACXF71-17611	←	O
	90	MODEL LABEL	2	ACXF87-13521	ACXF87-13531	O
	91	DISCHARGE GRILLE - COMPLETE	1	ACXE20C08791	←	O
	92	DISCHARGE GRILLE - COMPLETE	1	ACXE20C08801	←	O
	93	BAG	1	ACXG86-04431	←	
	94	BASE BOARD - COMPLETE	1	ACXG62C02960	←	O
	95	SHOCK ABSORBER	1	ACXG70-15320	←	O
	96	SHOCK ABSORBER	1	ACXG70-15330	←	O
	97	C.C. CASE	1	ACXG50-63070	←	O

Note:

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SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-WDG09LE5	REMARK
	1	BASE PAN ASSY	1	ACXD52K05240	O
	2	O-RING (COMPRESSOR)	4	B811017	
	3	ANTI - VIBRATION BUSHING	4	ACXH50-00480	
	4	FLAT PLATE ASSY	1	ACXD64K00031	O
	5	NUT	4	H561049	
	6	O-RING (COMPRESSOR)	3	B811017	
	7	ANTI - VIBRATION BUSHING	3	H50055	
⚠	8	COMPRESSOR	1	ACXB09-09340	O
	9	NUT	3	H561049	
	10	CONDENSER COMPLETE	1	ACXB32C27490	O
	11	MANIFOLD TUBE ASSY	1	ACXT07K10850	O
	12	HOT WATER COIL - COMPLETE	1	ACXB90C01880CZ	O
	13	PARTICULAR PLATE	1	ACXD90-28980	O
	14	PARTICULAR PLATE	1	ACXD90-28990	O
	15	PARTICULAR PLATE	1	ACXD90-29010	O
	16	FOAMED POLYSTYRENE	1	ACXG07-07800	O
	17	EXPANTION VALVE	1	B051029	
	18	2-WAYS VALVE	1	ACXB02-03960	O
	19	RECEIVER	1	ACXB14-00760	O
	20	STRAINER	2	B111024	
	21	4-WAYS VALVE COMPLETE	1	ACXB00C03121	O
	22	4-WAYS VALVE	1	ACXB00-01520	O
⚠	23	HIGH PRESSURE SENSOR	1	ACXA50-06450	O
⚠	24	HIGH PRESSURE SWITCH	1	ACXA10-00710	O
	25	MULTIBENT TUBE	1	ACXT31-23920	O
	26	HOSE BAND	1	4090023	
	27	SOUND PROOF MATERIAL	1	ACXG30-13340	O
	28	SOUND - PROOF BOARD	1	ACXH15-03931	O
	29	SOUND PROOF MATERIAL	1	ACXG30-13380	O
⚠	30	FIXED INDUCTORS	1	G0C392J00060	O
	31	L-SHAPED TUBE	1	ACXT20-14160	O
	32	O-RING (TUBE)	1	ACXB81-06910	O
	33	L-SHAPED TUBE	1	ACXT20-14150	O
	34	O-RING (TUBE)	1	ACXB81-06910	O
	35	FILTER COMPLETE	1	ACXB51C00130	O
	36	TUBE ASSY	1	ACXT00-78420	O
	37	O-RING (PRESSURE RELIEF VALVE)	2	ACXB81-06820	O
⚠	38	RELIEF VALVE	1	ACXB62-00740	
	39	TUBE ASSY	1	ACXT00-78430	O
	40	O-RING (PRESSURE RELIEF VALVE)	1	ACXB81-06820	O
	41	TUBE CONNECTER	2	ACXT29-02620	O
	42	O-RING (TUBE)	2	ACXB81-06910	O
	43	HOLDER - COUPLING	1	ACXH35-02320	O

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-WDG09LE5	REMARK
	44	PLUG	2	ACXH35-02320	
	45	O-RING (PLUG)	2	ACXB81-06770	
	46	RETAINING RING	2	H581007	
	47	RETAINING RING	1	H581038	
	48	RETAINING RING	5	ACXH58-00370	
⚠	49	SENSOR - CO. WATER IN OUT (CN-TH4)	1	ACXA50C18860	O
	50	O-RING (SENSOR)	2	ACXB81-06780	O
	51	O-RING (TERMINAL COVER)	1	ACXB81-07090	O
	52	TERMINAL COVER COMPLETE	1	ACXH17C00091	O
	53	NUT	2	H561049	
	54	SOUND PROOF MATERIAL	1	ACXG30-13360	O
⚠	55	V-COIL CO. (4 WAY VALVE)	1	ACXA43C07580	O
⚠	56	V-COIL CO. (EXPENSION VALVE)	1	ACXA43C07590	O
	57	PARTICULAR PLATE	1	ACXD90-29000	O
⚠	58	SENSOR - CO. OUTDOOR DISCHARGE (CN-DIS)	1	ACXA50C18870	O
⚠	59	SENSOR - CO. OUTDOOR HEAT & AMBIENT (CN-TH1)	1	ACXA50C18880	O
⚠	60	SENSOR - CO. EVA OUTLET (CN-TH3)	1	ACXA50C18900	O
⚠	61	SENSOR - CO. REF TEMP (CN-TH4)	1	ACXA50C18910	O
	62	FAN MOTOR BRACKET	1	ACXD54-05030	O
⚠	63	FAN MOTOR DC 120W 3PH	1	L6CAYYYL0195	O
	64	FAN ASSY	1	ACXH03K01200	
	65	WASHER	1	H571075A	
	66	NUT	1	H561112A	
	67	CONTROL BOARD ASSY	1	ACXH10K03170	O
⚠	68	ELECTRONIC CONTROLLER - MAIN	1	ACXA70C00120	O
	69	O-RING (ELECTRONIC CO. - MAIN)	1	ACXB81-07210	O
	70	SCREW	6	H551198	
⚠	71	BOX SHAPED PLATE - CO. (LEADWIRE)	1	ACXD66C00710	O
	72	O-RING (BOX SHAPE PLATE)	1	ACXB81-07140	O
	73	SCREW	6	H551198	
⚠	74	TERMINAL BOARD ASS'Y	1	A28K1213	
	75	SCREW	6	H551198	
	76	CONTROL BOARD COVER	1	ACXH13K00910	O
	77	O-RING (CONTROL BOARD COVER)	1	ACXB81-07230	O
	78	CONTROL BOARD COVER - COMPLETE	1	ACXH13C06430	O
	79	CABINET SIDE PLATE - COMPLETE	1	ACXE04C08840	O
	80	HANDLE	1	ACXE16-00230G	O
	81	CABINET SIDE PLATE	1	ACXE04-13370	O
	82	CABINET FRONT PLATE - COMPLETE	1	ACXE06-05250	O
	83	HANDLE	1	ACXE16-00230G	O
	84	CABINET TOP PLATE - COMPLETE	1	ACXE03C02370	O
	85	CABINET FRONT PLATE - COMPLETE	1	ACXE06C04900	O
	86	HANDLE	1	ACXE16-00230G	O

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-WDG09LE5	REMARK
	87	ACCESSORY - COMPLETE	1	ACXH82C24921	O
	88	INDICATION LABEL	1	ACXF71-17601	O
	89	INDICATION LABEL	1	ACXF71-17611	O
	90	MODEL LABEL	2	ACXF87-13541	O
	91	DISCHARGE GRILLE - COMPLETE	1	ACXE20C08791	O
	92	DISCHARGE GRILLE - COMPLETE	1	ACXE20C08801	O
	93	BAG	1	ACXG86-04431	
	94	BASE BOARD - COMPLETE	1	ACXG62C02960	O
	95	SHOCK ABSORBER	1	ACXG70-15320	O
	96	SHOCK ABSORBER	1	ACXG70-15330	O
	97	C.C. CASE	1	ACXG50-63070	O

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